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NESA EVOLUTION USER MANUAL

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1 Introduction

evolution represents the natural development of the already known and reliable TMF series of NESA dataloggers, with a product equipped with the most modern data acquisition technologies that combines the highest level of performance with improved and complete ease of use. A modular open datalogger with very high precision, equipped with a **Linux**, web server **Apache** and **PhP** interpreter operating system, , usable and configurable via the web without the need for any external or proprietary software.

evolution contains all the computing power of an **ARM® 32 bit processor,** managed across the knowledge of over twenty years of experience in **Linux Embedded** operating systems, and with a high technical ability in processing, calculation and data processing methodologies and algorithms.

evolution has a **superior ability to acquire signals** in a practically unlimited number of I/O and has different types of memory for the protection of data or copies of them, managed at different levels of security.

evolution offers transmission and data transmission functions that allow the use of all modern broadcast channels (modem, radio, mobile phone network, satellites, copper or fibre cable, etc.), also in redundancy between them (**transmission redundancy**); it makes available most of the widespread protocols between information sharing systems, for example RS232, RS485, Modbus, TCP-IP, SMTP, FTP, Socket TCP, SDI12, SNP and many others. The structure of the stored and / or transmitted record or data track offers a wide range of customisation, starting from the simple structure of the Nesa, being able to change in the sequences and contents, as well as in the separation characters.

Finally, evolution **has an innovative user management system**, dividing them between master user and normal user, allowing the sharing of machine resources, while creating different configurations for the user as if they were different dataloggers. In this manual a specific section is dedicated to the topic.
On-board software is multilingual with possibility of choice.

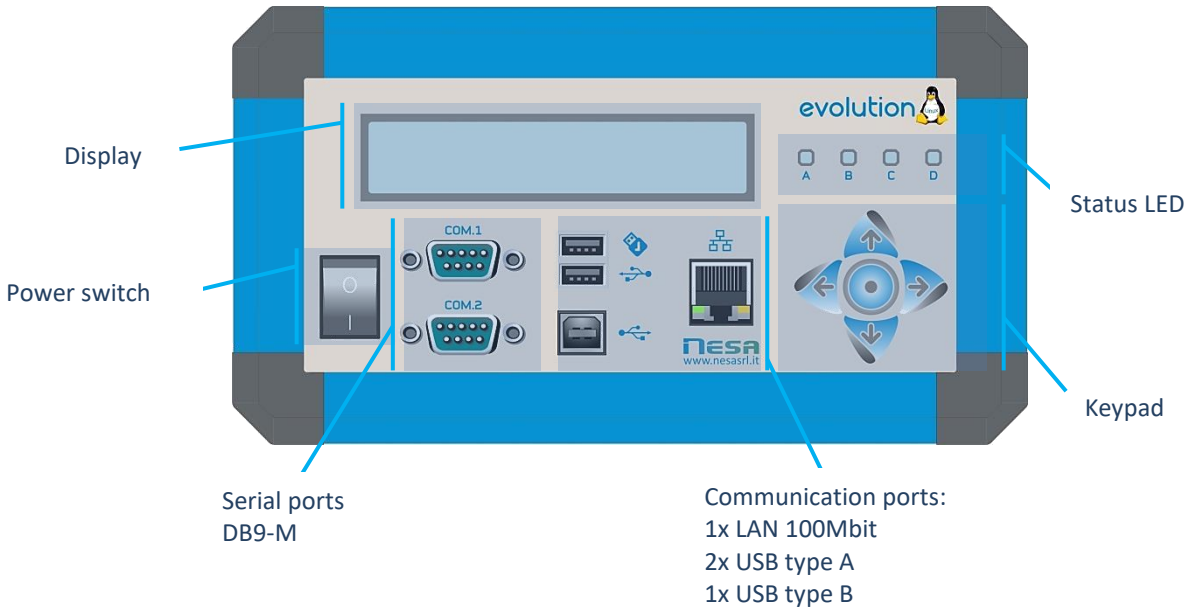
1.1 Contents of the package

Before the installation check that the contents of the package correspond to the following:

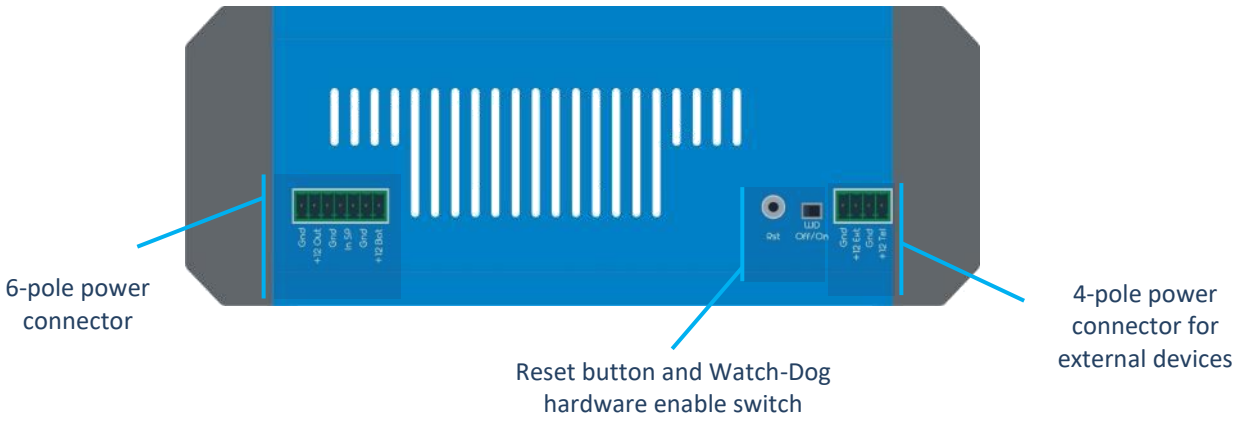
- no.1 evolution terminal complete with extractable Phoenix-type screw connectors
- no.1 USB dongle key minimum size 8GB or higher.
- no. 1 crossed RJ45 Ethernet cable (LAN cross cable).
- Calibration report with annual validity from the first active configuration.

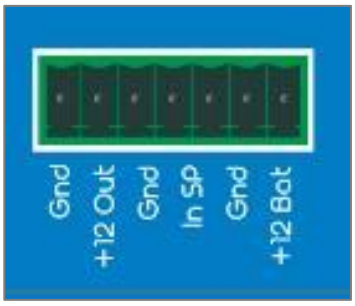
The datalogger is supplied calibrated, tested, and equipped with an example configuration for the acquisition of a set of measurements.

Front



Rear panel





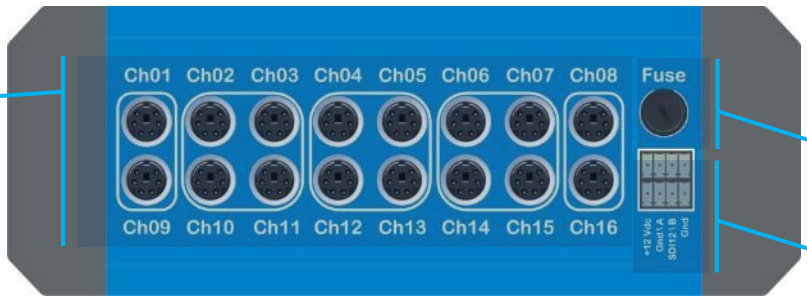
6-pole power connector	
Name	**Function**
+12 Bat	Battery-powered (+10.8Vdc÷ 15Vdc)
Gnd	Grounding clamp (3 terminals)
In.S.P.	Power input from solar panel or AC/DC power supply (220Vac/13.8Vdc)
+12 Out	Auxiliary power supply output (max 3A)



4-pole connector	
Name	**Function**
+12 Tel	Telephonic terminal power supply ONLY *(status is managed by datalogger automatically)*
Gnd	Grounding clamp (2 terminals)
+12 Ext	Auxiliary power supply output (max 3A)

NOTE: do not use external charger with internal charger, this can damage the datalogger. In case ask to NESA how to proceed

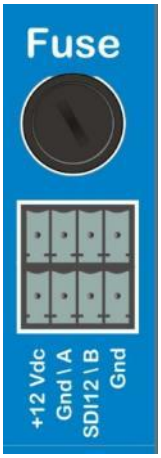
Front panel



6-pin DIN 6-pin interface connectors for analog and digital I/O channels

Fuse 5x20mm Capacity 5 A

RS485 and SDI12 input



4-pole I/O double connector		
Name	**Lower position**	**Upper position**
+12 Vdc	Powered output (1A)	Powered output (1A)
GND/A	Mass line SDI12	RS485 A
SDI12/B	Line SDI12	RS485 B
Gnd	Ground clamp	Ground clamp

1.2 First start-up

Before connecting the datalogger, make sure that the power button is in the 0 (Off) position.

Connect the power supply to the datalogger through the special 6-pole connector, using a battery (12Vdc minimum 1A / h), or a 220Vac/12Vdc wall power supply according to the following procedure:

a) If you have battery-only, connect its terminals to the **+12Batt** and **Gnd** inputs.

b) If you only have a wall power supply unit , connect the terminals to the **+12Batt** and **Gnd** inputs.

c) If you have a wall-mounted power supply unit + battery, connect the battery terminals to the **+12Batt** and **Gnd** inputs, and the power supply unit to the **InPS** and **Gnd** inputs. In this configuration the battery is automatically recharged (max. charging current 5A) from the datalogger.

NOTE: *As the datalogger has an internal battery charger, the supply voltage from external power supply should be at least 13.5-13.8Vdc.*

Turn the power button to the 1 (ON) position. 

In this phase it is possible, through the sequence of the status LEDs, to verify the correct ignition of the machine.

In particular:

1. When the power button is set to ON the LED **A** is on. 

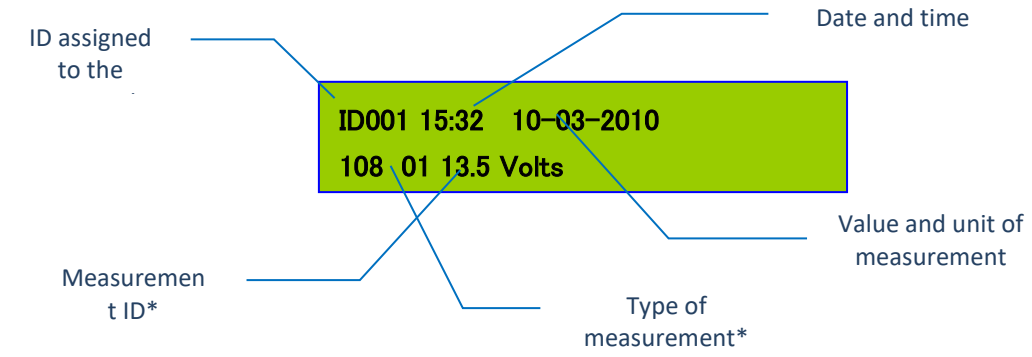
2. Depending on the type of power supply (battery, or battery plus panel or battery + power supply) the LEDs **A** and **B** will light up, which remain lit to signal the presence of the power source. In particular LED B will only be on if there is battery connected and charging.

3. The **C** and **D** status LEDs are normally off, only flashing if there is an anomaly in the system start-up phase, signalling an operating system kernel anomaly.

Status Led		
Name	**Colour**	**Function**
A	Green	Power on of Led (always on)
B	Red	Battery charge Led (only during charge time)
C	Yellow	Boot anomaly signalling Led
D	Yellow	Boot anomaly signalling Led

The ignition sequence is also shown in the display, where the machine starts and diagnostics messages appear (see paragraph 9); of these the first one represents the serial number of the datalogger.
The entire start-up sequence of the datalogger takes approximately 90 seconds.

Once the start-up is complete, the datalogger starts the standard acquisition cycle and the display of the instantaneous data in the display has the following form:



* For decoding see appendix A and B

The measurements are displayed cyclically at intervals of approximately 2 seconds, passing from one measurement to another depending on the configuration of the machine.

1.3 Connection to the datalogger

The datalogger can be immediately connected to a computer, laptop or PDA, through the Lan network interface. It leaves the factory with a pre-set IP address, and subsequently modifiable, to which reference must be made for connection.

LAN interface:			
IP:	**192.168.1.115**	Subnet mask:	**255.255.255.0**



To connect to the machine it is necessary to use the supplied Ethernet cable or any cross (cross) type cable following the procedure below:

1. Leave the datalogger on or switch it on.
2. Connect the cable to the LAN port of your computer.



3. Connect the other end of the cable to the datalogger Ethernet port.
4. Turn on the computer and set an IP address of your network card congruent with that of the datalogger, for example 192.168.1.10 and subnet mask 255.255.255.0.

At this point the connection between computer and evolution is established and it is possible to access configuration pages or data visualization pages as explained in the next paragraphs.

NOTE: *The data logger gets off the connection from browser after 30 minutes if there aren't any input from user. This time interval depends on the Browser used.*

2 Web interface

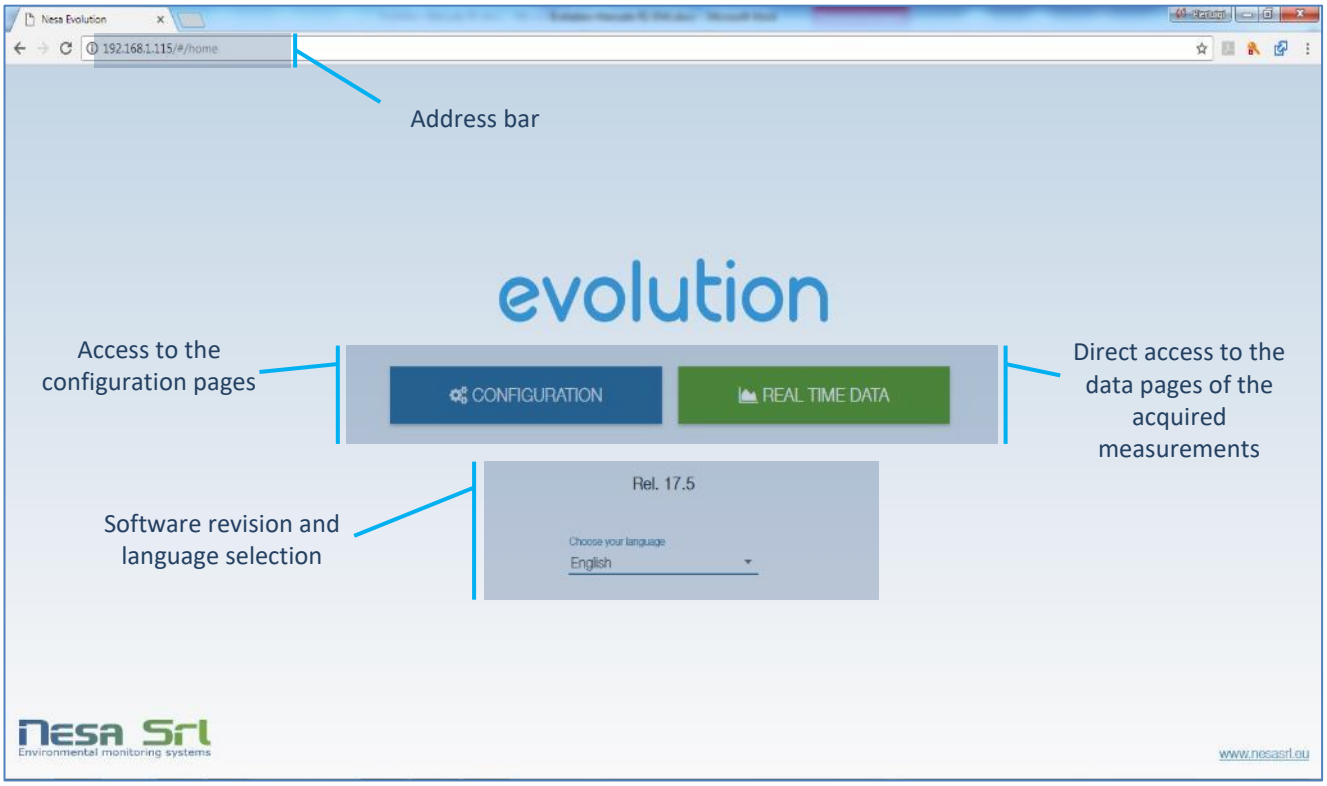
To access the configuration and/or display data page of the machine, it is sufficient, after establishing the connection as in the previous paragraph, to open one of the following Internet browsers for which the datalogger is compatible:

- ✓ **Chrome** (of Version 63.0.3239.132)
- ✓ **Firefox** (of 57.0.4)
- ✓ **Edge** (of 41.16)

The configuration procedure of an evolution terminal only takes place through the use of web pages, while the display of data is also possible on the display.

For correct display of the pages, enable javascript scripts and the option to ***search the most recent versions of the stored pages*** in the tools or options menu of each browser

With the browser active, type in the address bar the IP of the datalogger: http://192.168.1.115 (factory address) and wait for opening of the main page that allows the general user to choose between viewing the instantaneous data and configuring the datalogger.



2.1 Authentication

Access to any datalogger resource always requires initial authentication.

Authentication can take place as a primary user (**MASTER user**) or as a secondary user (**NORMAL User**), depending on whether you have full access to all functions or only a more limited part.

A MASTER user has access to all the resources of the datalogger and to the management of the credentials of secondary users, in particular they can:

- Change the IP and the Subnet Mask of the datalogger
- Change Master user password
- Create, edit or delete up to a maximum of 5 NORMAL users
- Access the advanced functions of the datalogger

A NORMAL user can use all the functions of the datalogger except for those of the MASTER user, that is, they can generate, modify, delete configurations associated with that NORMAL user, view and download data, and program the datalogger.

This advanced mode of evolution allows generating virtual dataloggers on the same machine, as if they were completely independent acquisition systems that use the same hardware resources. It is possible, for example, to create two or more different configurations that acquire data from different sensors, processing them, treating them and storing them (with or without remote sending), in a manner different from each other. For each configuration, the datalogger associates a different memory area for the configuration, for the data and for the management of the transmission buffer.

Every NORMAL user can only see and manage their own configuration and data, without knowing the existence of other users on the same machine; only the MASTER can see it all indifferently.

From the factory the datalogger always exits with a pre-enabled **MASTER user** to which up to **5 secondary users** can be associated. The factory credentials of the MASTER user are:

username: **root** password: **root**

At the touch of one of the two keys, the 

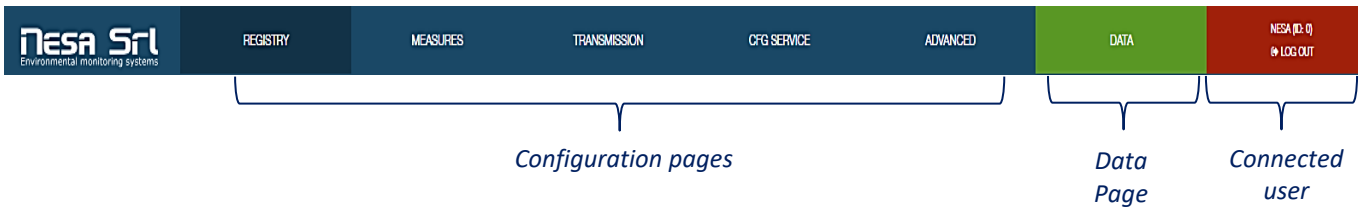
authentication page opens, where, by entering the credentials of the selected user, you have access to the pages relating to the acquired data, or to the configuration ones, according to the authorisation associated to the logged user.



2.2 Access as MASTER User

The MASTER user has all the access privileges to all the functions of the machine.

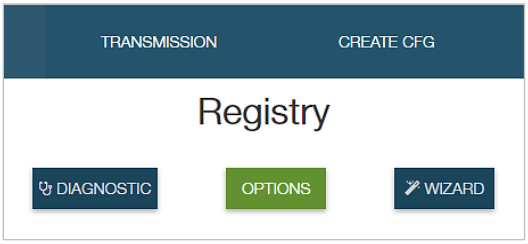
The web pages of evolution have in common a simple and efficient menu divided into three coloured areas:



- The Blue zone pages are used to configure the datalogger and can be differentiated according to whether the connected user is MASTER or NORMAL.
- The pages of the Green area allow the visualisation of data in numerical and graphic format
- The Red zone identifies the user connected with an ID.

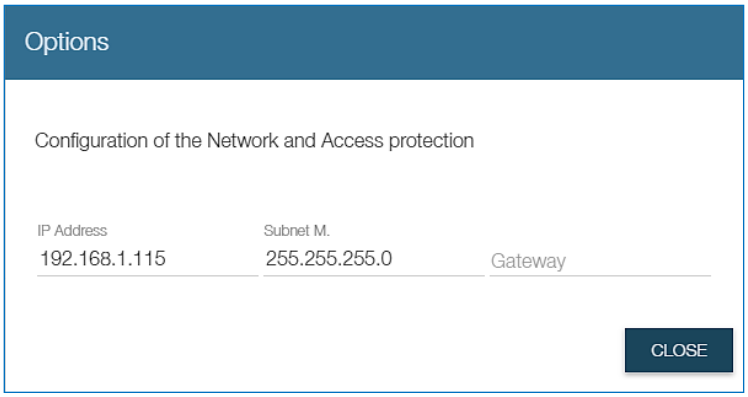
The pages accessible only to the Master user will be highlighted below, while in the next section the pages shared by all NORMAL users will be explained with standard access privileges, pages also common to the MASTER user.

The MASTER user on the *MASTER DATA* page has the Options button not visible to NORMAL users enabled.



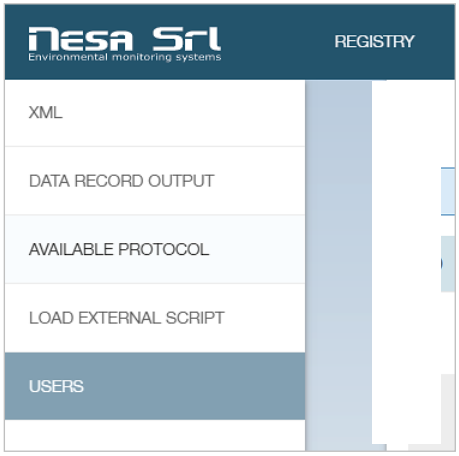
Page ***Master data -> Options***

The *Options* menu contains information on the *IP address* of the datalogger, of the *Subnet Mask* and allows entering of a possible *Gateway* (if any) for exit of the network communication packets.



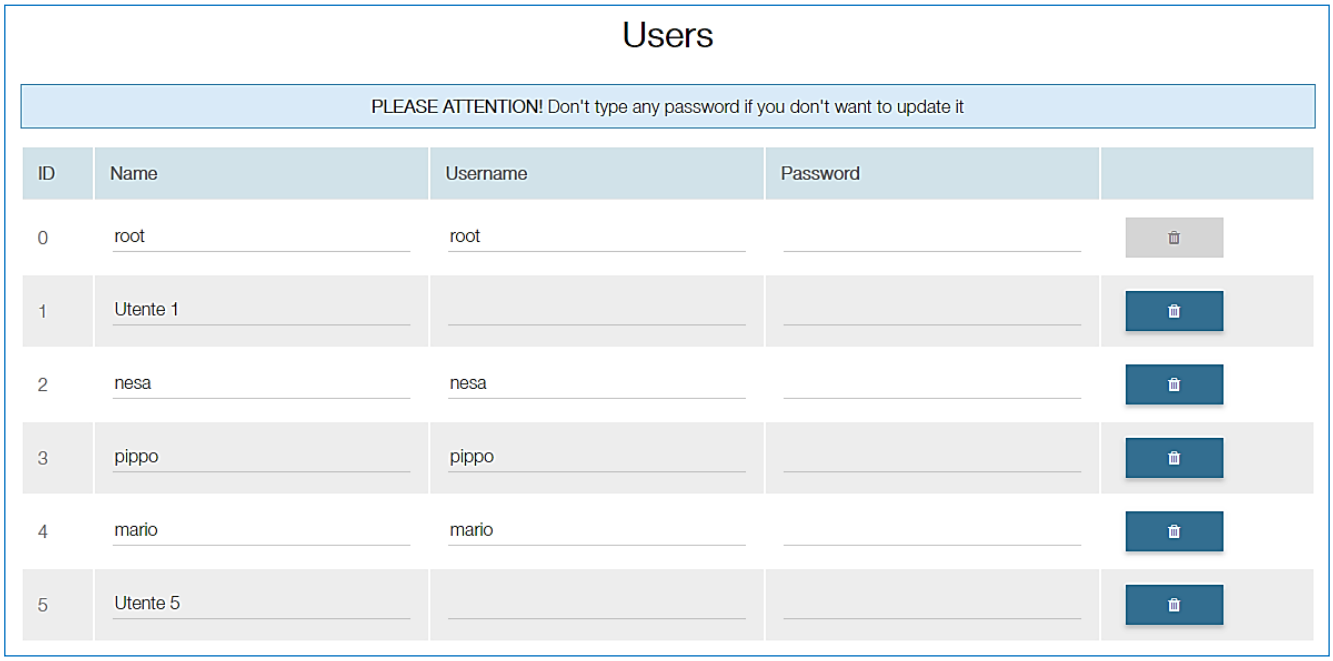
These are clearly unique features for the hardware and can be uniquely set by the MASTER user.

All modifications become operative by pressing the SAVE PAGE key button. On the *ADVANCED* page the master user has available the Users section not visible to NORMAL users.



Page ***Advanced -> Users***

It allows the MASTER user to manage NORMAL users, here it is possible to add, delete users or modify their credentials from 1 to 5; user 0 identifies the MASTER.



The password of the same MASTER user (ID = 0) is editable on this page, a pop-up appears after saving to prompt keeping track of the credentials.

Each user can also be disabled through the special switch called "*Enable / Disable cfg*". In case it is not enabled, all the actions associated to the user (memorization, sending data ...) will be suspended.

NOTE: *MASTER password recovery occurs only by contacting NESA, which will send the recovery procedure.*

Changing the password is done simply by changing the field relative to the chosen user. If this field is left blank, the password will remain the one previously assigned.

At the end of the changes, it is necessary to confirm them by pressing the SAVE PAGE button at the bottom of the page, while they will become effective after the restart of the datalogger through the corresponding button CREATE AND APPLY in the Create CFG page.

At the end of any modification to the configuration, is necessary to confirm them by pressing to SAVE PAGE make them effective.

2.3 Log in as a NORMAL User

Every evolution's user, from the operational point of view, has:

a. a dedicated space in the file-system of the datalogger, with folders and configuration files only associated with that user.
b. completely separate management of the data records created.
c. independent management of the process of sending the created files.
d. The possibility to create specific customised configurations.

Access to the datalogger takes place either via the configuration button or through the data display button:



It is always possible to move from one page to another.

2.3.1 Viewing data in Real Time

If the authentication has been carried out by pressing the key you 

are redirected directly to the page of instantaneous data acquired by the datalogger according to the set configuration. Otherwise, you can always reach this page from any other point by clicking on the green button at the top of the menu as shown in the paragraph.2.2.

The page opens on measurements in tabular format, whose fields and values are shown and updated in a few seconds. On the **left** there are 3 sections (**MEASUREMENTS, GRAPHS and MAP**) which correspond to similar manners of displaying information.

At the top of the page, common to each section, is some information of the datalogger for its identification: the name, the ID, the date of the client computer connected to the datalogger (User Data) and the date and time of the measurement (to recognise any time zone). Also through the indicated link, it is possible to **download the data in Excel format (.csv) of the current month or previous month** with respect to the datalogger date.



The file that is downloaded is a **CSV** (**C**omma-**S**eparated **V**alues) file type, which can be opened with a spreadsheet, and whose name shows the following format:



user_id – terminal_id-current_month.csv

user_id – t erminal_id -previous_month.csv

where *user_id* is the numerical identifier of the user who has logged in, *terminal_id* represents the numerical identifier associated with the machine in configuration and *current* and *previous* are respectively the file of the current month and that of the previous month.

For example, the file *4-101-current_month.csv* is the file containing the data of the current month of the datalogger with ID 401 created by the user 4.

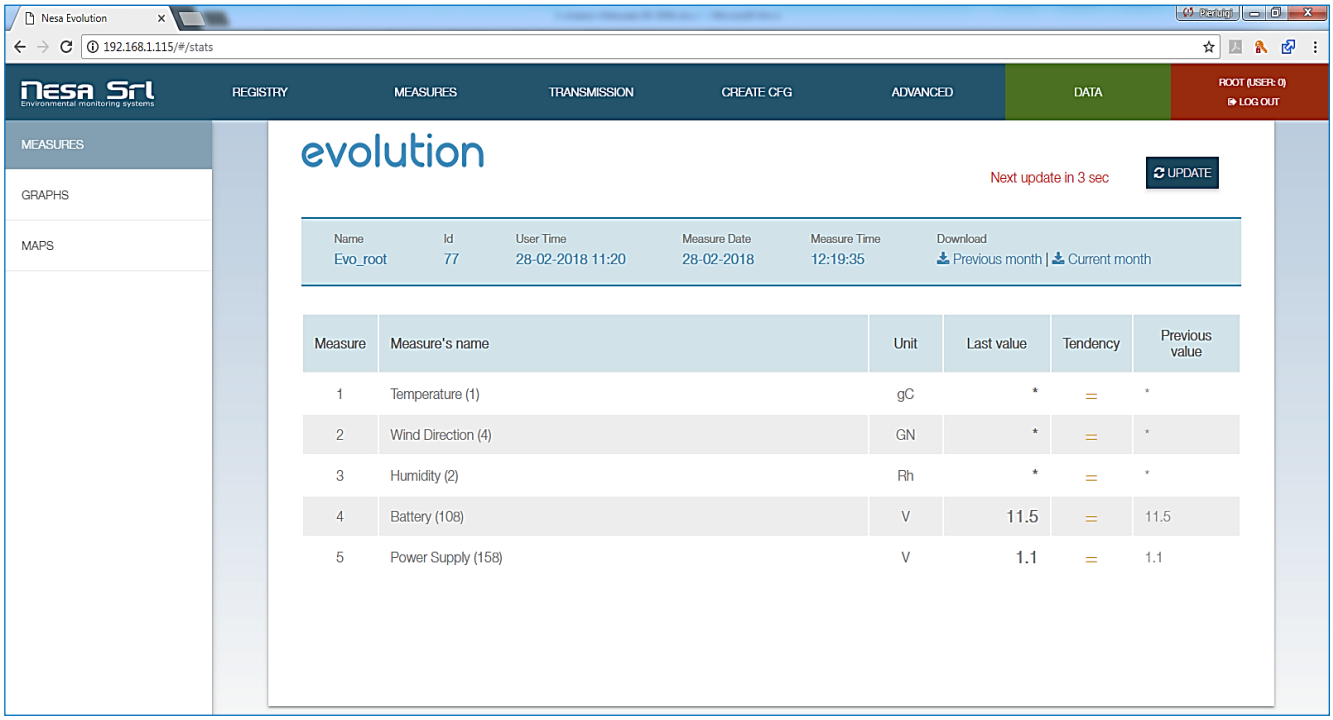
Open in a spreadsheet, the file contains column labels to ascertain the date and time of storage (every row) and the measurement with relative processing associated with the data in the column.

An example:

	A	B	C	D	E	F	G	H
1	Dati	Dal 14-02-2018	Al 15-02-2018					
2	Date	Hour	Snow Level Avarage	Snow Level Min	Snow Level Max	Batteria Instant	Alimentazione Instant	
3	14/02/2018	16:38:00	1795.5	1794.5	1796.5	12.7	0.8	
4	14/02/2018	16:42:00	1741.7	6.9	1795.0	12.7	0.8	
5	14/02/2018	16:44:00	1794.5	1794.3	1795.0	12.7	0.8	
6	14/02/2018	16:46:00	1794.9	1794.6	1796.2	12.7	0.8	
7	14/02/2018	16:48:00	1795.3	1794.4	1795.9	12.7	0.8	
8	14/02/2018	16:50:00	1795.5	1794.6	1795.7	12.7	0.8	
9	14/02/2018	16:52:00	1795.5	1795.1	1795.9	12.7	0.8	
10	14/02/2018	16:54:00	1795.4	1795.0	1795.9	12.7	0.8	
11	14/02/2018	16:56:00	1795.2	1794.9	1795.6	12.7	0.8	
12	14/02/2018	16:58:00	1795.5	1795.1	1795.9	12.7	0.8	
13	14/02/2018	17:00:00	1795.4	1794.6	1795.8	12.7	0.8	
14	14/02/2018	17:02:00	1795.6	1794.9	1795.9	12.7	0.8	
15	14/02/2018	17:04:00	1796.4	1795.1	1797.7	12.7	0.8	
16	14/02/2018	17:06:00	1794.9	1793.5	1797.8	12.7	0.8	
17	14/02/2018	17:08:00	1795.2	1793.8	1795.8	12.7	0.8	
18	14/02/2018	17:10:00	1795.0	1794.2	1796.3	12.7	0.8	
19	14/02/2018	17:12:00	1795.5	1795.1	1795.8	12.7	0.8	
20	14/02/2018	17:14:00	1795.8	1794.4	1796.3	12.7	0.8	
21	14/02/2018	17:16:00	1795.3	1794.5	1796.0	12.7	0.8	

MEASUREMENTS section

This section summarises in a user-friendly format all the information concerning the measurements acquired and available to the user who has logged in.



In the central part of the page is the information associated with the acquired measurements:

Measure	Measure's name	Unit	Last value	Tendency	Previous value
1	Temperature (1)	gC	*	=	*
2	Wind Direction (4)	GN	*	=	*
3	Humidity (2)	Rh	*	=	*
4	Battery (108)	V	11.5	=	11.5

- ✓ **Measure** represents the sequence of measurements according to the saved configuration
- ✓ **Measure's name** Name of the measurements according to the saved configuration with possible ID
- ✓ **Unit** engineering unit associated with the measurement according to the saved configuration
- ✓ **Last value** represents the last value acquired by the datalogger
- ✓ **Tendency** Change in the last measurement compared to the previous value
- ✓ **Previous value** represents the value of the previous acquisition cycle of the datalogger

The page values are updated automatically every 10 seconds or can be forced by pressing the button UPDATE or chosen on the upper right menu.

NOTE: *the Nesa dataloggers, calculate the instantaneous value of a measurement (in real time) through a Gaussian average on a very high number of samples, in other words the data per second is not the result of an analogical digital conversion of the electronics, albeit of the highest level, but the result of a weighted average over a Gaussian bell, of an average number of over 100 samples.*

***GRAPHICS* Section**

The graphs section allows representation with a powerful graphical engine of the values acquired according to one or more Cartesian diagrams based on time, auto-adaptive on the scale of the acquired values, and independent for each measurement.

Each diagram is automatically populated in real time and cyclically over periods of 10 seconds, displaying the value acquired at the time instant with a dot. All measurements in the configuration are displayed in different colours.



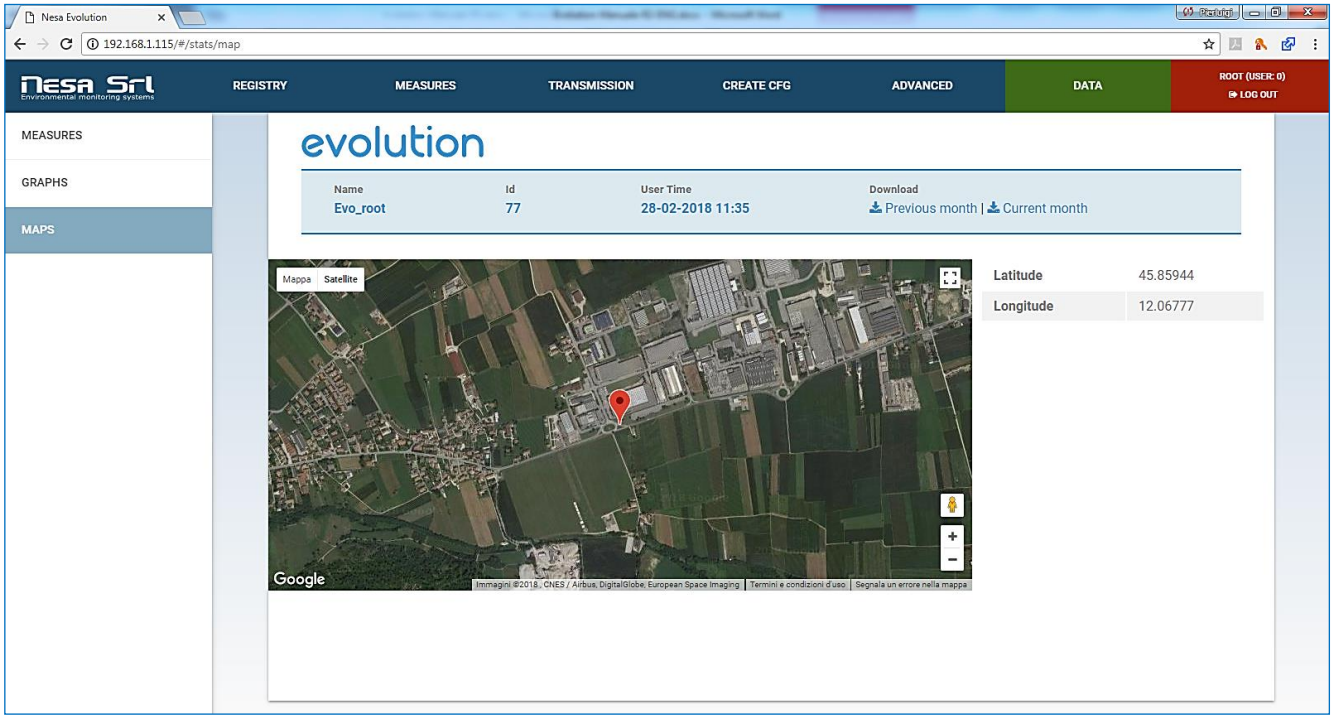
On the right the identification label of the measurement is always present.

NOTE: *Even if not present in the configuration, are always shown both in the Measurements section and in the Graphics section, two measurements called "Voltage" (in Volt) and "Power" (in Volt), which represent the two additional analogue channels of the datalogger, used in a native way for measurement of the primary power value, typically the battery, and the secondary one, typically the photovoltaic panel or a direct voltage power supply or other source.*

MAP Section

The Map section allows geo locating of the datalogger. It uses the GPS Latitude and Longitude coordinates entered during configuration, or acquired from a GPS antenna connected to the same datalogger.

The map representation passes through a connection to the Google Maps® site and through a library of tools made available for this purpose. Obviously, the map only appears when the datalogger is connected to the network.



On the right are the coordinates in **WGS84 GD decimal format.**

2.3.2 *Configuration*

If the authentication has been carried out by pressing the key, you 

will be redirected to the first page (ANGLE) to start the configuration sequence of the datalogger.

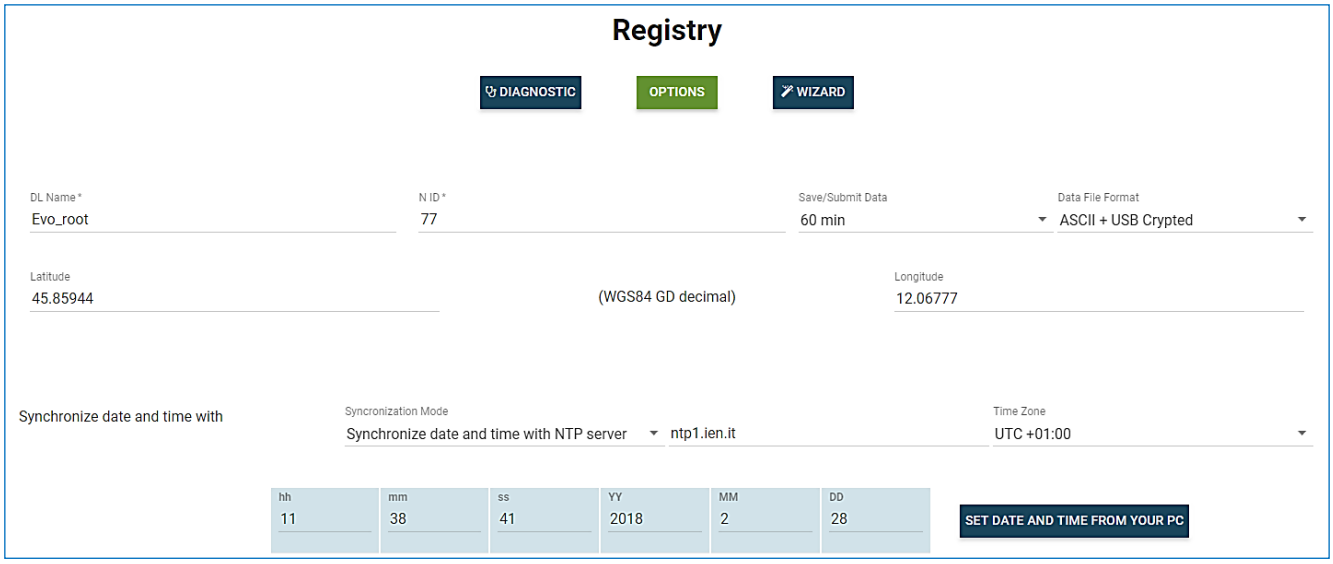
All configuration pages are similar and characterised by a pair of buttons in the bottom right:



The "Save Page" key allows the saving of any changes made to the individual page, automatically generating the new configuration without restarting the datalogger.

The "Reset" key allows resetting of the valuesof each field of the page.

2.3.2.1 ***MASTER DATE page***



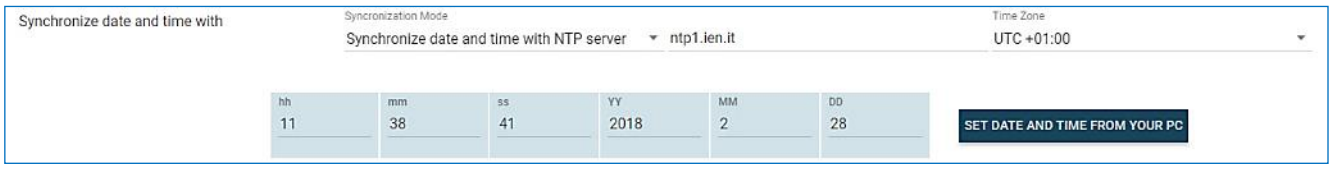
This page contains all the *information that characterises the datalogger* (it can be different for each user).

In the **upper part** we have:

- ✓ DL name — name of the alphanumeric terminal
- ✓ ID No. — numeric identification of the terminal (max 6 digits)
- ✓ Save/Submit Data — saving and/or sending of the file (data, images, etc.), from 1 to 1440 minutes
- ✓ Data file format — formatting applied to the data file: *NESA ASCII with comma as separator, or dot/semicolon, NESA ASCII with comma, dot/semicolon Encrypted, custom format*

In the **central part** of the page, it is possible to enter the *GPS latitude and longitude coordinates* in WGS84 GD decimal mode. If a GPS antenna is connected to the datalogger, the coordinates are read directly from the antenna, so you do not need to fill in these fields.

At the **bottom** of the page are the *clock settings* inside the datalogger, i.e. date and time, time zone and, if provided by the connection to the network or to an external device, the type of synchronisation to be applied, with indication of the NTP server on which to engage.



The clock synchronisation possibilities are as follows:

- ✓ None: the datalogger keeps the internal date and time since the last adjustment made.
- ✓ Synchronise date and time with NTP server it is necessary to insert the address in an NTP server
- ✓ Set date and time from GPS (if present) the type of GPS used, must be chosen from a drop-down list
- ✓ Set date and time from SATELLITE (if present): select the type of satellite used from a drop-down list
- ✓ Set date and time by synchronising them from PC. In this case, the data logger will automatically show the updated time on the display after a few seconds. This action also corresponds to an automatic restart of the acquisition process, as it is necessary to re-schedule all the activities that the data logger had in progress before the date and time was changed.

NOTE TO SET THE TIME ZONE CORRECTLY

Select the time zone to apply from the drop-down menu.

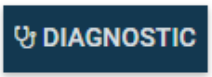
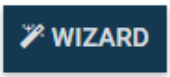
Press SET DATE AND TIME FROM YOUR PC

Press SAVE PAGE

The time zone change occurs at midnight and, in the case of synchronization via NTP, it will be automatically aligned.

Once you have entered the information of interest for this page, confirm it by pressing or cancel SAVE PAGE it with the button RESET otherwise, changing the page it will be lost.



There are also two buttons on the page:  

Pressing the "*Diagnostics*" key a procedure is launched in the datalogger which allows extraction of a text file representing the functional diagnosis of the datalogger operating system. It contains some important information that allows understanding of the operating status, including the amount of total memory, the memory used and the available memory, both program memory and data memory.
It also contains a list of all active processes, to immediately identify a blocked or shutdown process.

An example of how this file can appear:

```
Filesystem                Size      Used Available Use% Mounted on
ubi0_0                   53.9M     33.8M     17.3M  66% /
devtmpfs                 57.1M      4.0K     57.1M   0% /dev
tmpfs                    16.0M    360.0K     15.6M   2% /var/volatile
tmpfs                    57.1M         0     57.1M   0% /dev/shm
tmpfs                    10.0M     32.0K     10.0M   0% /home/httpd/sito
ubi0:nesa                49.2M      6.8M     39.8M  15% /mnt/nand
PID USER VSZ COMMAND
13355 daemon 83036 /usr/sbin/httpd
13597 daemon 80988 /usr/sbin/httpd
13587 daemon 80988 /usr/sbin/httpd
```

The system file also identifies the memories available for each user and is a file that can be transmitted remotely by setting in the datalogger, among the types of data sent, also the system file (see paragraph 0).

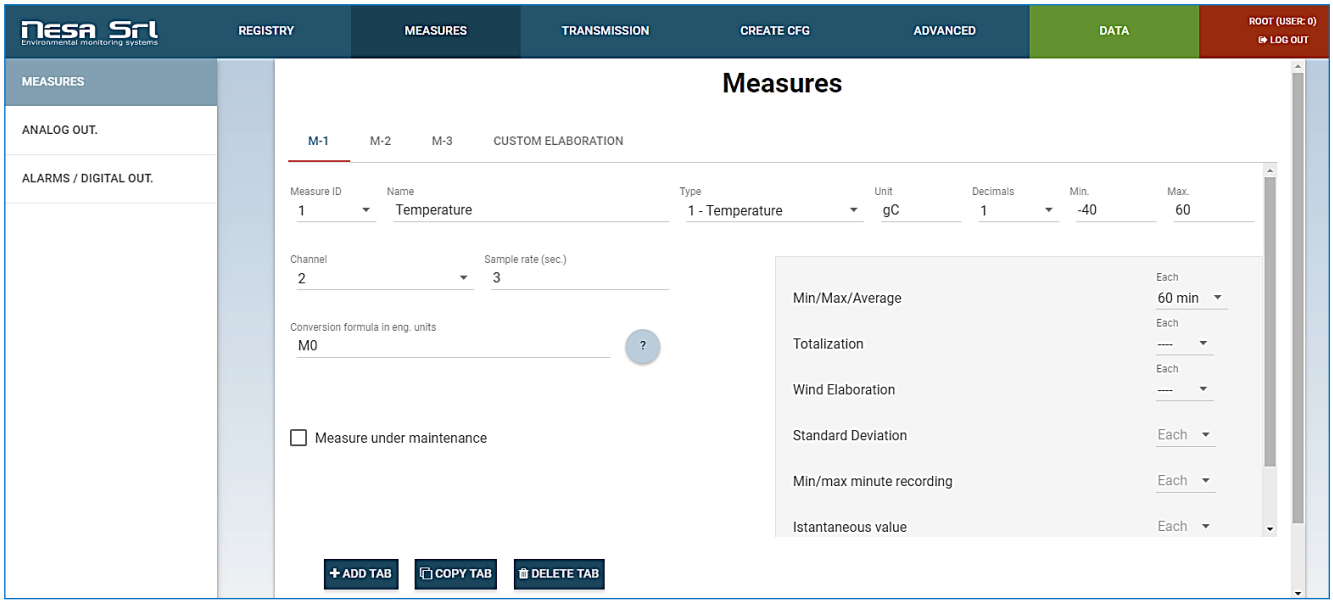
For the "Wizard" button it is not active in this version of evolution.

2.3.2.2 MEASUREMENETS page

From the *MEASUREMENTS* page it is possible to configure and customise all the digital or serial analogue acquisition functions of the datalogger, as well as to generate analogue or digital outputs, alarms, etc., by associating each of the specific preloaded or user-created elaborations. These configurations may be different for each user.
The page shows a series of sections on the **left** and is positioned upon opening on the homonymous one.

MEASUREMENT Section

This first section is organised into tabs, each corresponding to a measurement identifiable with M-1; M-2,…M-n, depending on the number of measurements inserted or to be inserted in the configuration. It is possible to switch from one measurement to another by clicking on the corresponding tab. (in the figure below are measurements identifiable by the tabs M-1, M-2, M-3).



At the bottom of the page there are three buttons:

+ ADD TAB it allows adding of a new measurement to be acquired or a new tab.

COPY TAB it allows cloning of an already existing measurement or cloning of a tab.

DELETE TAB It allows deletion of an already existing measurement or deletion of a tab.

The tab of each measurement presents a series of fields that are used to characterise it during the data acquisition process, transforming the input signal into a processed piece of data. The always present fields that must be filled in are:

- ✓ Meas. ID

  It uniquely identifies **the index of the family to which the measurement belongs** , according to a specific table of ID wired in the datalogger (See Appendix B Tables of measurements and processing). The value can be selected from a pre-loaded dynamic list. Selecting an ID, on the left, the name of the family to which it belongs is proposed to the name field. If the name of a measurement is not in the list, it can be added. It is also possible to modify it. For example, ID = 1 corresponds to the family of temperatures.



For measurements of the same family (that is, for measurements that would have the same ID), the datalogger associates an **incremental ID in steps of 50**. In other words, it is possible to configure two or more measurements of the same type, and, for example by selecting two measurements belonging to the "temperature" family, for the first (Tab M-1 an ID = 1 will be proposed, for the second one) -2), an ID = 51 will be proposed, for a third ID = 101 and so on. For the humidity family we will have the first with ID = 2, the second with ID = 52 the third with ID = 102, Etc. Some IDs over 2000, are fixex and unchangeable, because used for specific measures inside the logger (if used as options), example: pressure, temperature ecc.

✓ Name

It is the **name associated with the measurement family** (e.g. temperature, humidity, etc.) which corresponds to a specific ID as shown in the previous point (see Appendix B Tables of measurements and processing). The proposed name can be modified and saved, for example by matching the mnemonic name "temperature 1", "temperature 2" or similar, to two temperatures, leaving the user free to choose.

✓ Type

It allows selecting, from a pre-configured list, the **type of measurement** intended as the **physical quantity to be acquired**, among those that the datalogger is able to discriminate, i.e.:

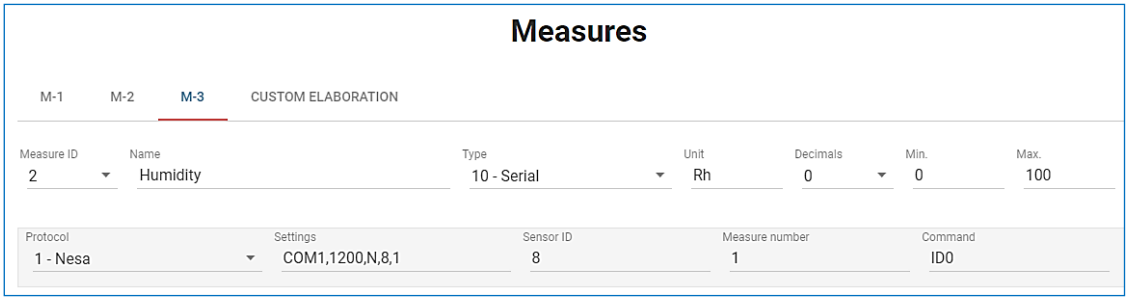
- Temperature: *analogue measurement* that allows the measurement of resistance on Pt100, through pulse current generator for a correct measurement of the resistive value (normally temperature sensors).
- Volt /mA: *analogue measurement* that allows acquiring of signals in a voltage range between 0 and 2,000,000µV.
- Amplitude-100mV: *analogue measurement* that allows acquiring of very weak signals in a voltage range between 0 and 100mV (or 100,000µV).
- Frequency: *digital measurement* that allows acquiring of periodic signals in a frequency range between 0 and 5000Hz.
- State: *digital measurement* that allows discriminating a state, 0 or 1, of an open-collector input digital signal, up to a maximum of 12Vdc.

- Counting: *digital measurement* that allows counting of the status changes of a digital signal in open-collector input, up to a maximum of 12Vdc.

- Calculated measure: *virtual measurement* which allows associating values acquired from different physical channels, combining them with a corrective formula to obtain a new measurement that does not correspond to a physical channel or to a specific type of measurement. It could for example represent the sum of two temperatures, or the calculation of the Dew-Point or other.

- Inside measure: *measure* from internal transducers as: *inside temperature, pressure, Humidity.* These measures are optional.

- Serial: *serial measurement* that allows connection on one of the datalogger input lines (RS232, RS485, SDI12, USB) of a digital instrument with a protocol that is implemented/can be implemented on board the datalogger.
  Selecting a serial measurement, within the same Tab, a line appears with the settings for the choice of the protocol and for the configuration of the same serial, in a simple and direct way.



In the first item, called "*Protocol*", through a drop-down menu, the protocols available and already implemented in the datalogger are listed (for a description of them, see paragraph 3).

Among the standard protocols always available there are:

- Nesa: Nesa serial protocol that requires the insertion of a string of *settings* in Posix format, the *Sensor Id* the number of *measurements* to be queried in the same call and the *reading command* for interrogation.

- Modbus: RTU standard protocol (using the RS485 port as Master only) and TCP-Ip, that requires inserting of the *settings* in Posix format, the*sensor Id*, the number of *measurements*



to be queried in the same call, the *type of register*, l' *the start address* of the reading and the *type of data encoding*. (ModBus standard).

- SDI-12 : (only slave mode) it requires the inserting of a string of *settings* in Posix format, the *sensor ID*, the number of *measurements* to be interrogated in the same call and the *reading command* for interrogation.

✓ Unit

It represents the engineering measurement unit that has been associated/can be associated with the measurement to be acquired. It can be changed.

✓ Decimals

It represents the number of decimals used in the representation of the numerical measurement, and used both in the display of the instantaneous data and in the data files saved and sent, or in the record of the processed data (statistics).

✓ Min.

It allows defining of a criterion for validating the measurement by imposing a minimum value of acceptable measurement, in engineering units, below which the measurement is not considered valid (it is then replaced with the asterisk character "*" in the data path or three "---" in the display and web visualisation).

✓ Max.

It allows defining of a validation criterion of the measurement by imposing a maximum value of the measurement acceptability, in engineering units, above which the measurement is not considered valid (it is then replaced with the asterisk character "*" in the data trace or three "---" in the display and web visualisation).

✓ Channel

It represents the channel number, both physical and logical, present in the datalogger (front side). It can be selected from a drop-down menu. The channels are numbered in ascending order and divided by Differential and Single Ended [i.e. **3D.(V+ V-)** indicates the differential channel No.3, while **3 S.E. (Va)** or **(Vb)** indicates always channel No.3 but Single ended). As a channel is used by adding measurements, the *software proposes only the remaining channels still available*. Depending on the type of measurement, be it, for example, voltage or frequency, the physical channels (i.e. present on the front of the datalogger) proposed, will be only those available for that type.

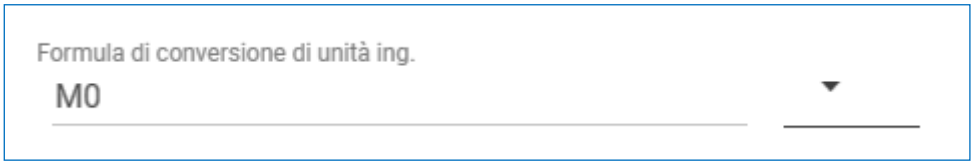
✓ Sample Rate (sec)

It indicates the sampling interval, in seconds, for the acquired channels. The minimum value is 1 second. The relationship between processing time and sampling interval determines the number of samples that will be considered for processing. Example: "processing of the average/min/max" every 60' and sampling every 3'', indicates 60'x60"/ 3" = 1200 as number of samples used by the datalogger to calculate the average.

NOTE: *the Nesa dataloggers, that produce the calculation of the instantaneous value of a measurement (in real time) through a Gaussian average on a very high number of samples, in other words the data per second (referring to "samples every"), is not merely the result of an analogical digital conversion of the electronics, albeit of the highest level, but the result of a weighted average over a Gaussian bell, of an average number of over 100 samples.*

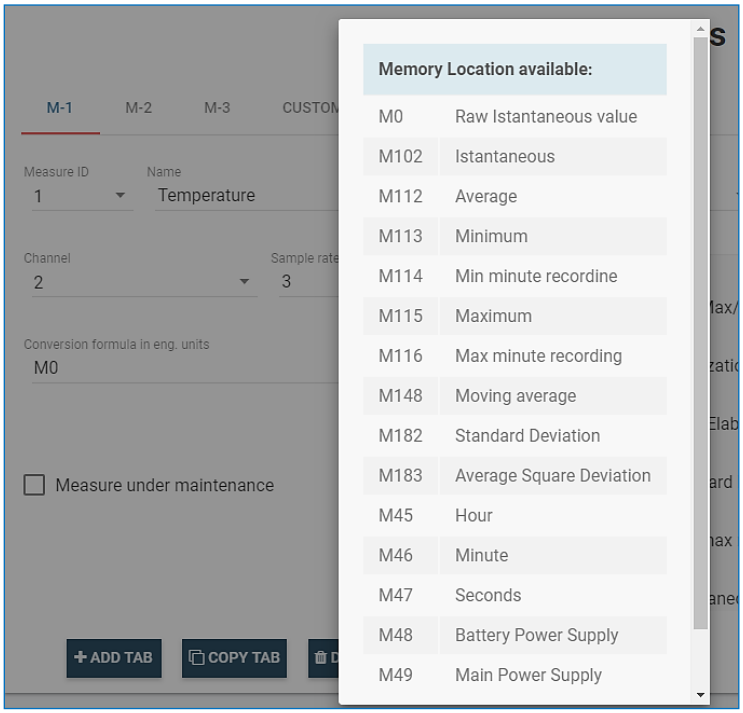
✓ Conversion formula

It is the algorithm that allows *"linearization"* of the electrical signal entering the chosen channel. It has a polynomial shape. It also allows conversion of the acquired electrical signal into an engineering measurement (example from volt to hPa). **The M0 value present by default represents the basic measurement acquired by the datalogger expressed in µV (microvolt, of 0÷2,000,000)** or for **measurements of the Pt100 type, the value directly expressed in °C of the temperature**. Thus, for example, a simple conversion from Celsius to Kelvin C° -> K° translates as **M0+273.15**.
Within this field it is possible to edit the conversion formula also choosing it from a list of conversion formulas on the right, that are described in the section Custom Elaboration.



The datalogger uses specific memory locations to manage all measurements, whether they are acquired or derived (More details in paragraph 2.3.2.3).

The definition of the locations is dynamic between the measurement channels, i.e. between the measurements and is presented by pressing the symbol ? next to the conversion formula, in the form of a table:



The meaning of the various locations is described in the same table. Through a drop-down menu, it is possible to integrate the conversion formula already memorized in the "*Custom Elaboration*" section.

✓ Elaborations associated with the measurement

It is a list of possible elaborations available for each acquired measurement, made on the value in engineering units, to which the user can add new ones upon choice.

- **Med in the selected time interval**: It represents the calculation of the average over the selected time interval. For example, by selecting 10 minutes in the drop-down list, the datalogger is programmed to acquire and process the measurement, providing an average every 10 minutes, calculated on n samples acquired every m second (*see sample every* above).
- **Min/Max in the selected time interval**: It represents the calculation of the minimum and maximum, in the selected time interval. For example, by selecting 10 minutes in the drop-down list, the datalogger is programmed to acquire and process the measurement, providing a minimum and a maximum every 10 minutes, calculated on n samples acquired every m seconds (see sample every above).

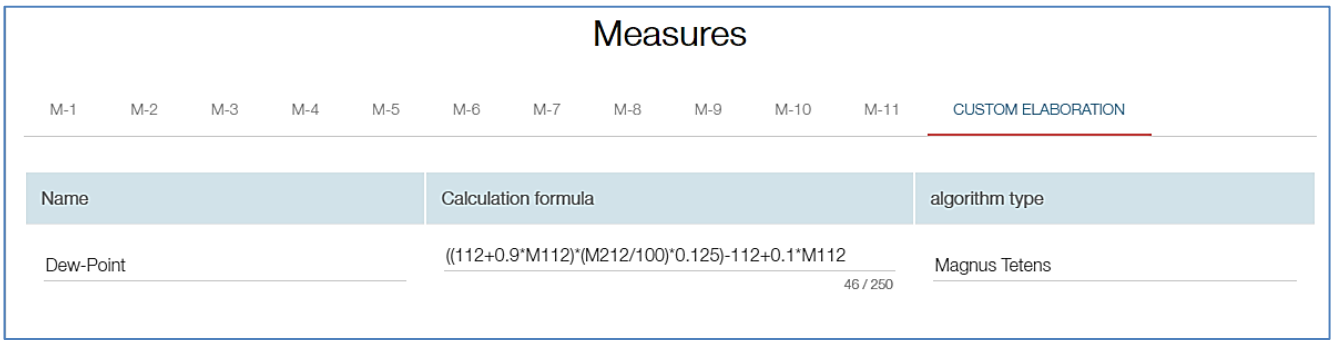
NOTE: *On some software versions, Min/Med/Max processing coincides with a single calculation.*



- **Accumulation in the selected time interval**: It represents the accumulation (sum or integral) of all the samples acquired for the measurement, in the selected time interval.
- **Wind direction calculation in the time interval**: It calculates the trigonometric mean of the wind direction *(sines and cosines method*) and automatically determines the **standard deviation STD** and the **mean square deviation** MSD or the **Turbulence** of the measurement. This processing is only valid if both the wind speed and the direction are present.
- **STD deviation in the selected time interval**: it determines the Standard deviation, the mean square deviation or the Turbulence of the measurement. It differs from the previous one because it is applied to the generic measure, not to the wind.
- **Min/Max minute recording in the selected time interval**: it activates the minute recording corresponding to the minimum and maximum measurement time in the selected processing time interval.
- **Instantaneous Data in the selected time interval**: It allows recording, in addition to the statistically processed values, also the last instantaneous value, that is the last sample acquired in the selected time interval.

NOTE: *If different elaboration intervals are set, for the storing of the processed files the datalogger synchronizes with the shortest time. In other words, if you want for example the average each 10 minutes and the accumulation or other processing, each 60', the files will be generated every 10' (the lesser of the two), and they will contain a valid value corresponding to the measure each 10 minutes and NULL for the measurement (s) with different intervals, until the multiple which provides them both, is reached. In this example up to 60 minutes, in this case the file contains valid values for both measures.*

CUSTOM ELABORATION: Customizable section, contains a list of linearization formulas, listed by name, which is loaded and displayed in the Conversion Formula section through the use of a drop-down list. The purpose of this table is to simplify with examples, many possible calculation algorithms that the data logger can process using the selected memory locations.



Each row inside this page is composed by:

- ✓ Name

  Customizable string that identifies the name of the linearization or processing formula, appears in the drop-down menu of the linearization formula of the measurement (i.e. Conversion 4÷20mA, Dew-Point, etc.)
- ✓ Calculation Formula

  It is the field used to insert the formula (i.e. (M0 / 400000) * 1600000), it has a maximum size of 250 characters.
- ✓ Algorithm type

  Simple text to explain the used algorithm.

Example: we want to insert the calculation of the Dew Point according to the Magnus Tetens method with the Bosen approximation, using the average measurement of the Temperature configured in the first channel (M-1) and of the humidity from the second channel (M-2). According to the table on pag.28 (visible by pressing the symbol i ? configuration page of each measurement), the relative memory locations to be used are M112 for temperature and M212 for Humidity, then we can write:

Used Algorithm	$Td = \left[(112 + 0.9T) \bullet \left(\frac{Rh}{100}\right) \bullet 0.125\right] - 112 + 0.1T$
Formula for datalogger	M0 =((112+0.9*M112)*(M212/100)*0.125)-112+0.1*M112

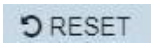
On the left side there is a selection box 

that allows insertion of **measurement in maintenance** without modifying its configuration. When the flag is activated, the contents of the Conversion Formula field are automatically replaced by the value **-9999**. If the flag is deactivated, the previously inserted linearization formula reappears.

To add or delete a measure, press one of the keys at the end of the section. The "Copy Tab" key allows easy repetition of a measurement configuration without having to resume the procedure from the beginning.

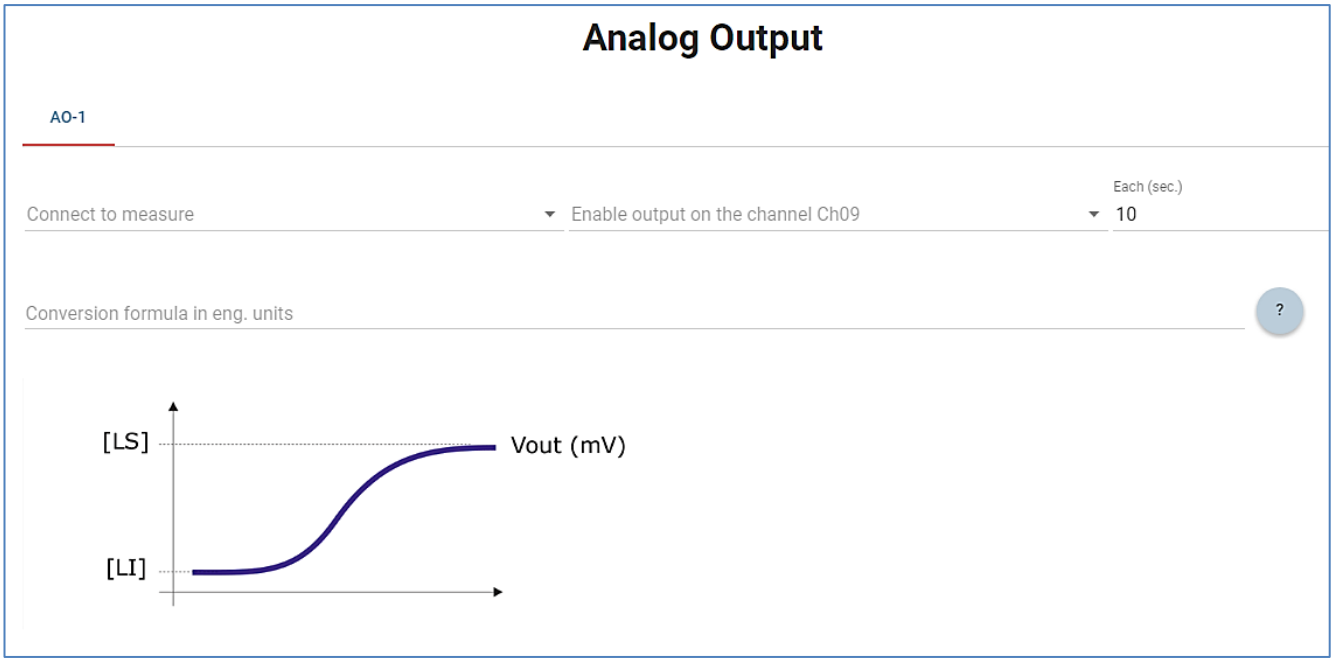


After having filled in all the fields of interest for each section size, it is necessary to save the page using the button 

Each time this button is pressed, the changes made are immediately active, without restarting the device, and the user can immediately check the result of the changes made.
It is also possible to reset all the fields of the measurement tab by means of the 

ANALOG OUTPUT Section

Through this section it is possible to activate the 4 analog outputs of the datalogger on the **physical channel Ch.09**. It is possible to generate an output signal associating it with an input channel, proportionally or not, through a formula, and reconstructing it using an analog-to-digital converter with 12 bit precision, in a voltage range between 0 ÷ 2.000mV, equivalent to 0 ÷ 2V.



There are 4 analog output channels available on the datalogger, which can be associated to any input measurement.

The section requires that the following fields be filled in:

- ✓ Connect to the measurement:

  It associates the analogue signal in output to an input measurement, i.e. it takes this measurement as a reference for the calculation and for generation of the output.

- ✓ Enable output on the channel: ***Ch09***

  The physical channel on which up to 4 output signals are generated, is channel 9 (Ch9). On this same channel the outputs are from 1 to 4. Choosing which output to activate, the datalogger will build a signal between 0 and 2,000mV, taking the connected measurement as a reference, and calculating it according to the conversion formula (see below).

- ✓ Every (sec):

  It represents the frequency in seconds with which the datalogger takes the measurement input and computes the output signal (refresh rate).

- ✓ Input unit conversion formula

  It represents the calculation formula that allows building of the analog output signal based on the input measurement. It can take values from the various memory locations of the measurement in cycle (press button). ?

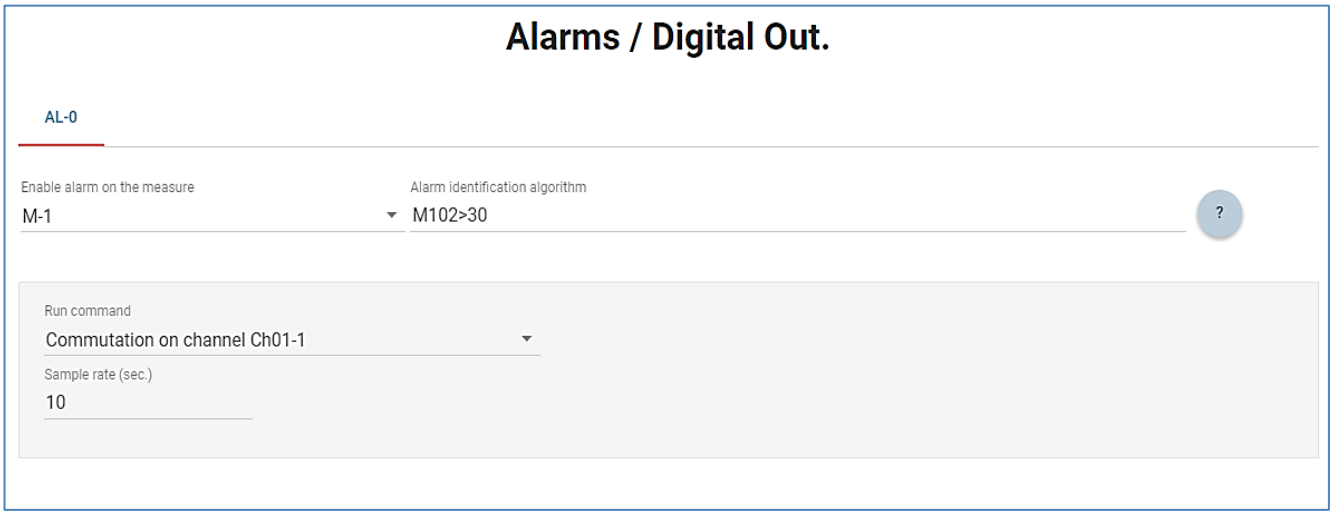
Like the previous section, at the bottom of the page there are three buttons to add other analog outputs, to copy or delete them.

***ALARMS (DIGITAL OUTPUTS)* section**

The Alarms section (Digital Outputs) is different from the previous ones because it is red.

It is possible to choose between different modes of activation of an alarm associated with a measurement, through a specific mathematical relation or algorithm. evolution manages different types of alarms, including the activation of a digital command or output on one of the 4 channels available for this purpose (for example by activating up to 4 relays @ 12Vdc) all physically present in the physical channel no.01.

The alarm condition or alarm return is always the result of a logic written in polynomial form and associated with one or more input measurements. (press ? button)





An alarm, if activated, always produces the memorization of its status into the data record; in addition to this is possible to associate the activation of a digital output (on/off command) or the transmission through the configured transmission channels such as: Gprs (FTP), sms and eMail.

The configuration of an alarm precedes the completion of certain fields:

- ✓ Enable alarm from the measurement

  It identifies the measurement that, periodically monitored, can generate the alarm condition alone or together with others. Take note that if you use a measure (i.e. M-1) to manage an alarm, it will not available for other alarms.

- ✓ Algorithm of alarm identification

  It represents the logic or formula or polynomial algorithm that identifies the alarm conditions (it is also possible to manage alert thresholds) on the basis of the input data whether they are the values of a single measurement or the combination of several measurements. For example a formula of the type M0>35 for the temperature, indicates activation of an alarm when the temperature, associated with the measurement M-1, exceeds 35°C.

  The datalogger manages only one type of alarm (the command can only be on or off), however creating a measure derived on the measurement to be monitored (i.e. M-1) it is always possible to create pre-alarm conditions to be used for example to send messages.

- ✓ Digital Output (relay)

  It identifies the type of alarm that will be activated if the conditions described in the identification algorithm occur. In this case, the digital outputs will be activated.
  You can choose from a drop-down list that allows you to:

  - ---: it doesn't enable any operation on the digital channel if an alarm occurs.

  - Switches the status of channel Ch01-1, Ch01-2, Ch01-3 or Ch01-4: When the event occurs, the channel's output switches its status and, if a relay is connected, it is activated/deactivated.

- ✓ Alarm message for sending

  By compiling this field correctly, the data logger prepares to send the corresponding message in the form that will be defined in the transmissions page, as a file attachment, or text. The syntax of the message always provides a separator between the Alarm and its Reset section as follows:



alarm message* / *message for alarm reset

The texts of the two messages, alarm and reset, must always be separated by a "**/**". It is possible to omit the alarm text or re-entry, but the separator must be always used, for example "Temperature Alarm". The maximum total length of the two messages is set in 200 characters. This text is inserted into an alarm file or a voice file .wav, and available for all ways to send it, so: SMS, eMail (attachment) and via serial (file sending) . If the syntax is incorrect, the message is not sent.

- ✓ Sample Rate (sec)

  It represents the frequency in seconds, with which the datalogger checks the alarm condition and updates the status of the outputs/transmissions.

Like the previous section, at the bottom of the page there are three buttons to add other analog outputs, to copy or delete them.



2.3.2.3 Memory location and correction formula

The evolution datalogger the logger manages the processing of data from the acquisition to their processing through an internal database where it stores information temporarily, from raw data to data processed to any intermediate steps.

The database is dynamic, therefore dimension and name of the fields (from now are memory locations M) depend very much on the complexity of configuration and of the process of acquisition and elaboration, but the basic structure, useful for programming, contains references to locations that can always be recalled at any time, even from different pages during web programming:

Location	*Description*
M0	Primary sample
M24	Totalization
M45	Hour
M46	Minutes
M47	Seconds
M48	Battery voltage
M49	Power supply voltage
Mx02	Istantaneous data
Mx12	Average
Mx13	Min
Mx14	Minute of Min
Mx15	Max
Mx16	Minute of Max
Mx48	Dragged average
Mx82	Standard Deviation
Mx83	Square quadratic difference



The first 6 locations are unique for the system, while from the seventh onwards, they depend on the number of measures configured in the datalogger, for this reason they are indicated in the table with an "x", in particular the first measure will be M102, M112, ..., M183 , the second measure will be M202, M212, ..., M283 etc.
The meaning of each location is described in the table and is possible to refer to it in any calculation formula in the datalogger, it is enough to recall it with its abbreviation (eg M112 to refer to the average value of the first measurement).

In order to understand, in every evolution page, where the use of memory locations is foreseen, there is a button (?) which recalls the table for the specific measurement.

Let's see how they are used in the correction formulas:

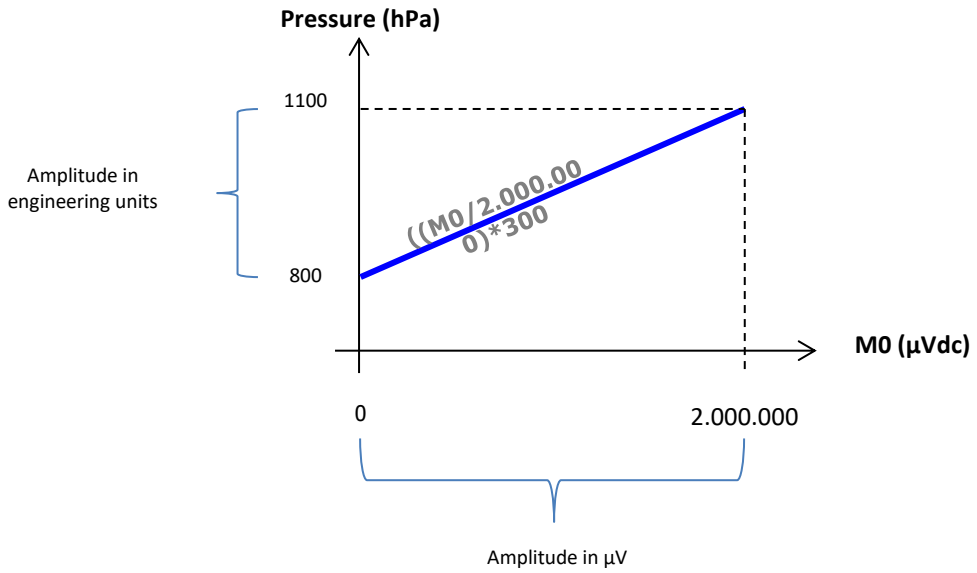
For analog signals, the **correction formula** is an algorithm written in polynomial form with the normal mathematical and logical operators, that is necessary both to linearize a measurement and to convert the electrical measure into engineering units..

evolution acquires signals in µV in a scale from 0 to 2.000.000µV that become corresponding to a measurement in engineering units (° C, mA, weight,% etc.) according to calculation algorithms that use as a basis, the value in µV. This value is called the **raw primary sample** and is always stored in a **memory location called M0**. With the sole exception of the temperature for which M0 already contains the value expressed in ° C, in all other cases, M0 is a value in µV between 0-2,000,000.

To determine the algorithm used to linearize a given measurement or to convert it from a µV to an appropriate engineering unit, we first need to know the **signal amplitude** on which the measurement of the instrument we are going to connect to our datalogger, is moving.

Take for example a Barometer, or a measure of atmospheric pressure with a range 800 ÷ 1100hPa in engineering units and, suppose the corresponding electrical signal is 0 ÷ 2Vdc. We therefore have a signal width 300hPa (1100-800) of amplitude, and an initial measurement point at 800hPa.

The algorithm we are looking for, is none other than the linearization line of the measurement, whose angular coefficient will be given by the ratio between M0 and the maximum value in µV (2,000,000) multiplied by the amplitude of the signal (300), all in engineering units. To this, an initial offset of 800 should be added, which represents the minimum value measured by the sensor when its electrical output corresponds to 0V.
By putting the data in a chart we will have:



Obtaining the linearization formula of this line is simple:

hPa = ((M0/2.000.000)*300)+800.

Similarly for a measure of net solar radiation, with a measurement range between -500 ÷ 1500W / m2 always with 0 ÷ 2Vdc output signal, for the same considerations we will have:

W/m^2 = ((M0/2000000)*2000)-500.

Finally for a 4-20mA signal, the simple correspondence is:

mA = ((M0-400.000)/1.600.000)

and applying it to a barometer like the previous one (800-1100hPa) but with 4-20mA electrical output instead of 0-2Vdc, finally:

hPa = (((M0-400.000)/1.600.000)*300) +800

2.3.2.4 TRANSMISSION page

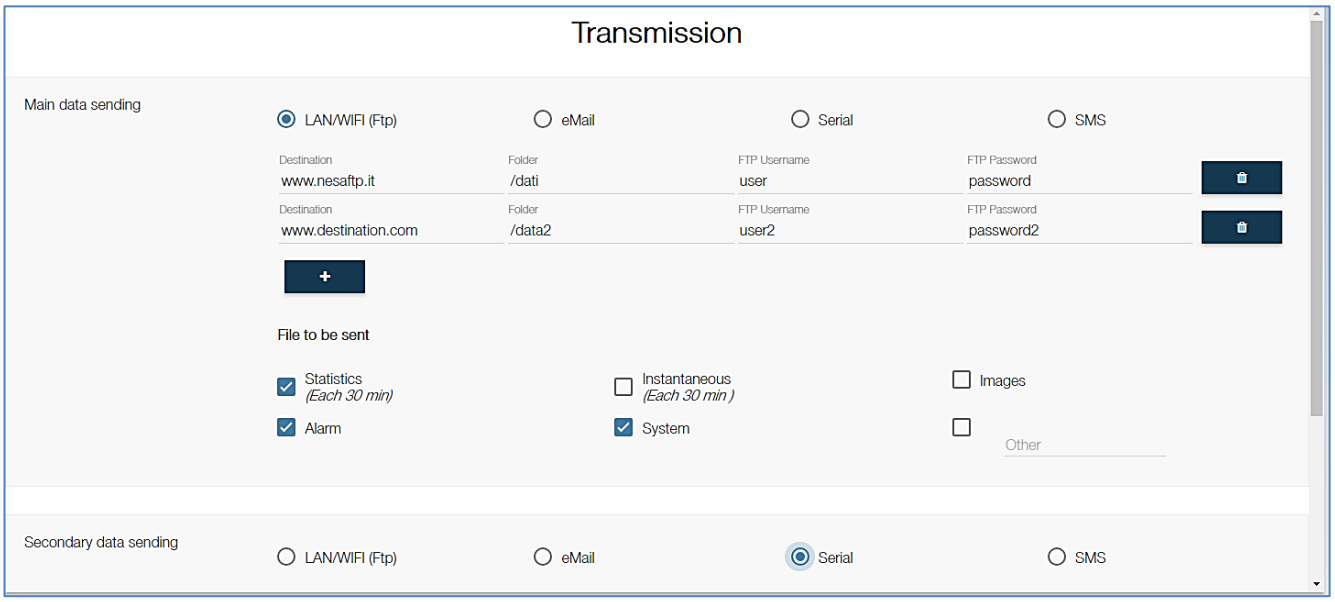
On this page it is possible to define the conditions that allow the transfer of data acquired and stored by the datalogger to remote areas or external storage (remote servers, serial lines in listening, etc.).

evolution manages the transmission with a dedicated scheduling process that is independent of data acquisition processes. It is sufficient to fill in all the parameters of the type of transmission that is of interest and automatically the datalogger performs, optimises and verifies the transmission of the acquired and processed data (data record file).

It is possible to activate a communication redundancy at the same time on two different physical lines and with different methodologies and protocols (e.g. GPRS and serial or satellite).

It is possible to enter an unlimited number of recipients (e.g. email addresses) or destination servers (FTP areas).

In sequence, first select the physical mode of data transmission (FTP, Serial, SMS (only alarms) etc.) and then the type of data to be sent. Some settings being unique to the data logger, are selectable only by the MASTER user, as explained below.



The two physical lines of data communication are called *Principal* and *Secondary*.

The ***Main Data Sending*** allows the following modes:

- ✓ GPRS (ftp)

  With this selection it is possible to set the parameters of an FTP connection made through the use of a GPRS/UMTS serial terminal connected to the COM1 of the datalogger (whose management, both on/off and power supply, is totally dependent on the data transmission scheduling process) and of a telephone SIM (possibly bidirectional), to be activated with the aid of the chosen telephone provider. The datalogger is able to transmit data with FTP protocol to multiple destinations (FTP areas), simply by adding them with the key 

  Selecting the GPRS fields appear in two lines:

  - *APN (Access Point Name)*: it represents the point of access to the data network through the GPRS protocol. It is closely related to the chosen telephone provider (ask the SIM provider).
  - *Username e Password*: these are any credentials required for the APN connection

    NOTE: *This first line can only be filled in by the MASTER user of the datalogger and will not appear on the NORMAL user pages.*

  The next line contains information about the data destination areas:

  - *Destination*: the ftp address of data destination (e.g. 188.123.123.1, or www.ftpadd.com)
  - *Folder*: data destination directory in the format (/name1/name2/)
  - *FTP username*: ftp user credentials associated with the data destination folder
  - *FTP password*: the password of the ftp user

    NOTE: *The FTP protocol intrinsically provides that, after a successful transfer of a file, it is deleted from the source. It is advisable to associate no more than three destinations for each type of file. For example, to transfer the statistical data files .txt, it is advisable to do this on up to three different destinations, for each user. This limitation is given in order not to slow down the transmission process which has, in the case of a GPRS transmission channel, a limited bandwidth and also to the limit in memory usage for replication of data memory. Each destination is managed by the datalogger through the use of a separate data folder; this means that, if the data transfer on one or more destinations fails, the scheduling process will try again sending only the incorrectly transferred files.*



✓ LAN/WIFI (ftp)

Here it is possible to set the parameters of the FTP connection made through a wired 10/100MBit LAN connection or WI-FI.

In this case the necessary information is the same as the GPRS connection relative to the FTP destination (Folder, FTP Username, FTP Password), while any Gateway has been previously defined in the Personal Data -> Options section and uniquely definable only by the MASTER user (see paragraph2.2).

✓ eMail

Here you can enter the addresses of the recipients to whom the information should be sent via e-mail (separated by ";"), the subject of the message and the text. The subject of the message has a predefined maximum length and the same for the body. The file to be attached may be one or more among the selections described below.

✓ Serial

Here it is possible to set the parameters for a serial connection between the datalogger and the listening device, through the use of COM2 or a serial port connected to a USB/Serial converter that can be connected to the machine, which allows data transfer (example in the case of a 4-port converter) from the COM3, COM4, COM5 and COM6 ports.

NOTE: *To the logger can be connected additional serial ports using natively USB ports that automatically recognise multiserial devices connected to them. Using serial transmission, some files cannot be transmitted (i.e. images) due to the wide band available.*

In the *Settings* box define the serial string of the port programming , according to the ***standard Posix***, for example:

COM2,9600,N,8,1 defines a serial connection that opens a channel on the COM2 port, at a rate of 9600 baud, without parity, with 8-bit word and a stop bit.

The *Autosend* box shows a list of times, in seconds, which represents the frequency with which the data are sent by serial link according to the settings used. As default there is a 10” (min 1 sec).



- ✓ SMS

  Here you can enter the mobile numbers to which you want to send an SMS (max 5 and only for alarms). After entering the first number it is possible to insert the successive ones simply by separating them with a semicolon separator ";"

  It is advisable to carefully insert the telephone numbers and not to exceed the maximum of 10 numbers, in order to not slow down the process of forwarding information via SMS. All types of files can be sent except for images.

At the bottom of the *Main Data Sending* section are all the **types of information (files)** that can be transferred according to the used way (GPRS, LAN, SERIAL, SMS (alarms), Etc).

The files are contained in the file system inside the *//mnt/nand/Data* folder associated with the specific user, in the case of the administrator user it will be the folder with path of */mnt/nand/0-Data,* for first USER in */mntnand/1-Dati*.

The information or files contained in the specific folder are divided into:

- ✓ Statistics:

  all files with the "*.txt*" extension contain the processed data, written according to the path defined in the *Master Data -> Data File Format* page, (NESA record as default, described in the appendix to this manual). These data are processed according to a time interval defined on the tab of each measurement, and the transmission is timed according to the timing indicated on the *Master Data -> Data Save/Send page*.

- ✓ instant

  all files with the "*.ist*" extension contain the primary samples obtained according to the frequency set in the measurements section - *sample every*). In the case of different sampling times, each file with the most extensive sampling is recorded.

- ✓ Images

  these are all files with the extension "*.jpg*", deposited by a screen device (e..g IP camera) that can be connected to the LAN port of the datalogger, capable of storing images exploiting the potential of the **FTP server** integrated in the datalogger.



- ✓ Allarm

  all files with the extension *".eaf"* (evolution alarm file), generated by a measure that went into alarm. Inside the file there is the alarm and return message, the value that generated the alarm or its return.

- ✓ System

  these are the files with the extension "*.dat*" that contain the dynamic IP address associated to the GPRS connection, the amount of free memory and other information about the status of the processes in progress.

- ✓ Other

  evolution allows the sending of personalised files. It is sufficient to place them in the same data folder mentioned above, and, by inserting in this text box the extension of these files (generated for example by external users to the main processes), they will be sent to external servers according to the modes chosen. Among them the *.wav* file

Secondary Data Sending allows the same way of transferring of the information of the Main one, with the distinction that, if the GPRS transmission was chosen as the *Main Data Sending* channel, it will no longer be possible to also choose it as *Secondary Data Sending*, channel, being the unique resource.

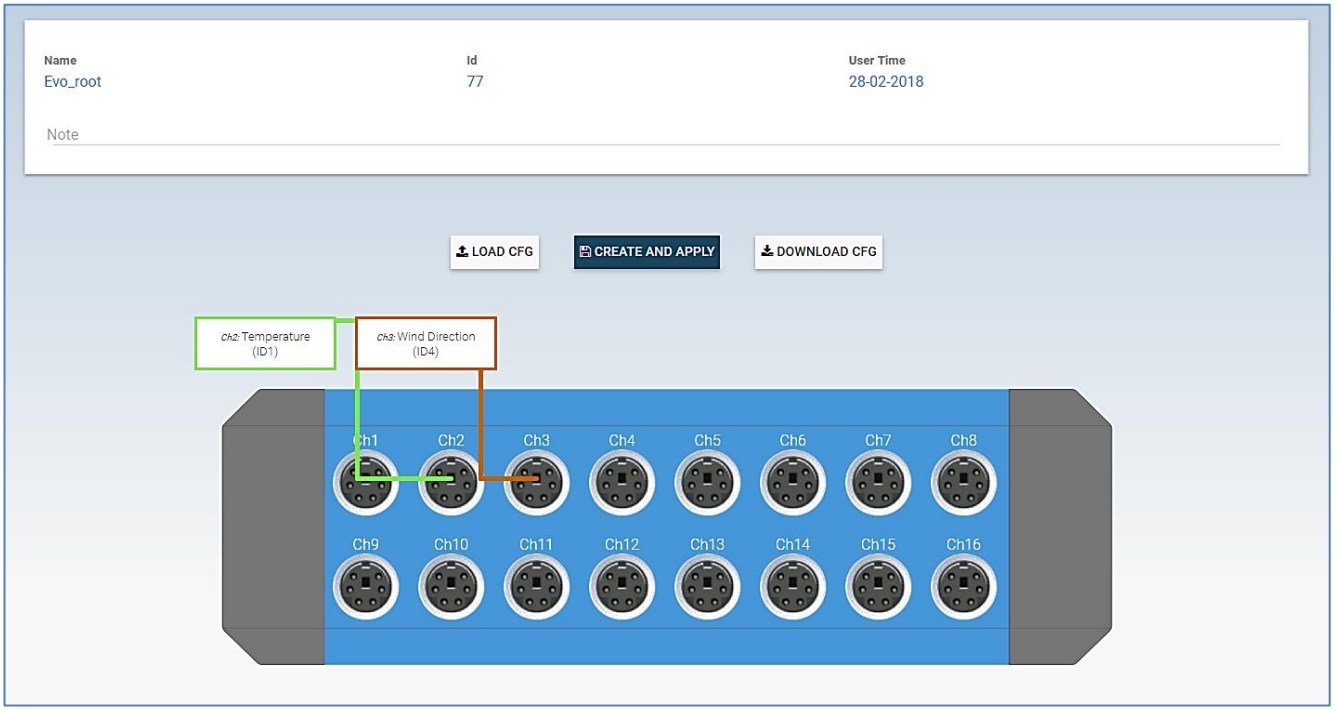
After completing compilation of the fields, it is necessary to save the page with the button 

Each time this key is pressed, the changes made to the page are immediately operative, in this way the user can make an immediate check on the result of the changes made, by observing the behaviour of the data transmission process.

It is also possible to reset the measurement page by pressing the button 

2.3.2.5 CFG-SERVICE page of the configuration check

This is the page allows a visual check of the physical configuration of the data logger.
Compared to all other configuration pages, it summarizes in graphics and text what was created when saving the configuration.



At the top of the page are the configuration *Name* , datalogger *Identification*, and creation *date*. There is also an editable *note field* that can be used by the user to add some comments.

At the bottom of the page are shown, in graphical form, the programmed physical channels with indication of the channel, type of measurement associated with its ID and a stylized connection to the connector that physically shows where the physical channels used are to be connected.

The configuration of a machine is easily replicable on other dataloggers of the same series. It is in fact possible to download the configuration (DOWNLOAD CFG EVOLUTION -> PC) or to load an existing one from a PC or from an external device(LOAD CFG PC -> EVOLUTION

Saving or loading a configuration means updating the files of the machine that supervise its operation. To facilitate and accelerate the transfer, during the loading (or unloading) phase, the configuration files are compressed into a single file (with extension ***.nsa***) that represents a complete "snapshot" of the status of the datalogger at that moment. In the same way, loading a configuration means loading a single file, which will then be decompressed and interpreted by the process that takes care of the machine set-up. At the end of the loading phase, the data logger always performs a restart; the user is redirected to the home page at the end of the reboot

The loading and unloading times may take some time depending on the browser used. The configuration files downloaded are normally named ***evolution.nand.nsa***.

NOTE *Warning: the management of a download through a browser is highly dependent on the browser used and the settings on the browser used. Using for example Google Chrome the file is present in the Download folder of the PC used for the connection.*

It is always suggested to enable the pop-up on the used browser

3 Communication processes

The datalogger of evolution has different modes of transferring information with external devices, using the physical lines of LAN and serial communication (RS232 and RS485).

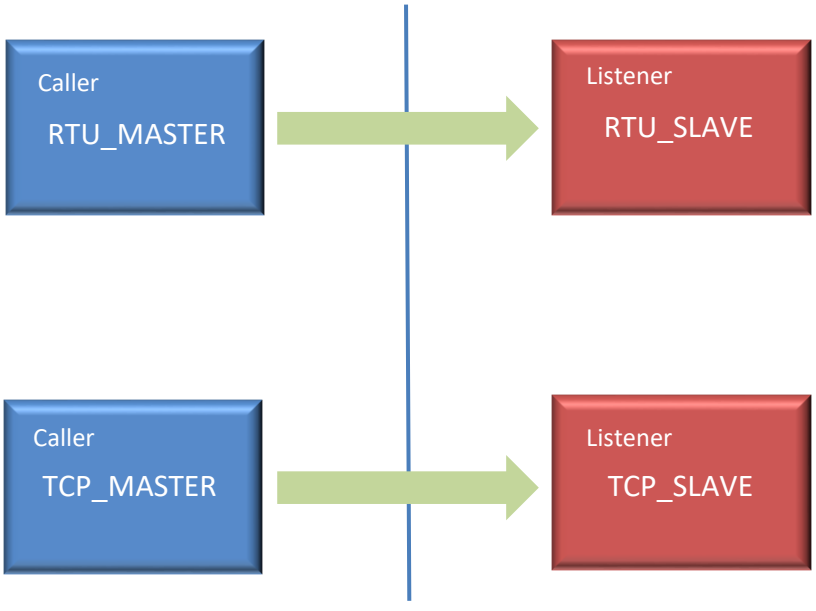
All the communication processes are completely independent (thread) modules and share the memory area containing the data collected by the logger process.

3.1 Modbus

The Modbus protocol is a standard protocol for communication between devices, that in evolution uses both the **RS485 serial** port in **Master** mode, Ther **serial port RS232 (COM2)** in **Slave** mode for data transfer (Modbus RTU - Remote Terminal Unit), and the **network Lan port** (Modbus TCP - Transmission Control Protocol) both for **Master and Slide mode.** For details of the structure, refer to the abundant literature present that describes the protocol in detail; this paragraph will provide all the information related to the interface mode and the structure of the months records available from the device.

It is a communication protocol with a **Master / Slave** structure, where Master means the calling device ("caller"), and for Slave we mean the listening device ("listener") that makes the data available.

In the Modbus communication module of evolution two communication tasks are always active, the Master and the Slave, which operate, depending on the requirements and programming carried out, on the serial communication lines or on the Ethernet line (**modbus RTU and modbus TCP-IP**).



3.1.1 "Listener" mode - Slave

In Modbus Slave mode, the evolution datalogger establishes communication with a Master of the network in which it is inserted, receives the request message and transmits the response. The information that is made available is all the measurements configured and acquired by the datalogger, in instant data, with the update frequency defined in the configuration. Its order is the same as that is defined in the configuration.

The measured values are read by using the **04h (read Input Register)** function defined by the standard starting from address **30001**.

Up to a maximum of 1000 16-bit registers are made available starting from address 0x30001.

The measurements acquired by the datalogger are always in the 4-byte floating *point* format in *little endian* coding (also known as float reverse mode), so the maximum exportable measurements via ModBus is **500** because two ModBus registers are required to contain the measurement.

The exported value is always in engineering units, as defined in the configuration.

As an example, for a datalogger that acquires two measurements, temperature and humidity, the associated ModBus map is as follows:

Address	Size	Measurement unit	Format
0x30001	Temperature	°C	Float reverse 32 bit
0x30002			
0x30003	Humidity	%	Float reverse 32 bit
0x30004			
0x30005	Battery Voltage	V	Float reverse 32 bit
0x30006			
0x30007	Power supply voltage	V	Float reverse 32 bit
0x30008			

The battery voltage and supply voltage measurements are always present.

3.1.1.1 TCP Slave

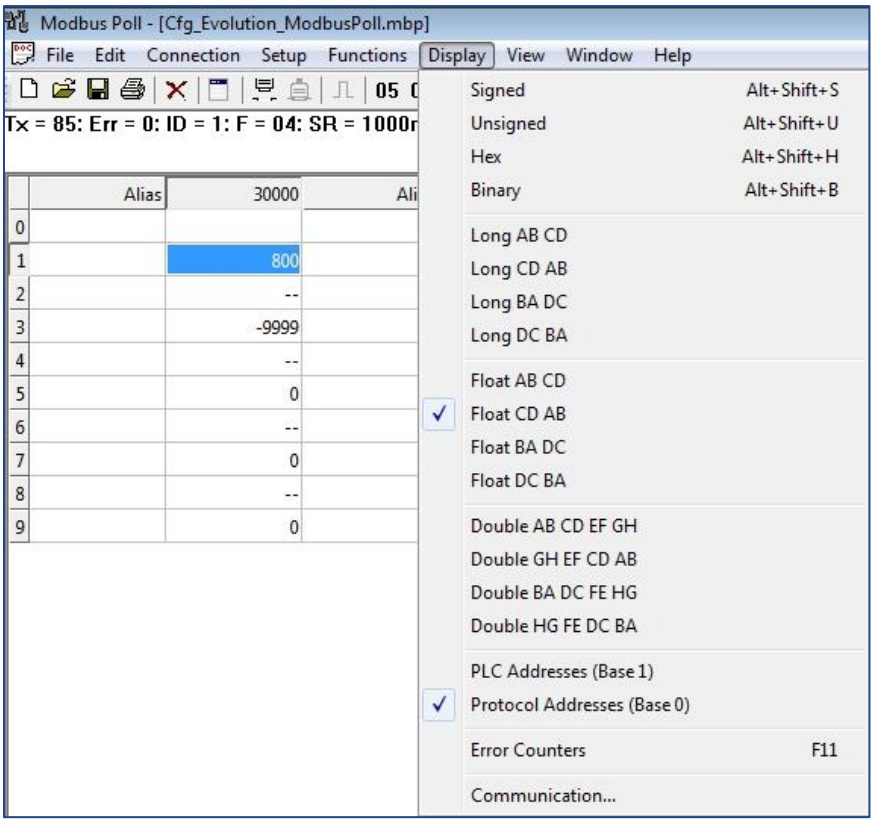
In Modbus TCP slave mode, the call from the master is on a LAN cable and must be addressed to the IP address of the datalogger, on **port 502**. The call function is **04h** (Read Input Register) starting from address **0x30001** in READ mode. Datalogger answers on the TCP-Ip port.

The number of registers to be read, up to a maximum of **100**, must be included in the call.

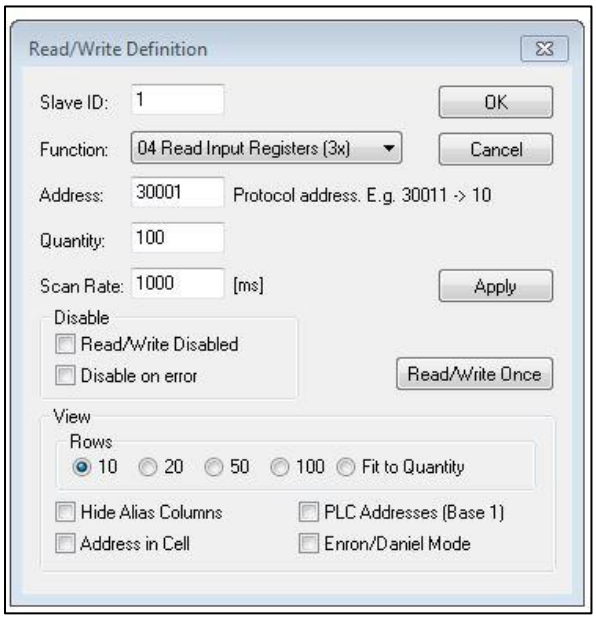
The measurements acquired by each user are mapped in 100 16-bit locations, correspondent at a maximum of 50 measurements. Recapping, the memory locations user by user:

User	Register at the begin of reading
root	0x30001
User 1	0x30101
User 2	0x30201
User 3	0x30301
User 4	0x30401
User 5	0x30501

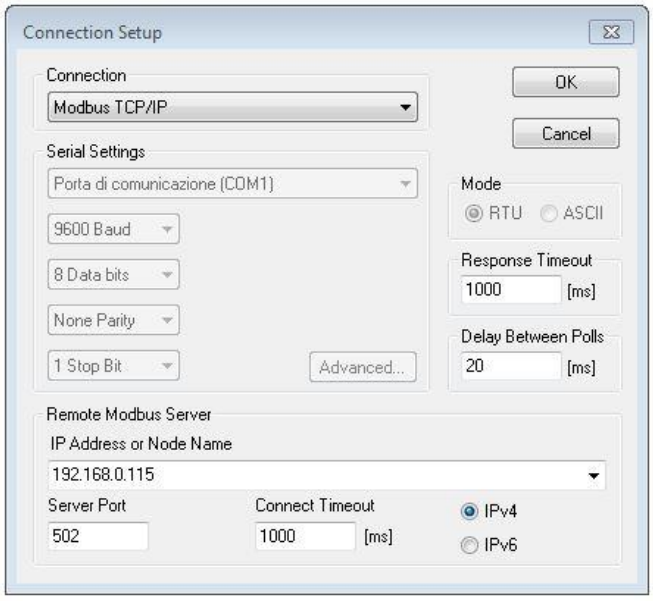
As an example of a device that connects via Modbus TCP, we report the configuration of a common software, called Modbus Poll, configured to connect to an evolution datalogger via Modbus TCP and to read its registers. The following image shows the type of data representation, inverse *floating point*, and the registers that contain the measurement. In the example is the measurement in register 30001 (the first in configuration) with a value equal to 800.



The following image shows the call function, the number of registers read and the address of the initial register.



The last part of the setting concerns the interface mode, Modbus TCP, the address to be called and the port.



3.1.1.2 RTU Slave

In Modbus RTU slave mode, the call from the master is on a serial cable (RS232) and must be addressed to the ID of the datalogger, which coincides with the ID of the terminal. **Only RS232 on COM2 can be used**.

The serial connection must have the following characteristics:

Speed: **9600 baud**

Bit number: **8**

Parity: **None**

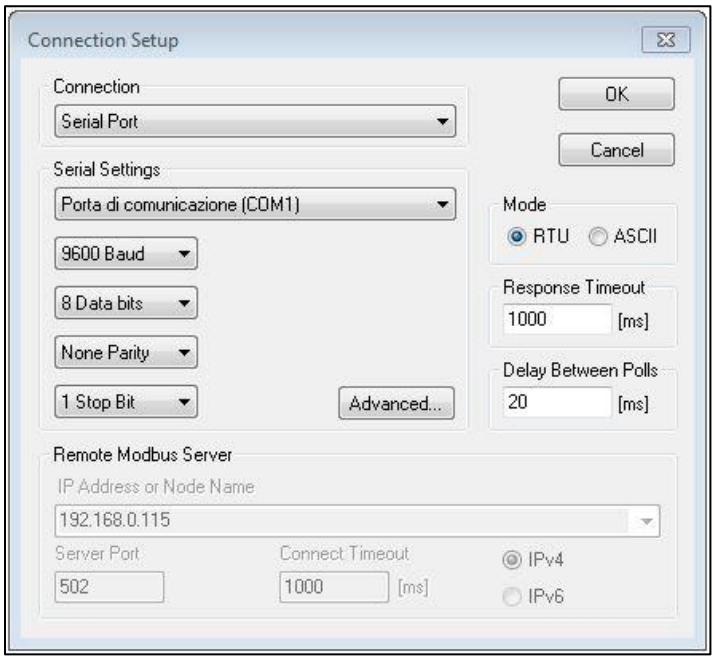
Stop bit: **1**

The call function is **04h** (Read Input Register) starting from address **0x30001** in READ mode.

The serial port to which the cable is connected is the **COM.2**

The number of registers to be read, up to a maximum of **500**., must be included in the call.

Similarly to what is described for Modbus TCP mode, the Modbus Poll call software configuration is absolutely the same, with the exception of the connection part, which is now serial, and must be filled in by inserting the previously mentioned connection parameters, as described in the image below.



3.1.2 "Caller" mode - Master

In Modbus master mode, the evolution datalogger establishes communication with one or more slaves of the network in which it is inserted, sends the request message to the slave, and only that specific device responds to the command. The information composing the message to the slave is that defined in the configuration phase (see paragraph 2.3.2.2).

The parameters to be set are different depending on the type of slave being interrogated, but they are defined by the protocol standard, and are:

- *Serial communication setting string*

  The setting string should be inserted in the format: *PORT, SPEED, PARITY, WORD BIT, STOP BIT*

- *Identification of the slave*

  It is an integer that identifies the slave within the network

- *Number of measurements*

  It is the number of measurements to be read by the slave

- *Type of register*

  It is an integer representing the function used during the query, currently the functions enabled are the following: 04 Read Input Register, 03 Read Holding Register. The

- *Departure address*

  It is the address from which to read the data

- *Type of data encoding*

  It is a numerical list that allows establishing the types of data to be read, defined by the following table:

ID	Data
1	FLOAT AB CD
2	FLOAT CD AB
3	FLOAT BA DC
4	FLOAT DC BA
5	LONG AB CD
6	LONG CD AB
7	LONG BA DC
8	LONG DC BA
9	SINGED INT
10	UNSIGNED INT

  Where the letters A, B, C, D represent an 8-bit word. They are grouped so that the modbus registers are 16bit. The coding with ID = 2 is the one that uses the datalogger evolution, i.e. inverse float, to represent the data.

The storage of data collected through this type of query complies with all the types of data acquired, i.e. it is stored as 32 bit float data.

3.2 SDI-12

SDI-12 is an asynchronous serial communication protocol used above all to communicate with environmental sensors. It is a protocol of type Master/Slave in which the sensor must be identified by a unique ID. The

communication, using only two lines, is half-duplex type, with a 1200 baud serial rate, 7-bit word and a parity bit with even parity, but sensors with speeds even higher than those defined by the standard.
The voltage levels are similar to those of an RS232 (5V) interface.

The main advantages of this solution are:

- Sensor interchangeability without reprogramming the logger that acquires them
- The sensor is powered by the same communication line (12Vdc)
- Variety of connectable sensors

The disadvantages are:

- Low information transmission rate
- Relatively short connection cable, cannot exceed 10 meters.

evolution has an SDI12 port (Slave only) in parallel to the RS485 port, so the use of one does not allow the use of the second (only one type at a time).

The communication interface in evolution, for SDI-12 sensors, is activated, from the measurement page, by setting the *Type* of communication, *10-Serial,* selecting the *3-SDI-12* protocol and filling in the fields:

- *Settings*

  It is the string that sets the serial port for communication. **Only COM0 is available for this function**.
- *IdSens*

  It is the numerical identifier of the sensor to be interrogated: Are excluded the ID: 1,2,4,8,16,32,64
- *Number of Measurements*

  It is the number of measurements to be acquired from the sensor
- *Reading Command*

  It is the reading command to be used to read the data from the sensor, supplied by the sensor manufacturer, for example aD0!

  The setting of an SDI-12 barometer, connected on COM0, is given as an example:



3.3 Socket

A socket, in the modern operating systems, indicates a software abstraction designed to use standard and shared libraries for transmitting and receiving data through one. It is a particular software object through which two processes residing on two different machines, connected to the same network, can read and write the data to be transmitted or received.

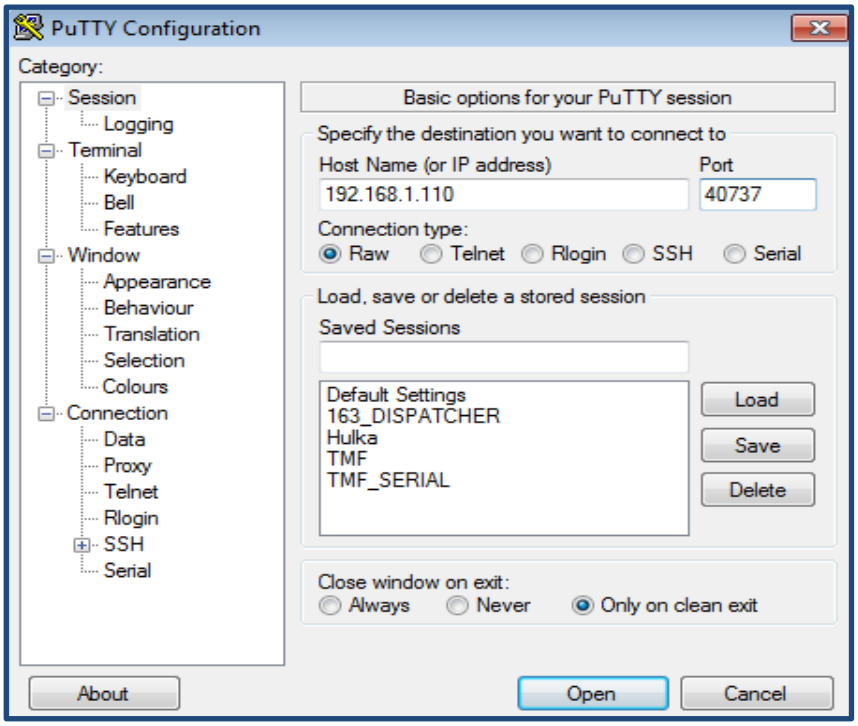
On the evolution datalogger there is always the possibility of a connection via **TCP/IP socket** listening on port 40737, with the possibility of:

- Access in reading/writing to the memory locations of the machine
- Reading files contained within the datalogger
- Sending files to the machine

Data is read by the use of cells (or memory locations), or MEM, within which the logger process stores data of the type float 32. The memory locations currently available are 10000 for each user, corresponding to a maximum of 100 measurements per user.

The first measurement occupies the leases from 1 to 100, the second one leases from 101 to 200 and so on.

The socket connection can be tested using the *putty* freeware software configuring with a raw connection as shown in the figure:



3.3.1 List of available commands

Multiple reading of data with or without value range

R|102|202|301:303 [CR]

response: W|102|0|202|1|301|0|302|1002.2|303|1002.0 [CR]

responds with the required values (locations 102, 202, 301 to 303) separated by pipes

Multiple writing with and without range

W|402|1|502|0|601:603|3 [CR]

writes the memory locations 402, 502 and from 601 to 603

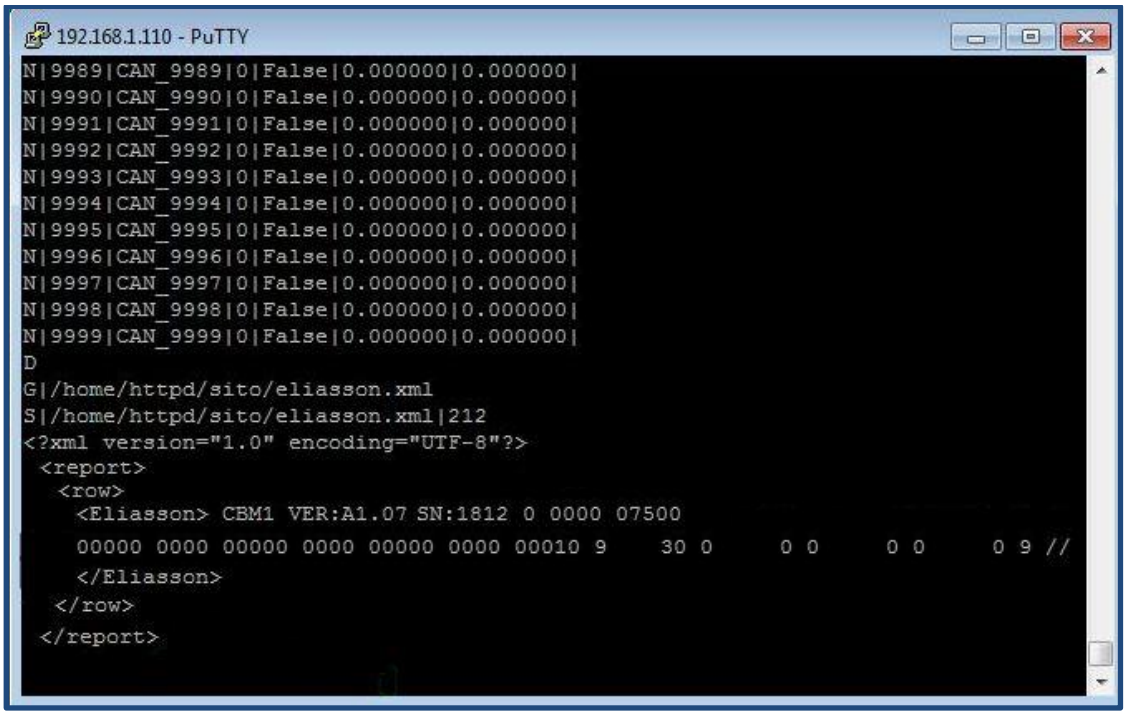
File reading

G|filename [CR]

E.g.:

G|home/httpd/sito/eliasson.xml [CR]

response: S|filename|len|[CR] + [file stream]



The response presents the name of the file followed by its length, and on a stream the content.



File writing:

S|filename|len|[CR] + [binary data len]

E.g.:

S|home/httpd/sito/test.txt|9|[CR]

Test file

Inside the folder /home/httpd/site/ there is now a file called test.txt whose content is the "Test file" string



3.4 FTP (File Transfer protocol)

File Transfer Protocol (FTP) (file transfer protocol) is a protocol for the transmission of data between host based on TCP and with architecture, type **client/server**.

The protocol uses separate TCP connections to transfer data, to monitor transfers, and requires client authentication by username and password, although the server can be configured for anonymous connections with dummy credentials. The data that FTP transmits both these credentials and any other communication, are uncripted.

On the evolution datalogger there are both functions, **server** and **client**.

3.4.1 Server FTP

FTP uses two separate connections to handle commands and data. The FTP server remains in listening on the **port 21** of TCP to which the client connects. The connection by the client determines the initialisation of the command channel through which the client and server exchange commands and responses. The actual exchange of data (such as a file) requires the opening of the data channel, which can be of two types: active or passive.

The mode implemented on the machine's FTP server is passive.

In a passive data channel the server opens a usually random port (above 1023); through the command channel it makes known the number of this port to the client and waits for it to connect. For this purpose, the PASV or EPSV commands can be used, depending on the network protocol used (usually IPv4 o IPv6).

An FTP server offers several functions that allow the client to interact with its file-system and the files that populate it, including:

- Download/upload di file.
- It resumes interrupted transfers
- Removal and renaming of files.
- Creation of directories.
- Navigation between directories.

The data connection from an FTP client to the FTP server of the machine takes place through the root unit, so there is therefore full access to the partition of the machine that contains the operating system with administrator privileges. This means that any modification made to the files of this partition, if incorrect, can affect the operation of the logger.

It is advisable not to delete or modify files during a connection to the ftp server if not fully aware of the effects of the changes.



For file upload transfer, especially if using a client on Windows machine, it is always advisable to run it by activating, before transferring, the *binary* directive.

To use the possibility of the FTP server of the machine, the user administrator password must be requested from NESA s.r.l.

3.4.2 Client FTP

Through the use of a terminal it is possible to use the functionality of FTP client, so it is possible to connect to an FTP server according to the methods and credentials of the server to connect to.

3.5 SSH (Secure Shell)

SSH (Secure SHell, shell secure) is a protocol that allows establishing of an encrypted remote session via command line interface with another one host present on the network. It is the protocol that replaced the similar, but non-secure, Telnet.

The ssh server of evolution works on **port 22.**

An ssh authentication process between a client and a server goes through these steps:

Negotiation of algorithms

Algorithm negotiation is one of the first steps in establishing an SSH connection. In order to determine which algorithms to use in the SSH connection, the client and server must exchange the list of algorithms they support for the connection. The list contains all the available algorithms in order of preference; the preference and the available algorithms are determined by the configuration of the client and server software. Once the list exchange is over, the protocols available on both machines are chosen giving precedence to the higher algorithms in order of preference.

Key exchange

After defining the algorithms to be used in the connection, one of the most important steps is taken in establishing the secure communication channel: the exchange of keys. Key negotiation takes place at the beginning of each connection.

In the current ssh clients, e.g. putty, when a client first connects to the evolution ssh server, a message appears prompting saving of the key in order to avoid subsequent negotiations.



Server authentication

Server authentication is used to prevent a malicious user from "tampering" with the having obtained the user credentials. For this purpose, a pair of asymmetric keys is generated for each server. The private key remains on the server. The public key must be known by the client, the client can obtain the key of a server by receiving it directly from the server during the first connection.

Authentication takes place during the exchange of keys, the server creates an encrypted message with its private key and sends it to the client, the client decrypts it with the server's public key verifying the identity of the server. If the message is decrypted correctly the client proceeds with establishment of the connection, otherwise it interrupts the procedure. As only the server should be aware of the private key, the client is able to determine the identity of the server it is communicating with.

Connection encryption

Having defined a secret key known exclusively by the client and the server, it is possible to use a symmetric cryptographic protocol to encrypt the communication between client and server.

A symmetric cryptographic algorithm allows the use of a single key to encrypt and decrypt information. In a symmetric key algorithm the shared key must be defined before initialisation of the connection using a method of communication of the secure key that is performed using the Diffie-Hellman algorithm in the SSH. Symmetric key algorithms guarantee a high standard of security and a low cost in terms of computing power (unlike asymmetric key algorithms such as the RSA algorithm).

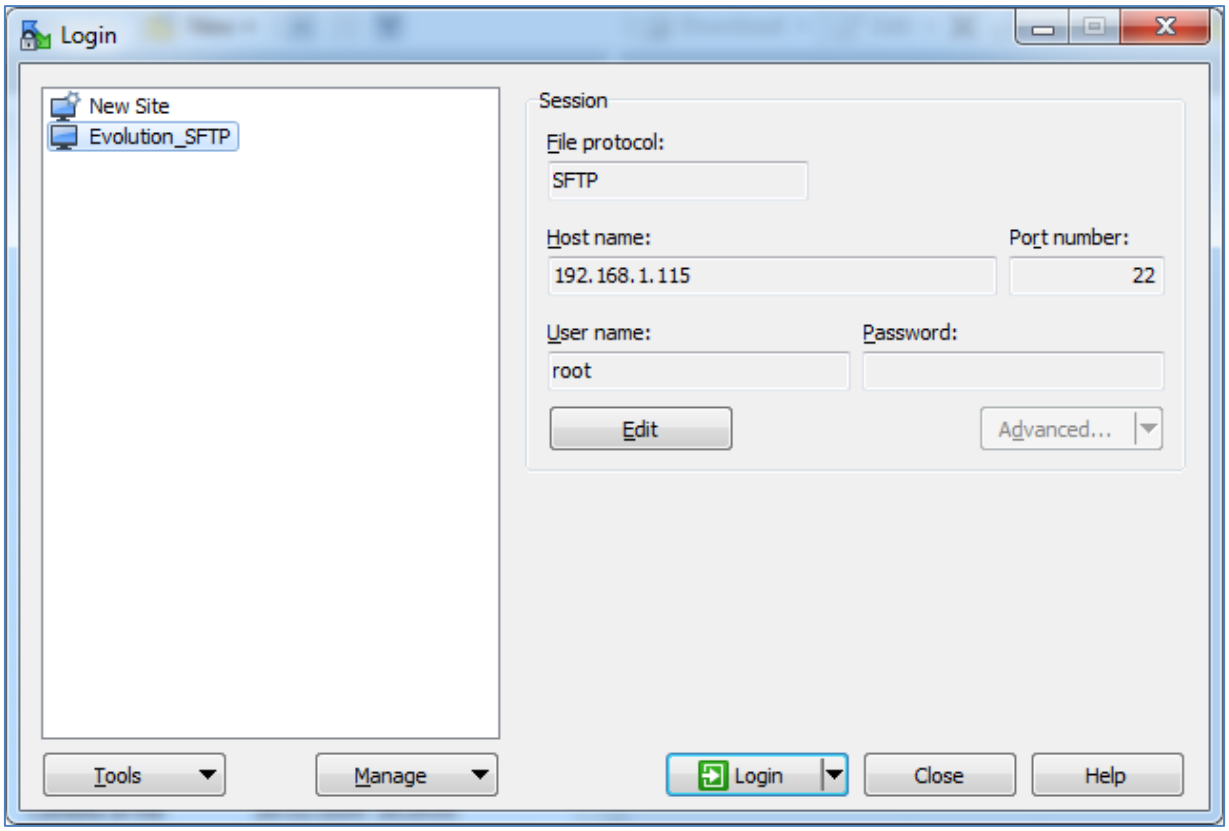
The user with whom a client can connect is the administrator user of the Linux system; the password of the administrator user must be requested from NESA s.r.l.

3.6 SFTP (Secure File Transfer Protocol)

The **SSH File Transfer Protocol** (or SFTP) is a network protocol that involves data transfer using the SSH protocol, so it is considered a secure file transfer.

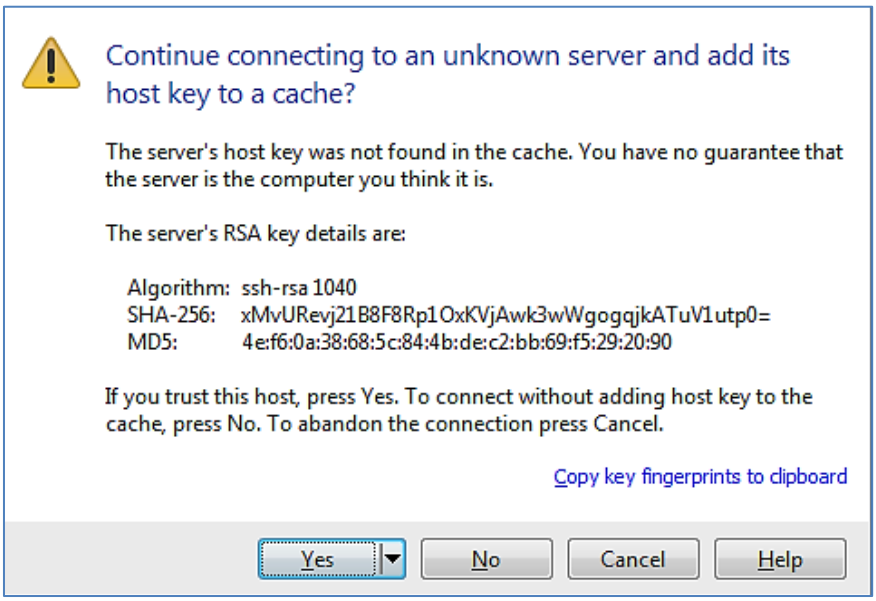
In the implemented version it allows data transfer using the standard port 22.

To connect to the datalogger server it is possible to use any client that supports the protocol, such as the **WinSCP** client freeware, described below.

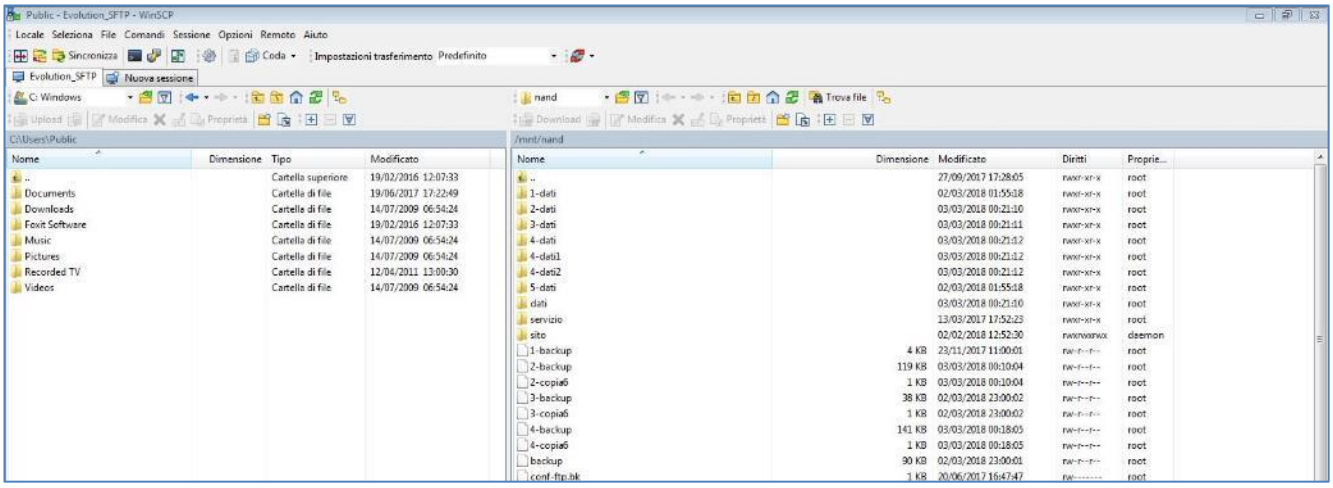


By selecting the protocol, setting the name and port of the server to be called and obviously User Name and Password (to be requested to NESA srl www.nesasrl.eu) it is possible to connect.

Being a secure connection, it is always advisable to save the key to be used for the connections:



And at the end of the transaction the window that allows the exchange of files between the connected client and the SFTP server inside the datalogger appears:



An updated list of available protocols is kept in *the Protocols Available* in *Advanced* section:



Protocollo	Porta	Linea di comunicazione
SOCKET	40737	TCP
MODBUS TCP	502	TCP
MODBUS RTU	---	Seriale
SDI-12	---	Seriale
FTP	21	TCP
SFTP	22	TCP
SSH	22	TCP
NTP	123	UDP

4 Connecting the sensors of evolution

Connection of the evolution sensors. A Nesa datalogger is able to manage, in its basic configuration, the following acquisition channels:

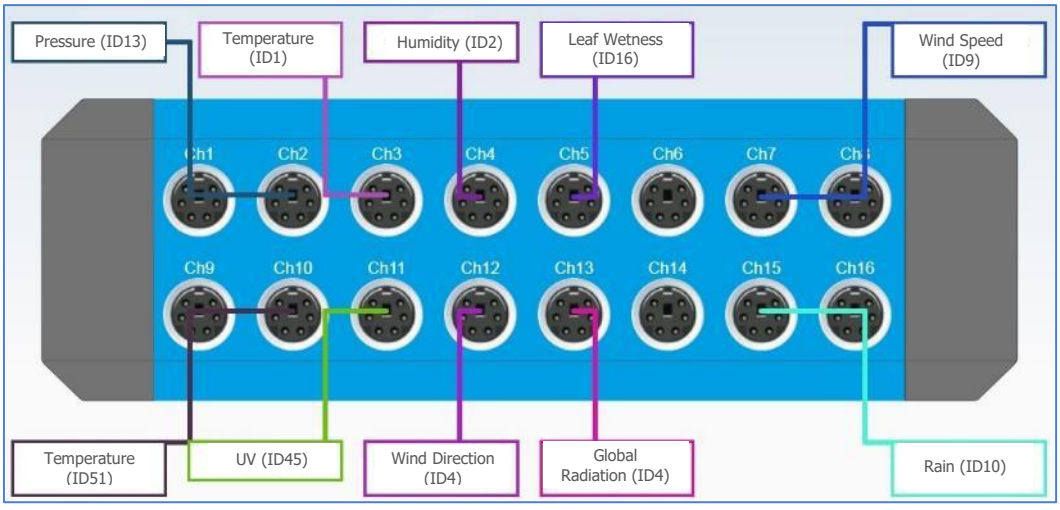
- **8 analog inputs** with 24 bit differential or **16** common mode precision
- **5 digital inputs** such as counters and frequency
- **4 analog outputs** in voltage with 12 bit precision
- **4 open drain digital outputs** for relay control
- **Measurements** of **battery voltage** and power supply voltage (**photovoltaic panel** or power supply), always present for a total of up to 18 analog inputs.
- **Serial connection** using **COM0** port if you would like to connect SDI-12 or RS485 sensors, but if connecting on USB port USB converter, are available serial port from **COM**3 to **COM9.**

  Note: on serial port COM 0 you can connect an RS485 sensor or an SDI-12, they cannot be connected simultaneously.

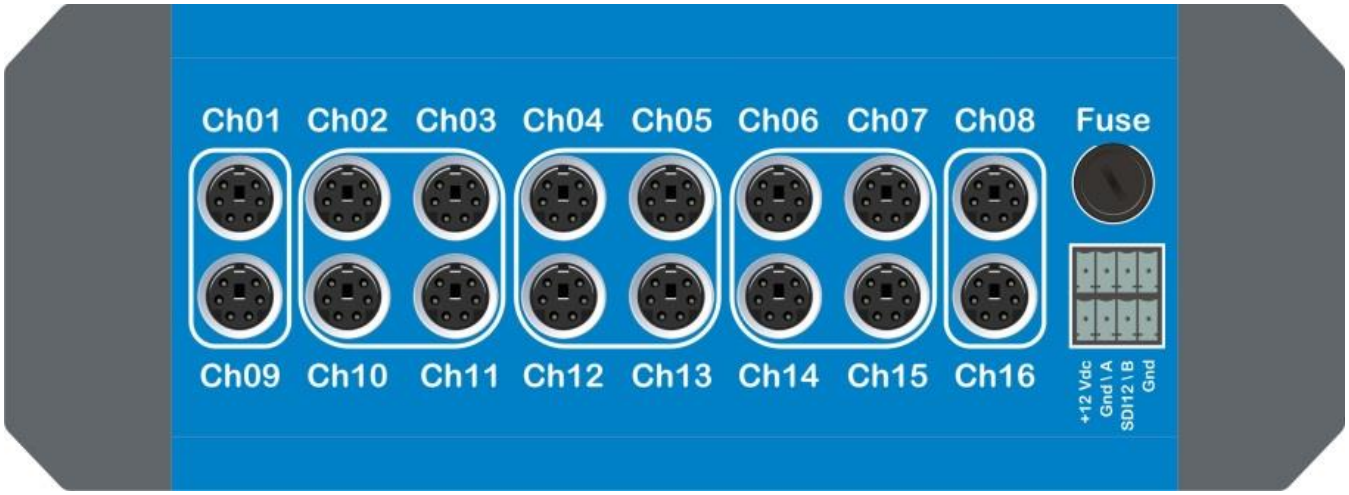
NOTE: *Before connecting a sensor to the datalogger, make sure that it is switched off, connect it and only then switch on the control unit to avoid electric shocks that could damage it irreparably.*

Each acquisition channel has a pinout according to the electric quantity to be acquired, as better specified below.

The connectors are of the Mini DIN 6-pole type (PS2 standard). If Nesa cables are used, they are already equipped with the relevant connector, otherwise it is possible to use the **M2Ch interface** tab that is equipped with overvoltage protections, or to use the schematic as show in next par.4.1 to realize your own cable with a PS2 connector. With reference to the paragraph 2.3.2.5, the relative configuration printout provides certain indications on the exact location of the PS2 inputs to be connected.



4.1 PS2 Connectors – Pin out



Channel	Pin	Acronym	Description
 **Ch 01**	1	**OC1**	Open Collector Output n.1 (max 50mA)
	2	**OC2**	Open Collector Output n.2 (max 50mA)
	3	**OC3**	Open Collector Output n.3 (max 50mA)
	4	**GND**	Ground
	5	**OC4**	Open Collector Output n.4 (max 50mA)
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 02**	1	**I1+**	Current pump for Pt100
	2	**V1+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V1-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I1-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 03**	1	**I2+**	Current pump for Pt100
	2	**V2+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V2-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I2-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt

 **Ch 04**	1	**I3+**	Current pump for Pt100
	2	**V3+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V3-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I3-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 05**	1	**I4+**	Current pump for Pt100
	2	**V4+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V4-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I4-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 06**	1	***nc***	Not connected
	2	**IN1 +**	First contact for digital input 1
	3	**IN1 -**	Second contact for digital input 1
	4	**GND**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 07**	1	***nc***	Not connected
	2	**IN2 +**	First contact for digital input 2
	3	**IN2 -**	Second contact for digital input 2
	4	**GND**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 08**	1	***CLK***	Link I^2Cbus (only service)
	2	***DATA***	
	3	***SDI12***	Link SDI12 (slave)
	4	**GND**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt

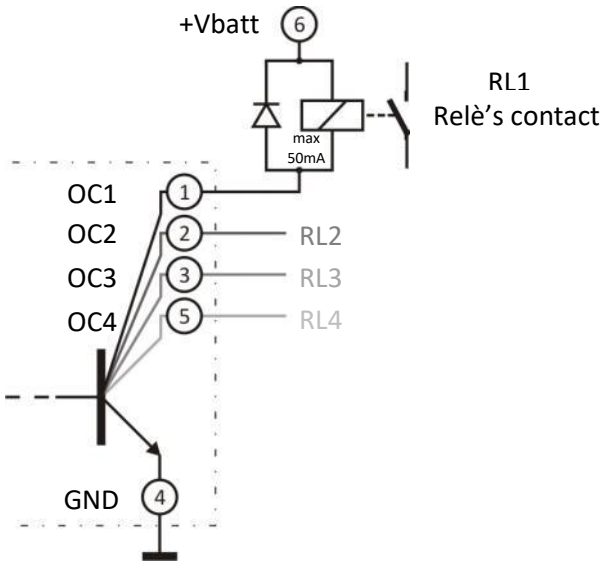
 **Ch 09**	1	***DAC-1***	1° Analog output 0÷2Vdc (min 100KΩ)
	2	***DAC-2***	2° Analog output 0÷2Vdc (min 100KΩ)
	3	***DAC-3***	3° Analog output 0÷2Vdc (min 100KΩ)
	4	***DAC-4***	4° Analog output 0÷2Vdc (min 100KΩ)
	5	***GND***	Ground
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 10**	1	**I5+**	Current pump for Pt100
	2	**V5+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V5-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I5-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 11**	1	**I6+**	Current pump for Pt100
	2	**V6+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V6-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I6-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 12**	1	**I7+**	Current pump for Pt100
	2	**V7+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V7-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I7-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 13**	1	**I8+**	Current pump for Pt100
	2	**V8+**	Voltage input V+ differential Voltage input Va+ single ended
	3	**V8-**	Voltage input V- differential Voltage input Vb+ single ended
	4	**GND (I8-)**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt

 **Ch 14**	1	***nc***	Not connected
	2	**IN3 +**	First contact for digital input 3
	3	**IN3 -**	Second contact for digital input 3
	4	**GND**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 15**	1	***nc***	Not connected
	2	**IN4 +**	First contact for digital input 4
	3	**IN4 -**	Second contact for digital input 4
	4	**GND**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt
 **Ch 16**	1	***Nc***	Not connected
	2	**IN5 +**	First contact for digital input 5
	3	**IN5 -**	Second contact for digital input 5
	4	**GND**	Ground
	5	***nc***	Not connected
	6	**+Vbatt**	Supply (out) + Vbatt

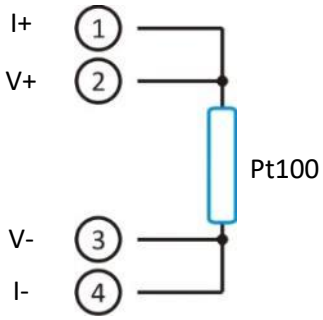
4.2 I/O connection on PS2 connectors

Following the pin-out shown in the table at par. 4.1, for the colors reported, is possible to connect:

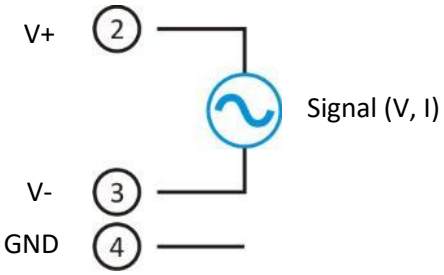
DIGITAL OUTPUT for commands. Configuration Open Collector: Ch 01



ANALOG INPUT. Configuration Pt100 and <u>differential input</u>: Ch 02 ÷ Ch 05; Ch 10 ÷ Ch 13

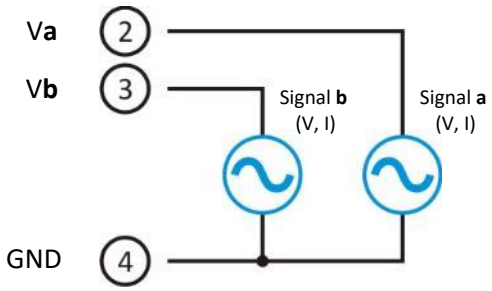


Temperature measure
Pt100

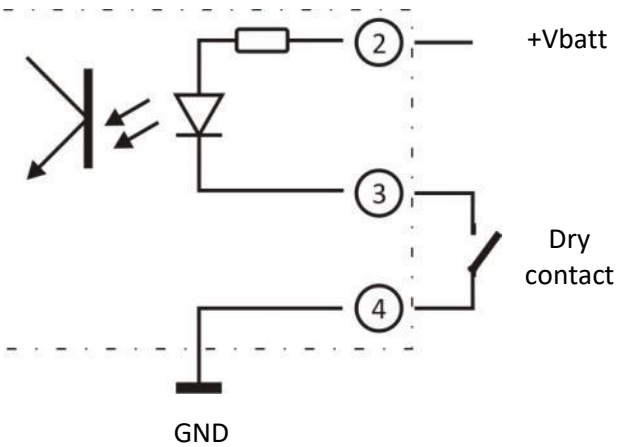


Differential signal's
measure

ANALOG INPUT. Configuration <u>Single Ended input</u>: Ch 02 ÷ Ch 05; Ch 10 ÷ Ch 13



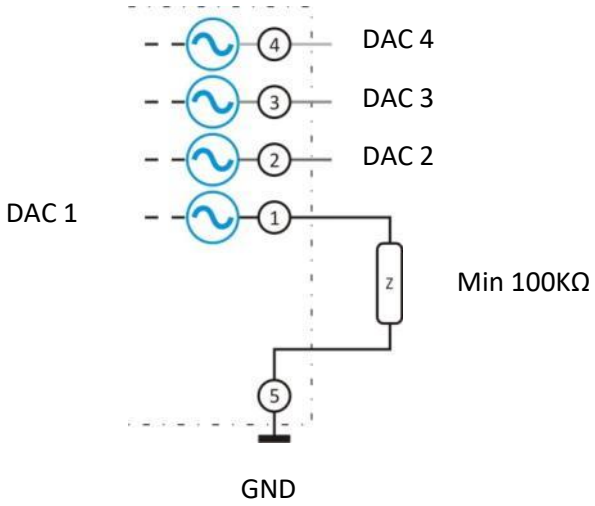
DIGITAL INPUT. Standard Configuration: Ch 06, Ch 07; Ch 14 ÷ Ch 16



Expansion: Ch 08

Dedicated channet to other devices as I^2CBus and SDI12

ANALOG OUTPUT: Ch 09



If using Nesa cables or assembled Ps2 cables, insert one at a time, with the machine off, in the relevant channel, pushing the connector all the way down, so the connector goes against the evolution terminal. Pay attention to the orientation of the connectors without forcing the insertion and without damaging the connectors themselves. It is advisable to number or mark the cable with a label that allows easy identification even after some time. Only after connecting all the sensors can the machine be switched on.

If no Nesa cables are used or special sensors are used, it may be necessary to have an interface between the PS2 and the sensor, which, in addition to facilitating the mechanical connection, also represents an electrical adaptation. More details in the next paragraph

NOTE: *If no Nesa cables are used or special sensors are used, it may be necessary to have an interface between the PS2 and the sensor, which, besides facilitating the mechanical connection, also represents an electrical adaptation and often an important overvoltage protection. More details in the next paragraph for using the Nesa M2Ch interface.*

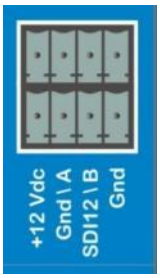
4.3 Connection of serial sensors

Into evolution you can connect serial sensors using:

- **Double a 4 pole connector**

This connector is located on the right side of PS2 connectors, and it allow to connect RS485 or SDI-12 sensors.

Note that you can connect or an RS485 sensor or an SDI-12 sensor, not both in the same time .



4-pole I/O double connector		
Name	**Lower position**	**Upper position**
+12 Vdc	Powered output (1A)	Powered output (1A)
GND/A	Mass line SDI12	RS485 A
SDI12/B	Line SDI12	RS485 B
Gnd	Ground clamp	Ground clamp

On the upper side you can connect an RS485 bus, instead on the lower side you can connect an SDI-12 bus.

Serial port's name is **COM0**.

- **Connection of a serial converter via USB**

You can also connect serial devices to logger using an usb to serial converter. Pay attention the chip of the converter must be only Prolific PL2303 or FTDI FT232.

In this case the serial communication ports made available go from **COM3** to **COM9**

4.4 I/O Connection trough M2Ch interface

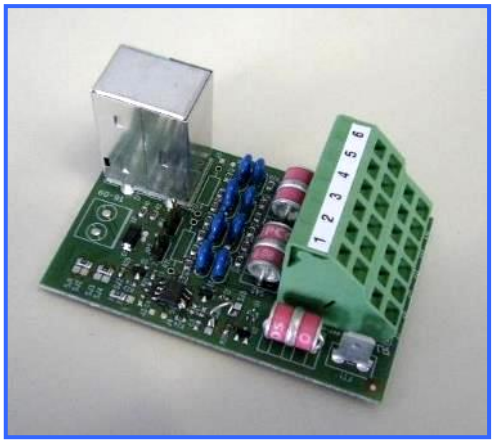
The NESA M2CH sensor protection board is configured as an accessory to the Nesa dataloggers, in order to make the wiring of any connected sensors more flexible, and to increase the protection against overvoltage in the data acquisition channels.

Using **a single tab** it is possible to acquire and protect **two acquisition channels**.

The tab consists of three main parts:

- double spring-loaded terminal block to allow easy wiring even of non-NESA sensors
- a double PS2 connector for direct connection to TMF family dataloggers
- a stage of protection against overvoltage by means of gas diodes and varistors

A voltage reference of 2,048 Volts was also introduced, available on pin 5 of each connector, in order to improve the interfacing of the non-Nesa potentiometric sensors; the minimum load on this pin **must be greater than 2KΩ**.



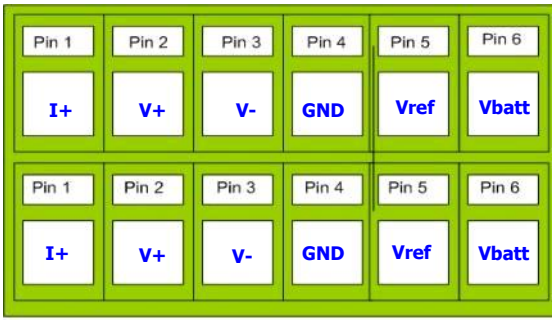
A connection to the system earth is also made available on the board, via the "faston" connector, necessary to make the protections against overvoltage effective.

The power supply of the board is supplied directly from the battery or power supply of the datalogger, or optionally, where it is necessary to use a separate power supply, through the use of a screw clamp.

The M2CH board is supplied on a tray for mounting on a DIN rail. The use of the M2CH protection tab is recommended in installations in sites particularly subject to overvoltage deriving from atmospheric discharges, such as for example the systems for the acquisition of anemometric data (wind towers) or for sensors that are otherwise difficult to connect.

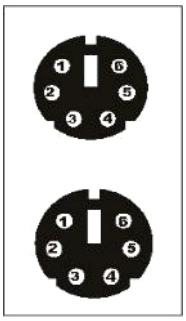
4.4.1 Electrical connection of the sensors

The pinouts of the spring-type connectors on one side and PS2 on the other and the signals present in them are



Upper Terminal

Lower Terminal



Pin	Signal
1	I+
2	V+
3	V-
4	GND
5	Vref (2Vdc)
6	+Vbatt

as follows:

4.4.2 Technical specifications

Downloader sizing voltage:	25V max
Test atmospheric current (10/350uS):	500A
Nominal current:	300mA
Temperature range:	-40÷+85°C
Insulation class:	IP20

4.4.3 Examples of connection

Sensor 4÷20mA loop power - 2 wires

Pin 1: not connected

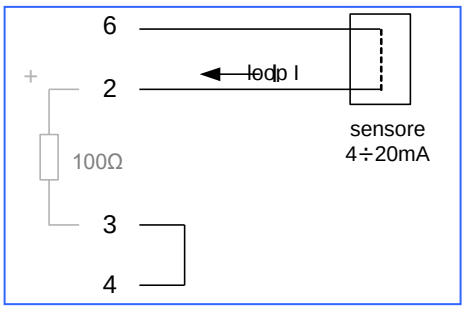
Pin 2: V+

Pin 3: V-

Pin 4: connector to Pin 3 + screen

Pin 5: not connected

Pin 6: +Vbatt



Red Label

In this case, request the module configured for current acquisition

Sensor 4÷20mA - 4 wires

Pin 1: not connected

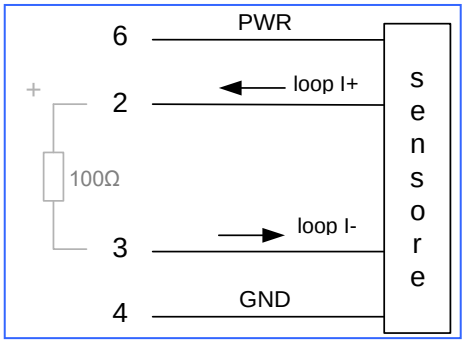
Pin 2: V+

Pin 3: V-

Pin 4: connector to Pin 3 + screen

Pin 5: not connected

Pin 6: +Vbatt



Red Label

In this case, request the module configured for current acquisition

Sensor 0÷2V

Pin 1: not connected

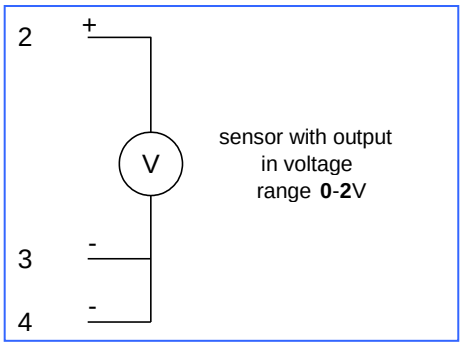
Pin 2: V+

Pin 3: V-

Pin 4: connector to Pin 3 + screen

Pin 5: not connected

Pin 6: non connected



White Label

Sensor 0÷5V

Pin 1: not connected

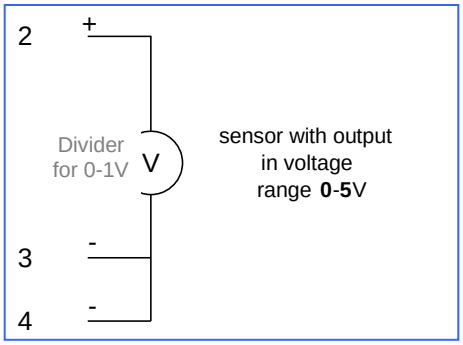
Pin 2: V+

Pin 3: V-

Pin 4: connector to Pin 3 + screen

Pin 5: not connected

Pin 6: not connected



Blue Label

In this case, remember that the signal is reduced 5 times and the configuration of the datalogger must therefore be updated.

Sensor Pt100 4 wires

Pin 1: Pt100 I+

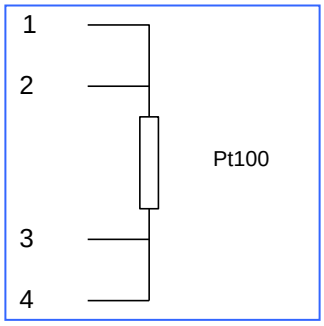
Pin 2: Pt100 V+

Pin 3: Pt100 V-

Pin 4: Pt100 I-

Pin 5: not connected

Pin 6: non connected



White Label

"Open collector" frequency sensor max 5000Hz

Pin 1: not connected

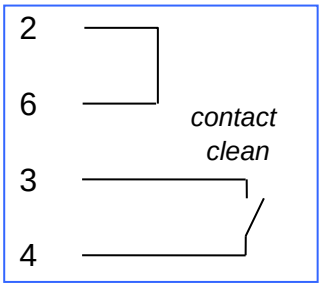
Pin 2: connected to Pin 6

Pin 3: frequency input

Pin 4: GND + screen

Pin 5: not connected

Pin 6: connected to Pin 2



White Label

Sinusoidal output sensor

Pin 1: not connected

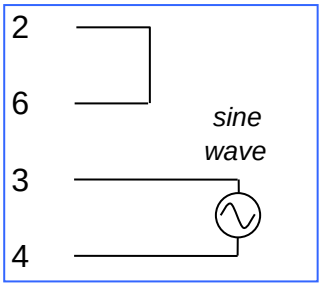
Pin 2: connected to Pin 6

Pin 3: input signal

Pin 4: GND + screen

Pin 5: not connected

Pin 6: connected to Pin 2



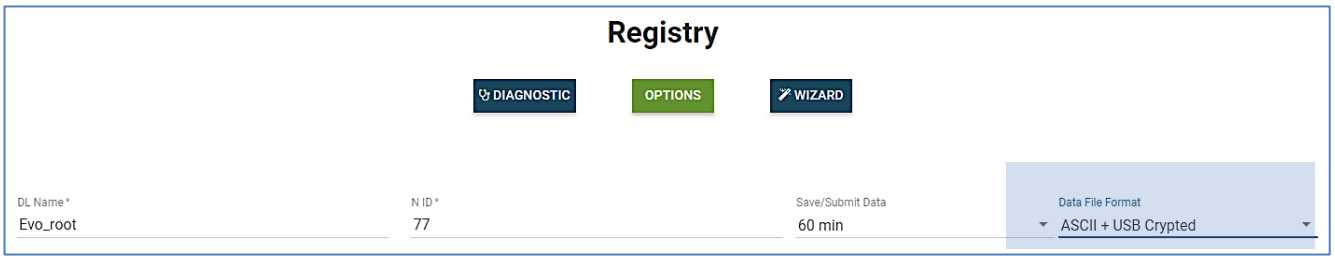
Yellow Label

5. Data encryption and protection

On the Personal Data configuration page, there is the format of the files that the machine can transfer. In particular:

- **ASCII + USB ASCII** : It stores data in the datalogger and in the USB memory data file readable ASCII format.
- **ASCII + USB Crypted** : It stores data in the data memory of the datalogger file in readable ASCII format and in the USB memory, data files in encrypted format with a 128bit key.

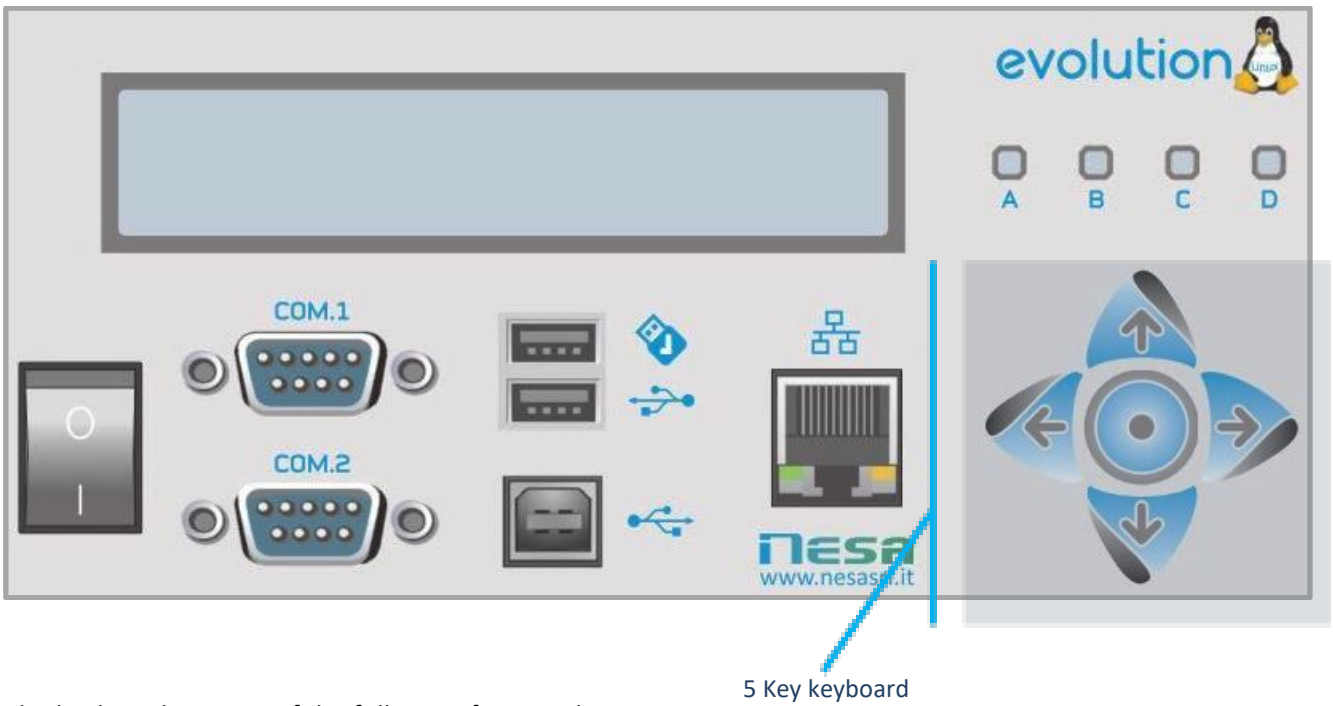
The data is also available in "**.csv**" format, which can be openable from any calculation sheet such as Excel ®
To select the encrypted data mode on the Registry page, select the ASCII + USB Crypted mode.



The **128 bit Hash** double-key encryption algorithm allows the generation of files that are completely unreadable without the decryption code and the corresponding program.
Both the program and the code, unique for each datalogger, are normally communicated and delivered with the terminal.
It is always possible to request a copy of this data by contacting NESA Srl at +39.0423.985209 or by e-mail to tecnico@nesasrl.it providing your personal details and the serial number of the product.
The program and the code will be sent by registered mail.

6. Using the keyboard

The evolution is equipped with a functional five-key keyboard that allows you to act on some parameters of the system and navigate through the acquired measurements.



The keyboard consists of the following function keys:



allows you to confirm the selection or enter the sub-menus



allow vertical scrolling of measurements or of menu items



allow navigation of measurements (associated to a each USER)

Display of measurements

Under normal operating conditions, the data logger presents on the first line of the display the ID associated with it (station ID), time and date, while, on the second line, it shows the measurements according to the configurations. The measurements are associated to each user starting from the MASTER (user 0)..

- By pressing the left /right arrows you can move to view the measurements of the specified user.
- By pressing the up/down arrow keys it is possible to move between any active users in the machine.



Setup

When the central button is pressed, the menu items relating to local configuration are enabled and available to the operator. Possible actions are:

`Date time setup`

Permits to modify time and data.

The menu shows sequentially all the date and time fields that can be modified using the up/down arrow keys and permit to confirm them using the center key.

At the end of the date/time change, the data logger restarts to apply the changes.

`Copy Bk file into USB`

It allows you to copy backup files inside the machine into a USB memory. At the end of the copying process, the output from the menu is automatic and the acquired measurements are presented again. (a memory of at least 8GB is recommended).

`Restore factory set`

It allows to restore the reconfiguration of the data logger to a default configuration stored in the factory during the production of the machine.

NOTE: *attention, in this case all the changes introduced by the user will be lost.*

`TEST Modem`

Allows you to perform a connection test with a serial modem using AT commands. To do this, follow the instructions given in paragraph 9.4

`Exit`

Exit from the menu.



7. ADVANCED Page: Data Record Output

In the Advanced configuration page, there are some functions that can be enabled or not depending on the version of the data logger.

In particular, there is the Tab **Data Record Output**, which allows you to customize the format of the file data (data record) of the stored and/or transmitted files for each user.

Starting from the NESA data record (see Appendix A: Extract of the data record layout) it is possible to write down the type of path that you want to build, combining the various elements of the original one.

More information available after specific training with Factory personnel.



8. CONFIGURATION EXAMPLE

Before making a configuration it is advisable to read this example section, building a configuration table that helps to avoid making mistakes, based on the suggested model below, and starting from simple considerations.

We provide an example configuration made by a user USER (or Master) that requires the acquisition of 11 mixed measures every 10 seconds, choosing between analog, digital, various serials, their processing in terms of minimum and maximum average value, each 30', the generation of an analog signal linked to a specific measurement and the generation of a threshold alarm given by the combination of two measures. Finally, the simultaneous sending of data and alarms to two different FTP areas and only alarms via email to different recipients.

It is possible to follow these steps:

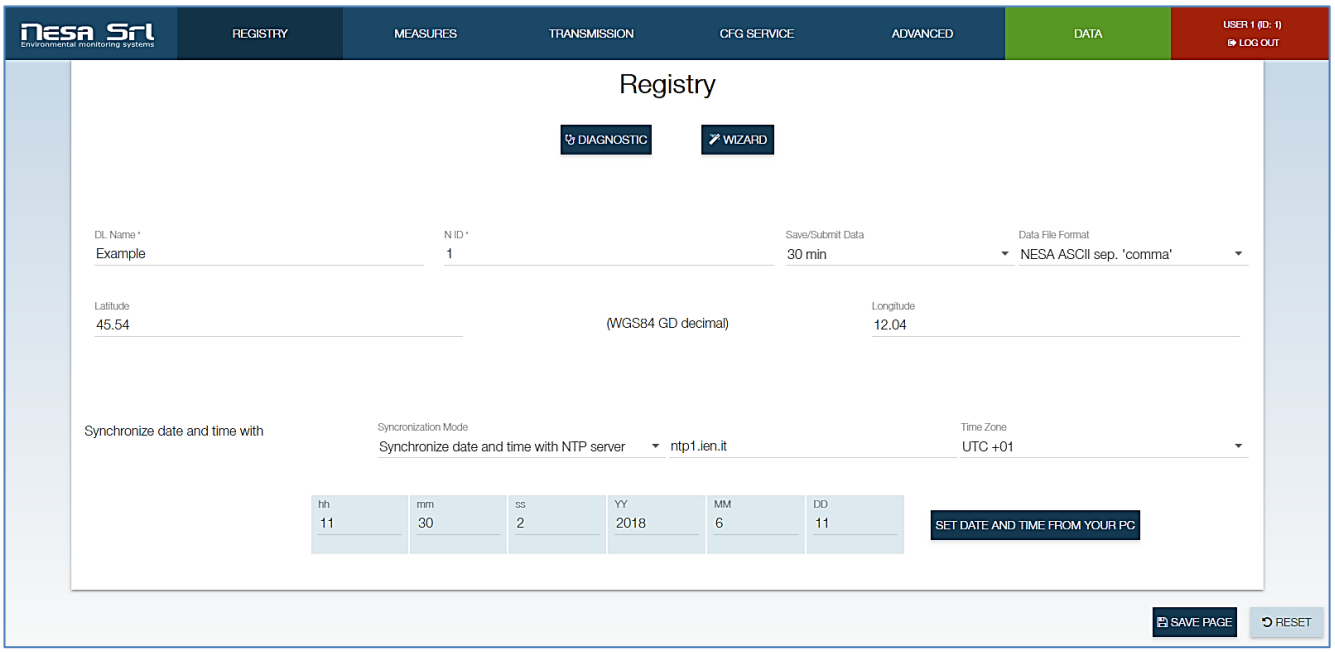
- Define the system registry
- Make a list of all measure to acquired
- Identify the type of signal available and the precision with which it is better to measure
- Identify the type of input to be configured
- Selects the needed elaborations for each measure
- Define from which measure and in which output to generate the analogue signal
- Define criteria (algorithm) and at which measure connect the alarm
- Program data and alarm transmission

Let this be the list of measures

List of measure to be acquired	Verify the measure each	Signal type	Input to be configured	Request elaboration each
temperature Pt100	10”	Pt100	Analogic Differential Temperature	Average min max 30’
pressure in Volt	10”	0÷2 Volt	Analogic Differential Volt/mA	Average min max 30’
Thermopile in µV	10”	0-100mV	Analogic Differential F.S. 100mV	Average min max 30’
discharge in Volt	10”	0÷2 Volt	Analogic Single ended Volt/mA	Average min max 30’
umidity in current	10”	4÷20mA	Analogic Single ended Volt/mA	Average min max 30’
Wind speed in frequency	10”	Hz	Digital – Frequency	Average min max 30’
Rain	10”	Accumulo	Digital – Counting	Totalization 30’

Contact's state	10"	Stato	Digital – State	instantaneous value 30'
Water level ModBus	10"	ModBus	Serial	Average min max 30'
Water level SDI12	10"	SDI12	Serial	Average min max 30'
Calculated measure	10"	Derivate	Derivate	Calculating dew Point 30'

8.1 *Step one: Registry*



Name:	**Example**	*simple mnemonic name*
ID N:	**000001**	*id that we want to give this Datalogger and that will be reported in all files transmitted and stored*
Save / Submit data:	**30'**	*time rate chosen for storing and sending data*
Data format:	**NESA ASCII Sep. ','**	*Nesa Data Record format with separator "," - see Appendix A: Data record track extract*
Lat. and Longitude	**45.54, 12.04**	*values chosen according to the position in WGS84 format*
Time syncronization	**ntp1.ien.it**	*via NTP on server ntp1.ien.it and UTC+1 as time zone in this example*

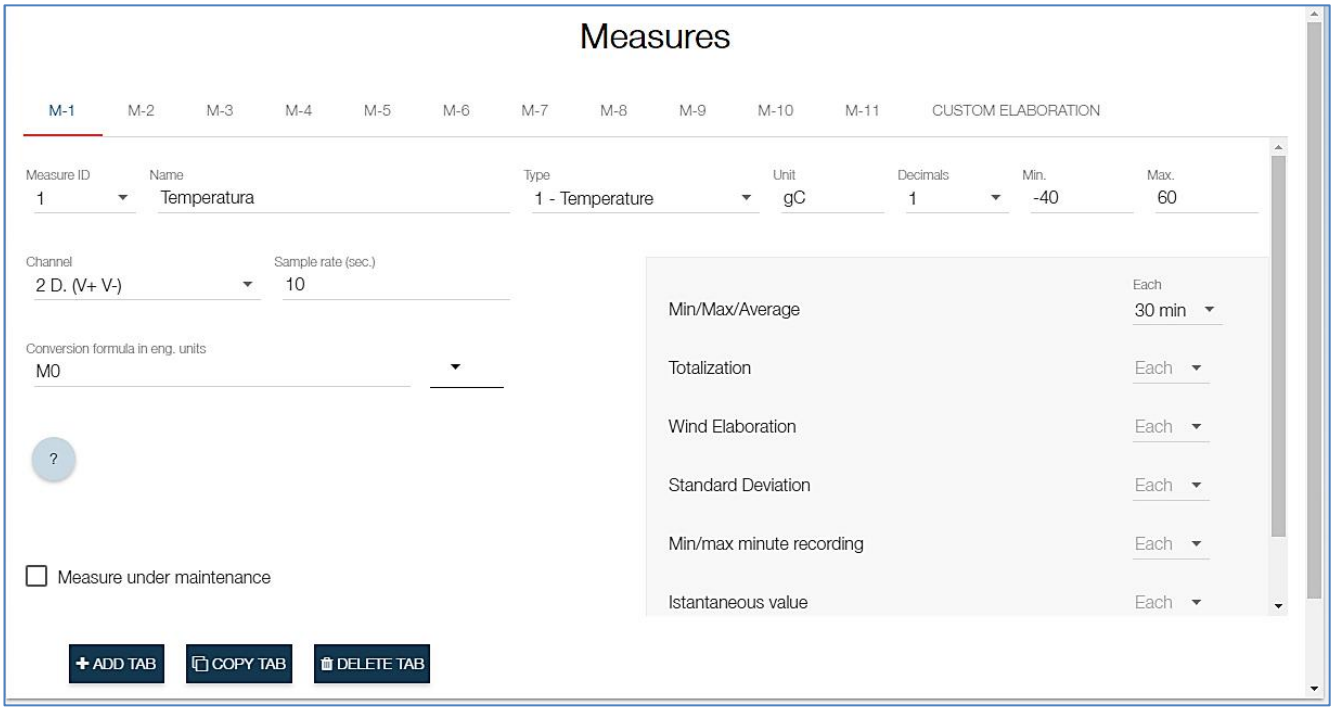
At the end of setting press SAVE PAGE

8.2 Step two: *Measure*

Referring to the table above, following the order, we configure each measure.

To save the configuration it is enough do it only once at the end of the configuration of all the measurements.

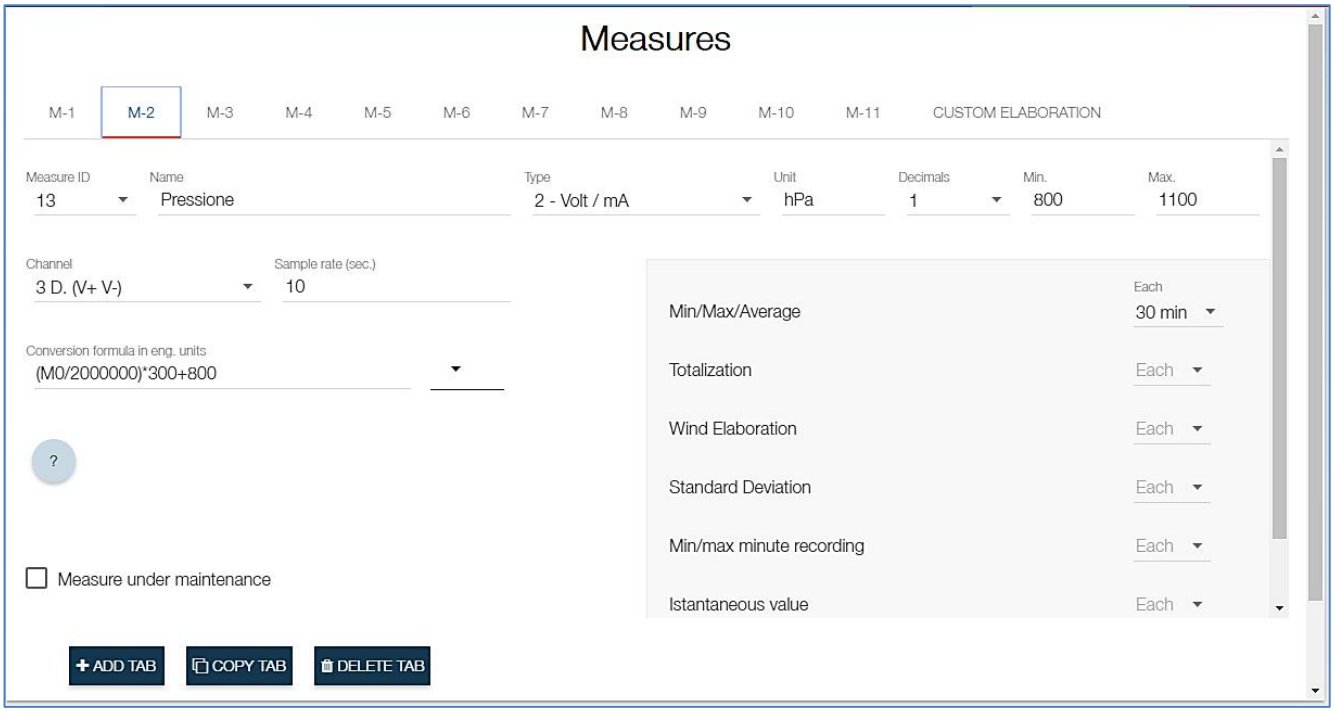
8.2.1 Temperature



Meas. ID/Name:	**Temperature**	*choose "Temperature" from drop-down menu*
Type:	**Temperature**	*Being a thermo-resistance it is necessary to specify that this is a temperature type measurement to prepare the data logger to the right way of acquisition*
Unit and decimals:	**°C 1**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**-40 , +60**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value (i.e. '*')*
Channel	**2 D.(V+ V-)**	*First analog channel available for differential measure*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample (see par. 2.3.2.2)*
Conversion formula	**M0**	*Latest raw instantaneous value (as per table ?)*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The first measure is configured, press the key + ADD TAB to go to the next measure.

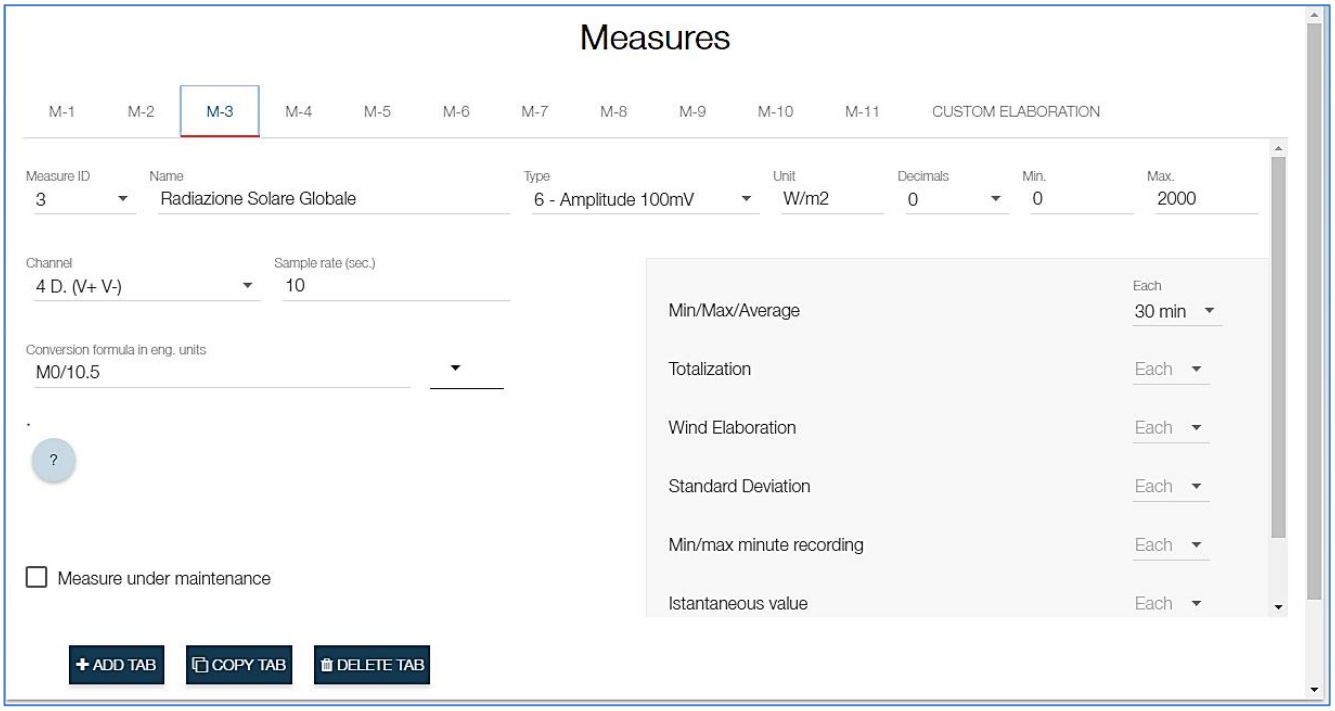
8.2.2 Pressure



Meas. ID/Name:	**Pressure**	*choose "Pressure" from drop-down menu*
Type:	**Volt/mA**	*Being a Voltage signal (see measures table)*
Unit and decimals:	**hPa 1**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**800, 1100**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**3 D.(V+ V-)**	*Second analog channel available for differential measure*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**(M0/2000000)*300+800**	*Conversion Formula for Barometer 800-1100hPa. The datalogger associates a measure 0-2Vdc or 0-2000000µV to the range of 300hPa (or else from 800 to 1100)*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The second measure is configured, press the key + ADD TAB to go to the next measure.

8.2.3 Thermopile or Solar Radiation

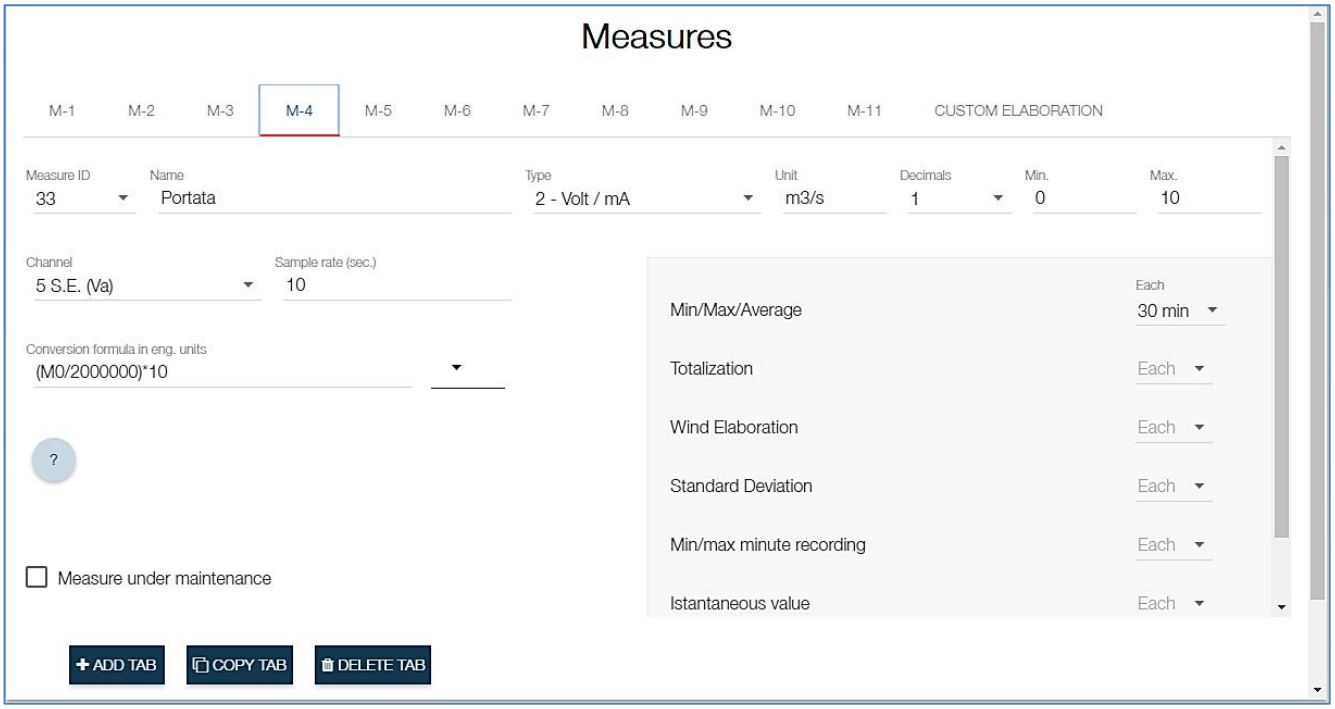


Meas. ID/Name:	**Solar Radiation**	*choose "Solar Radiation" from drop-down menu*
Type:	**Amplitude 100mV**	*Being a small signal in mV (see measures table)*
Unit and decimals:	**W/m² 0**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0, 2000**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**4 D.(V+ V-)**	*Third analog channel available for differential measure*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**M0/10.5**	*Conversion Formula for thermopile (10.5 is its constant of used thermopile)*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The third measure is configured, press the key to go to the next measure.



8.2.4 Discharge

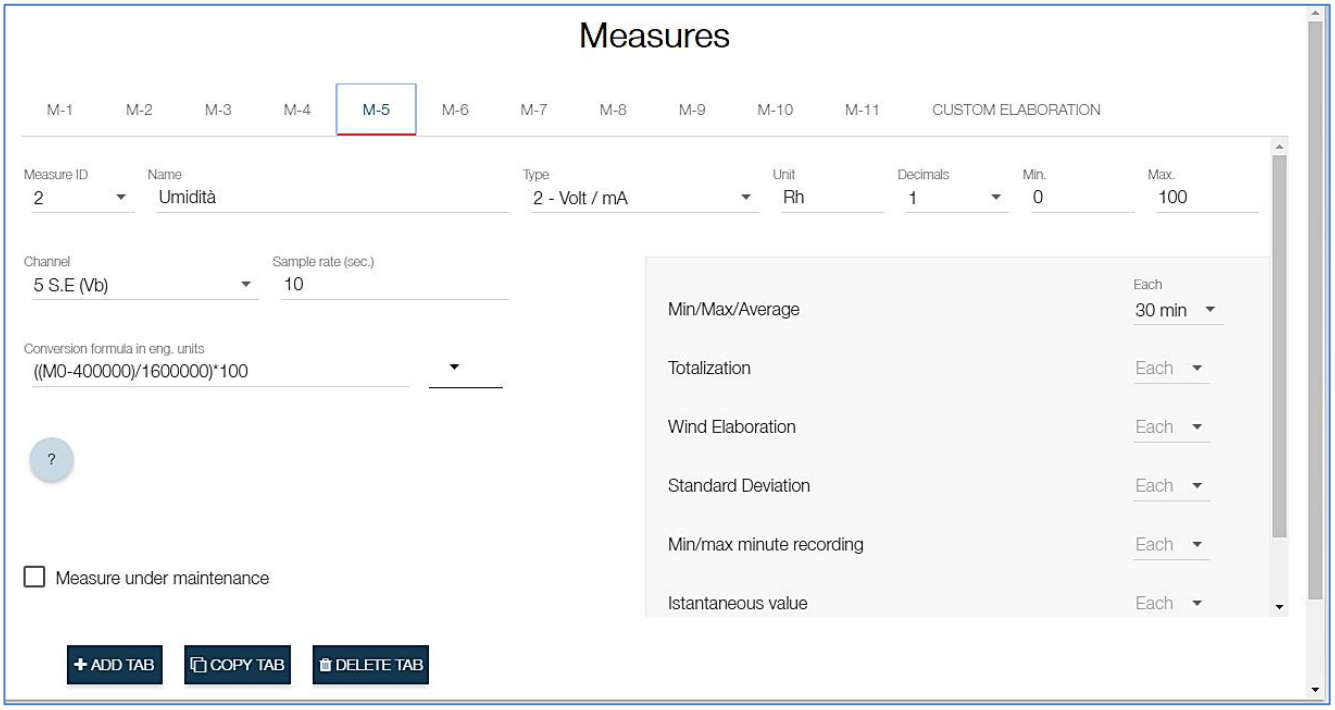


Meas. ID/Name:	**Discharge**	*choose "Discharge" from drop-down menu*
Type:	**Volt/mA**	*Being a voltage signal (see measures table)*
Unit and decimals:	**m^3/s 1**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0, 10**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**5 S.E.(Va)**	*First analog channel available for single ended measure*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**(M0/2000000)*10**	*Conversion Formula for a discharge instrument (full scale is 10m^3/s)*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The forth measure is configured, press the key to 

go to the next measure.

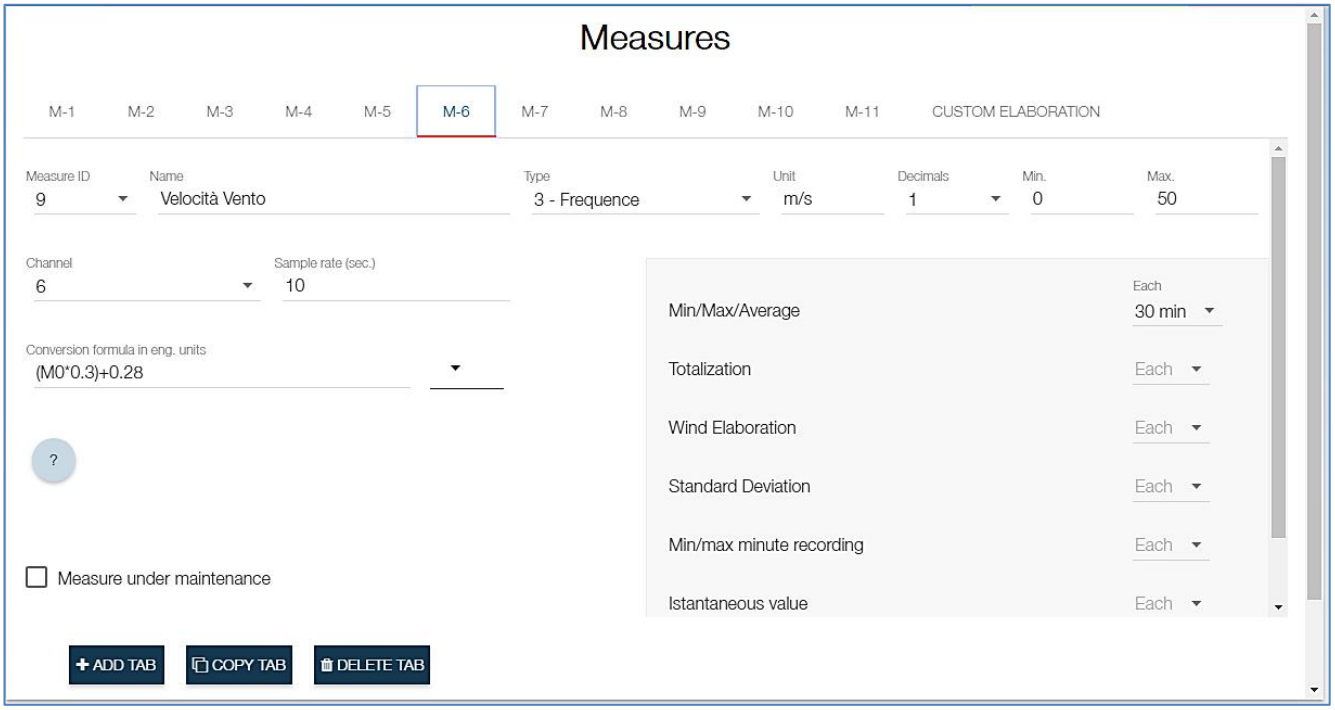
8.2.5 Humidity



Meas. ID/Name:	**Humidity**	*choose "Humidity" from drop-down menu*
Type:	**Volt/mA**	*Being a 4-20mA signal (see measures table)*
Unit and decimals:	**Rh% 1**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0, 100**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**5 S.E.(Vb)**	*Second analog channel available for single ended measure*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**((M0-400000)/1600000)*100**	*Conversion Formula for a 4-20mA signal (scale 0-100%)*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The fifth measure is configured, press the key + ADD TAB to go to the next measure.

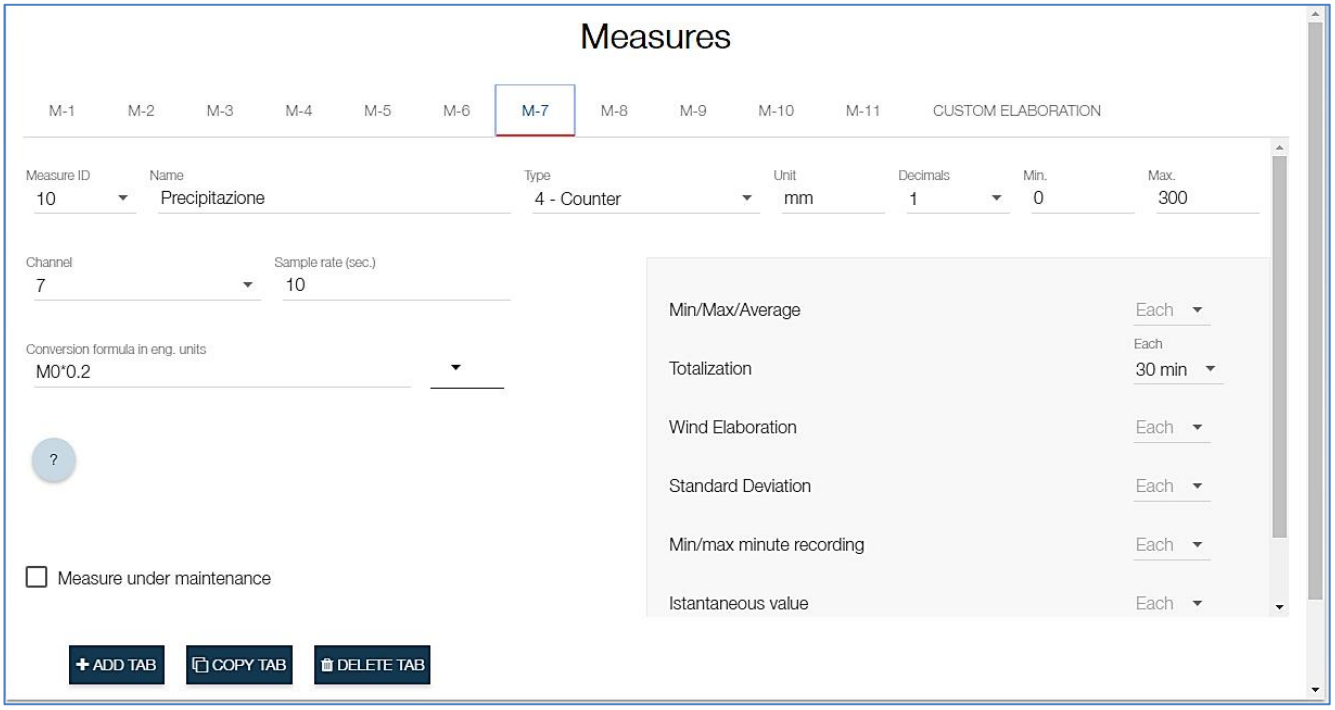
8.2.6 Wind Speed



Meas. ID/Name:	**Wind Speed**	*choose "Wind speed" from drop-down menu*
Type:	**frequency**	*Being a frequency signal (see measures table)*
Unit and decimals:	**m/s 1**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0, 50**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**6**	*First digital channel available for measures of frequency/state/counting*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**(M0*0.3)+0.28**	*Conversion Formula for a frequency signal with offset 0.28 and slope0.3 (specific for the used sensor)*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The sixth measure is configured, press the key + ADD TAB to go to the next measure.

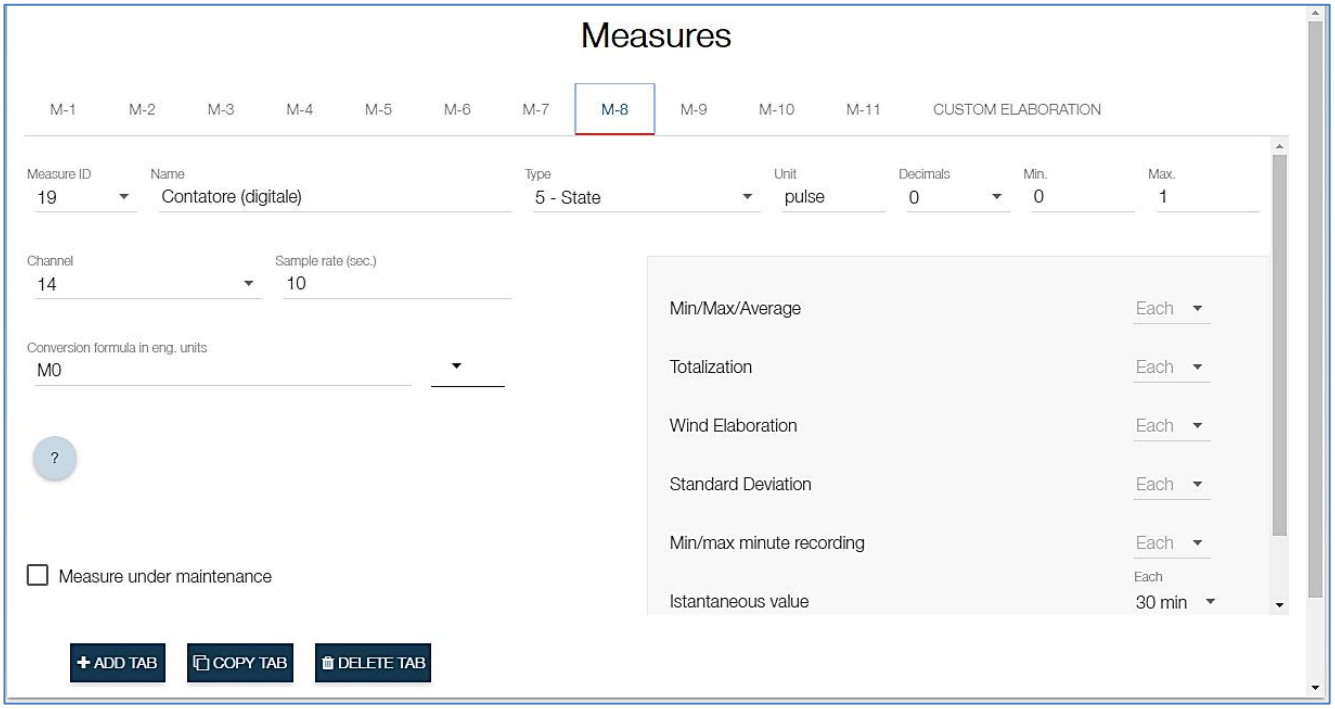
8.2.7 Rain



Meas. ID/Name:	**Rain**	*choose "Rain" from drop-down menu*
Type:	**Counter**	*Being a counter signal (see measures table)*
Unit and decimals:	**mm 0**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0, 300**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**7**	*Second digital channel available for measures of frequency/state/counting*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**M0*0.2**	*Conversion Formula for a counting signal with constant 0.2 (specific for the used sensor)*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The seventh measure is configured, press the + ADD TAB key to go to the next measure.

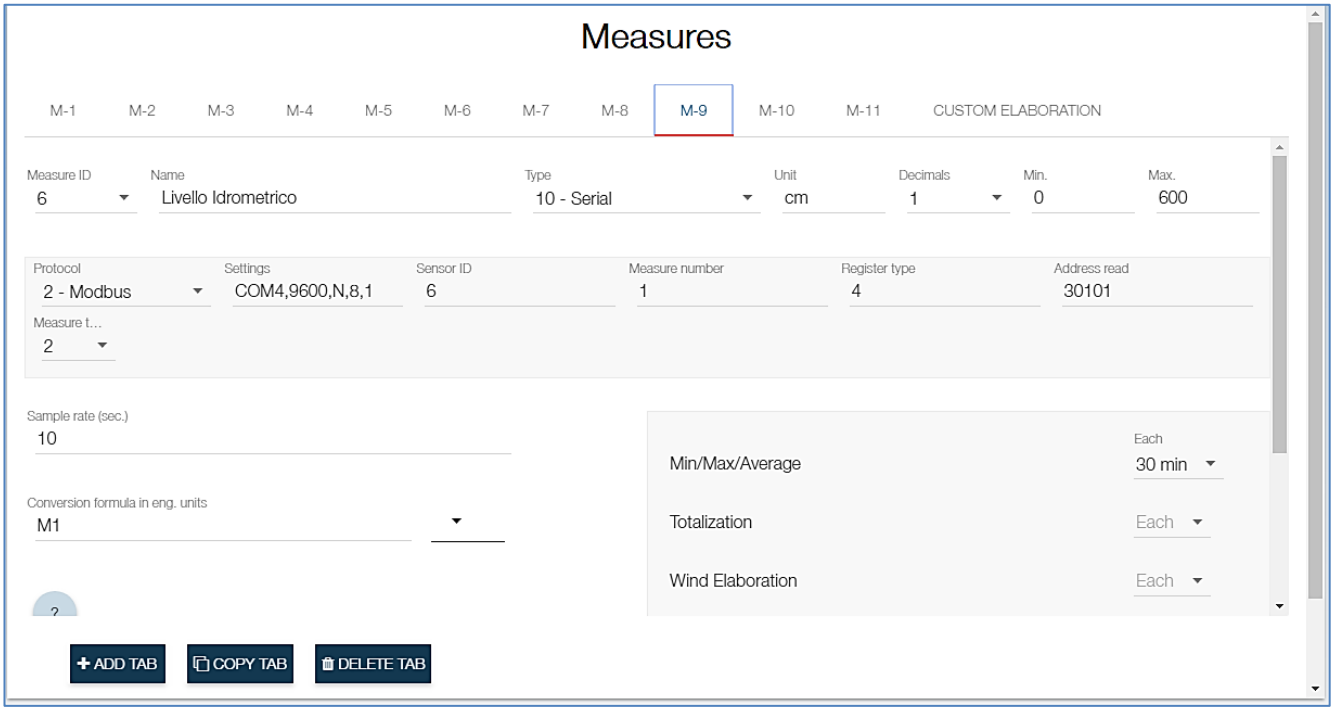
8.2.8 State



Meas. ID/Name:	**Counter**	*choose "Counter" from drop-down menu*
Type:	**State**	*Being an ON/OFF signal (see measures table)*
Unit and decimals:	**pulse 0**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0, 10**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**14**	*Third digital channel available for measures of frequency/state/counting*
Sample Rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**M0**	*State's value*
Min/Max/Average	**30'**	*choice for time interval to verify the state*

The eight measure is configured, press the key + ADD TAB to go to the next measure.

8.2.9 Water Level ModBus

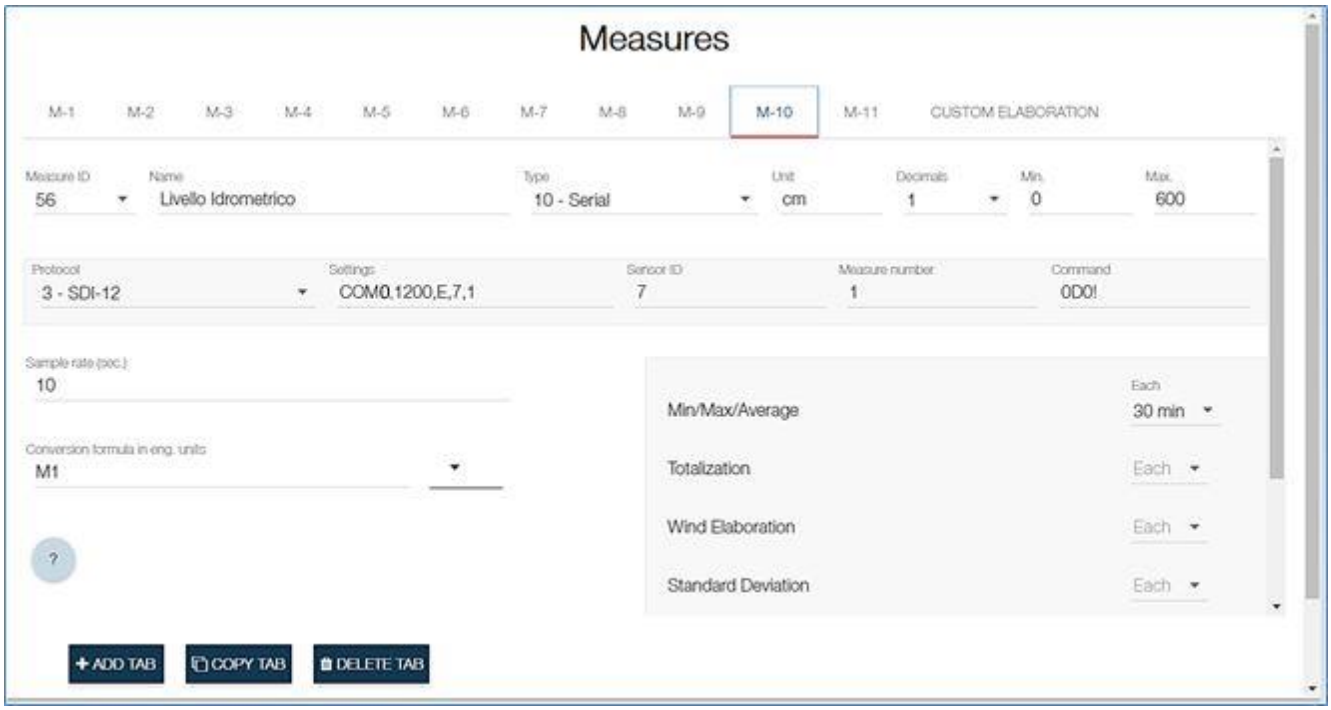


Meas. ID/Name:	**Hydrometric level**	*choose "Level" from drop-down menu*
Type:	**Serial**	*Being a "Serial" signal (see measures table)*
Unit and decimals:	**mm 0**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0 , 600**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value (i.e. '*')*
Protocol	**ModBus**	*Choice of protocol type*
Setting	**COM4,9600,N,8,1**	*Setting for serial communication, normally is COM4*
Sensor's ID	**6**	*Sensor's address on ModBus*
Measures' number	**1**	*Measures number to read (multi sensor case)*
Register's type	**4**	*Type of ModBus register to use*
Register address	**30101**	*ModBus address register*
Codification type	**2**	*According to Modbus specification for the used sensor*
Sample rate	**10"**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**M1**	*Latest read value (as per table ?)*

Med/Min/Max	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

The nineth measure is configured, press the + ADD TAB key to go to the next measure.

8.2.10 Water Level SDI12

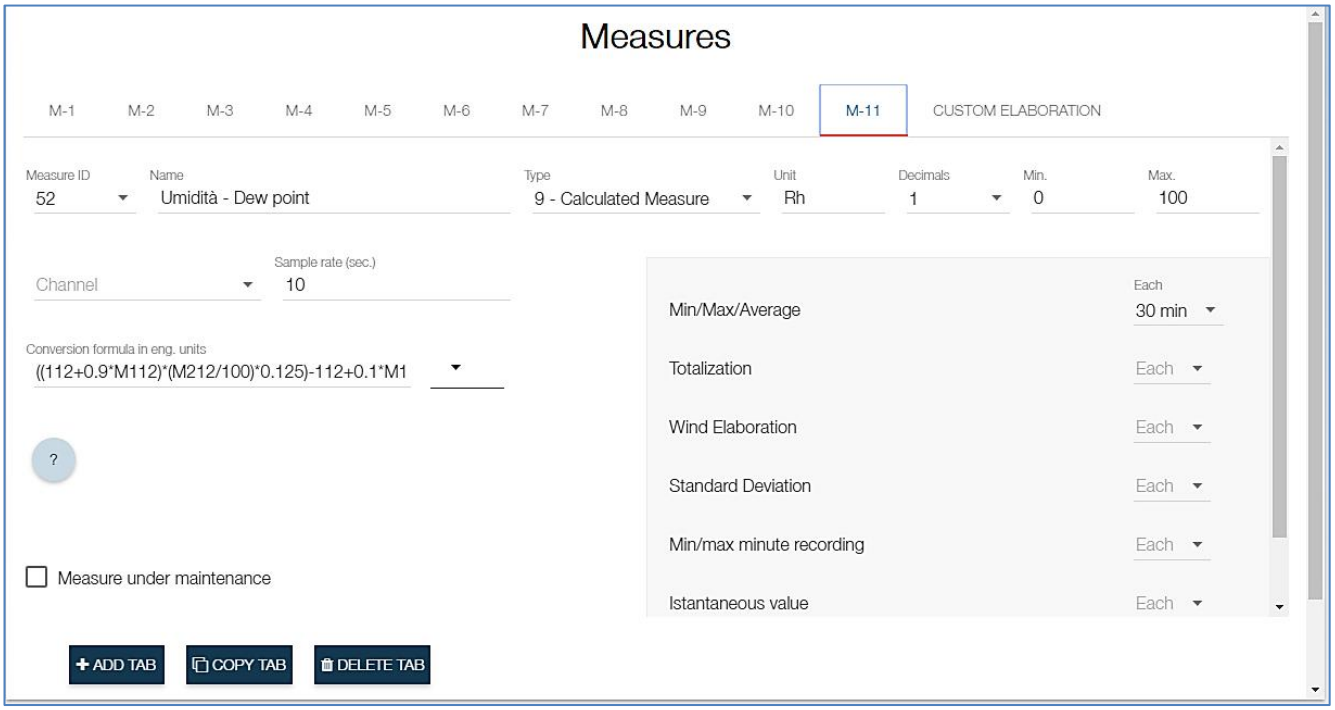


Meas. ID/Name:	**Hydrometric level**	*choose "Level" from drop-down menu*
Type:	**Serial**	*Being a "Serial" signal (see measures table)*
Unit and decimals:	**mm 0**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0 , 600**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value (i.e. '*')*
Protocol	**SDI12**	*Choice of protocol type*
Setting	**COM0,1200,N,8,1**	*Setting for serial communication, normally is COM3*
Sensor's ID	**7**	*Sensor's address on ModBus*
Measures' number	**1**	*Measures number to read (multi sensor case)*
Read command	**0D0!**	*According to the SDI version is possible to use all commands*

Sample rate	**10”**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*
Conversion formula	**M1**	*Latest read value (as per table ?)*
Med/Min/Max	**30’**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30’x60”/10”= 180 samples every 30' from which to calculate minimum and maximum average*

The tenth measure is configured, press the key + ADD TAB to go to the next measure.

8.2.11 Calculated Measure – Dew point



Meas. ID/Name:	**Humidity-Dew Point**	*Type the name that you prefer*
Type:	**Calculated measure**	*Being a derivate type (see measures table)*
Unit and decimals:	**Rh% 1**	*engineering unit and number of decimals with which you want to represent the data*
Min / Max:	**0, 100**	*Limit values (measuring range) of the sensor or measurement, out of which an out-of-scale symbol is substituted for the value(i.e. '*')*
Channel	**---**	*Being a calculated measure, it doesn’t use any physic channel*
Sample Rate	**10”**	*time interval in seconds for acquisition of primary sample(see par. 2.3.2.2)*

Conversion Formula	**((112+0.9*M112)*(M212/100)*0.125)-112+0.1*M112**	*Magnus Tetens formula for calculating del Dew point as per. 2.3.2.2*
Min/Max/Average	**30'**	*choice for processing data of samples acquired at intervals of 10". Putting 30' here, you will have 30'x60"/10"= 180 samples every 30' from which to calculate minimum and maximum average*

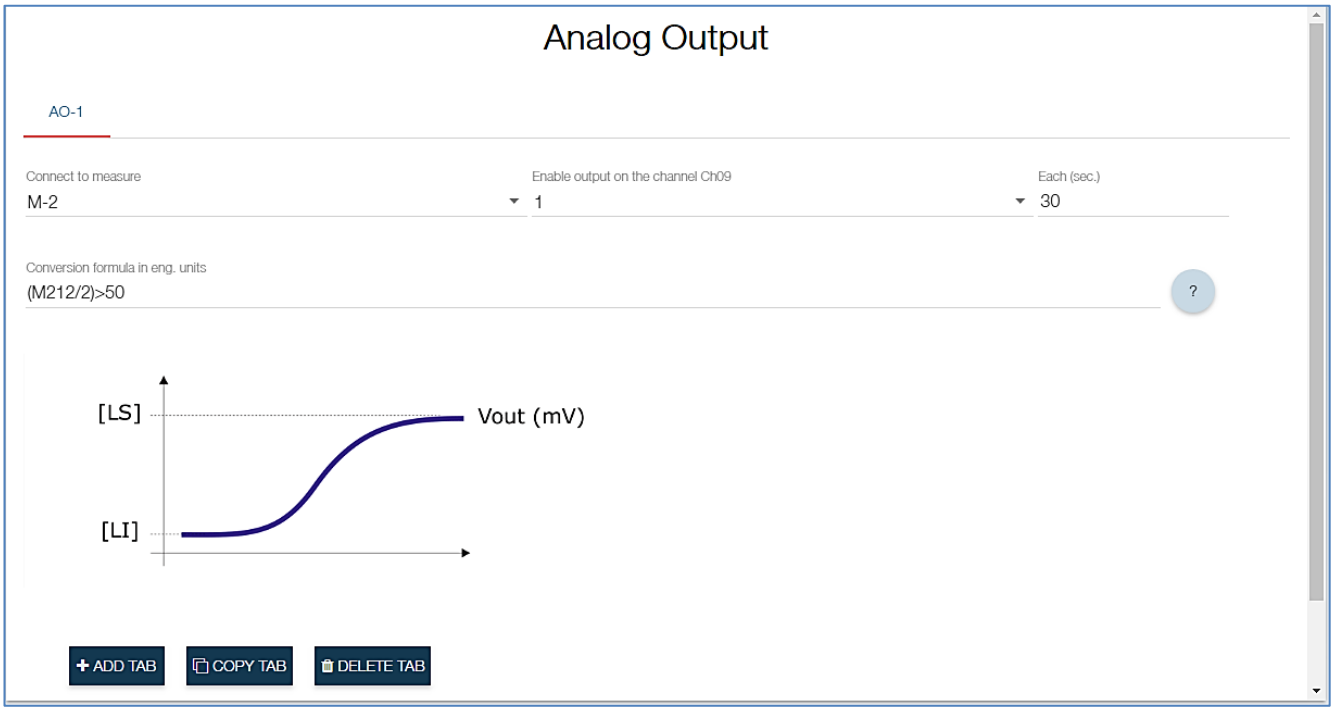
The eleventh measure is configured, press the key + ADD TAB to go to the next measure or to SAVE PAGE finish.

8.2.12 Generate analog output

As indicated at the beginning of the chapter, we must generate an analogue output linked, for example, to the measurement of pressure. Let's say we want a 0-2Vdc signal that follows the semi-value of the average pressure's value, only when it exceeds the 850hPa threshold.

The conversion formula will be: **(M212/2)>850**

Where **M212**, according to the table of the memory locations of measure n.2 (Pressure), is precisely the average value of the calculated pressure.



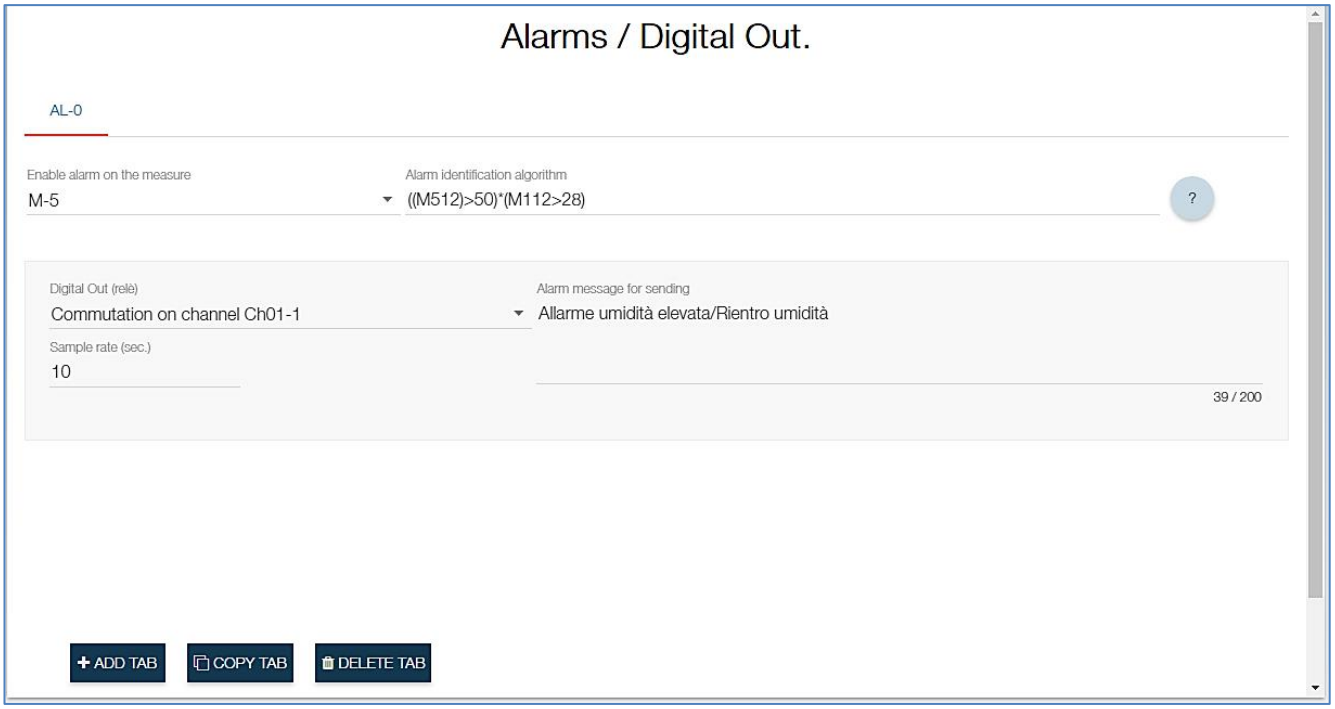
Connect to measure:	**M-2**	*choose from the drop-down menu the measure to generate from*
Enable output:	**1**	*select which of 4 outputs available we want to use (see par. 4.4.1) from channel no. 9*
Each:	**30”**	*time rate for updating the signal out*
Conversion formula	**(M212/2)>850**	*read above*

Press the key SAVE PAGE to finish.

8.2.13 Alarm management

We define as an alarm condition, the exceeding of the humidity threshold over 50% and with a temperature of 28 °.

The identification algorithm will be: **((M512)>50)*(M112>28)**



Alarm on measure:	**M-5**	*choose from the drop-down menu the measure to which hooked*
Alarm identification algorithm:	**((M512)>50)*(M112>28)**	*read above*
Digital output:	**Ch01-1**	*select which of 4 outputs available we want to use (see par. 4.4.1) from chanel no. 1)*

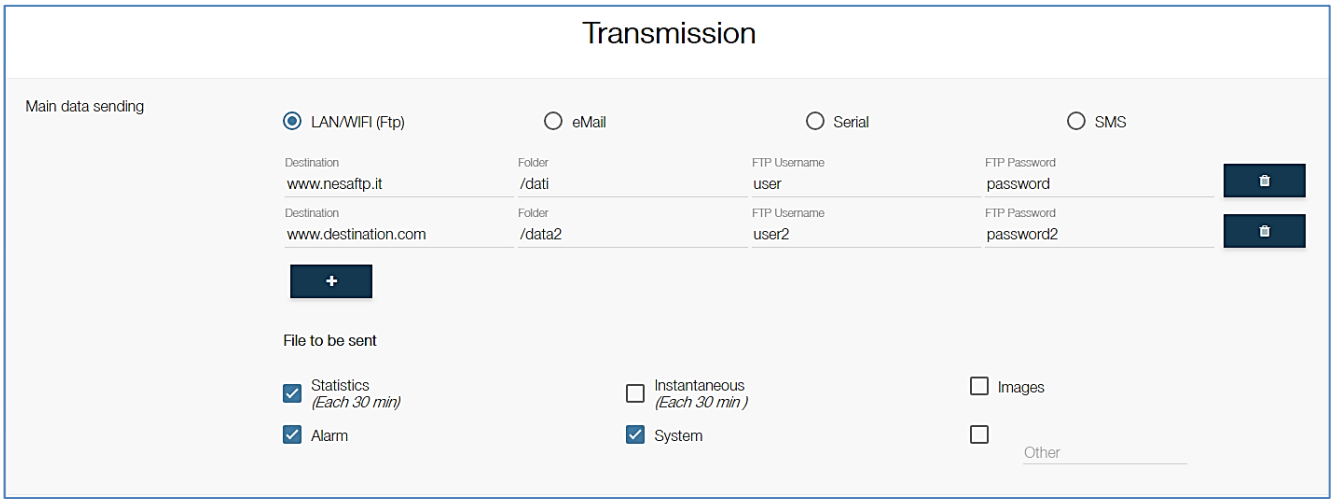
Alarm message for sending	**High humidity alarm/Reset alarm**	*alarm text and relative reset to be send via text message, email or file*
Sample rate:	**10”**	*time rate for updating the*

Press the key SAVE PAGE to finish.

8.3 Step three: Transmission

Once the measures are configured, we proceed to define the transmission of data.

In main data sending, we choose to send data via FTP to two different areas and recipients, including data files, alarms and system files. Each file will then contain the data processed for each measurement, calculated every 30' and sent / stored every 30'.

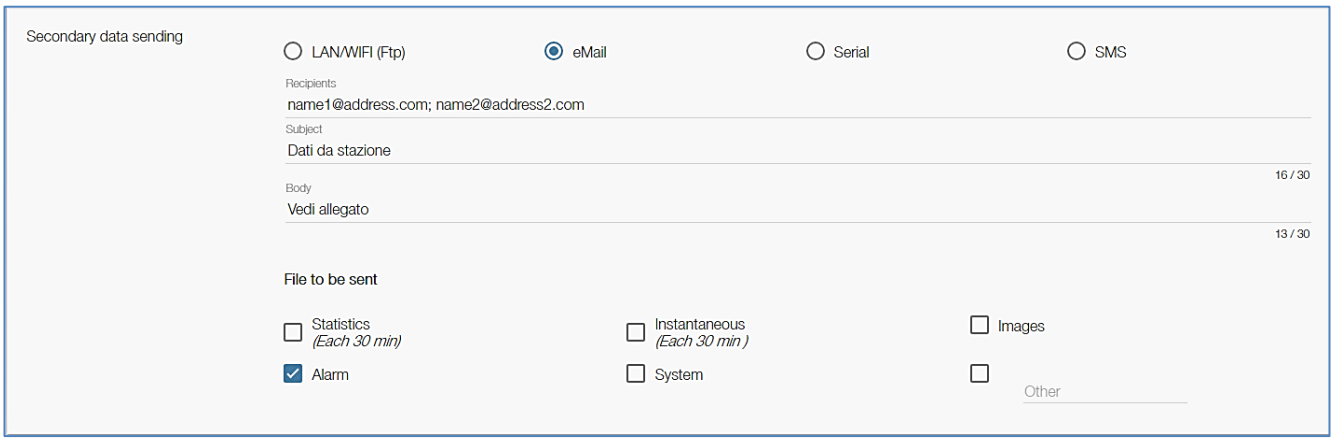


It is enough to fill in the fields of the chosen transmission type, in this case LAN/WIFI for the mode, addresses and destination credentials based on how many are the consignee.

Sending by e-mail (or text message or other)

Fill in the fields with the recipients by separating them with the ';' add the object and/or body of the message and choose the alarms as attachments.

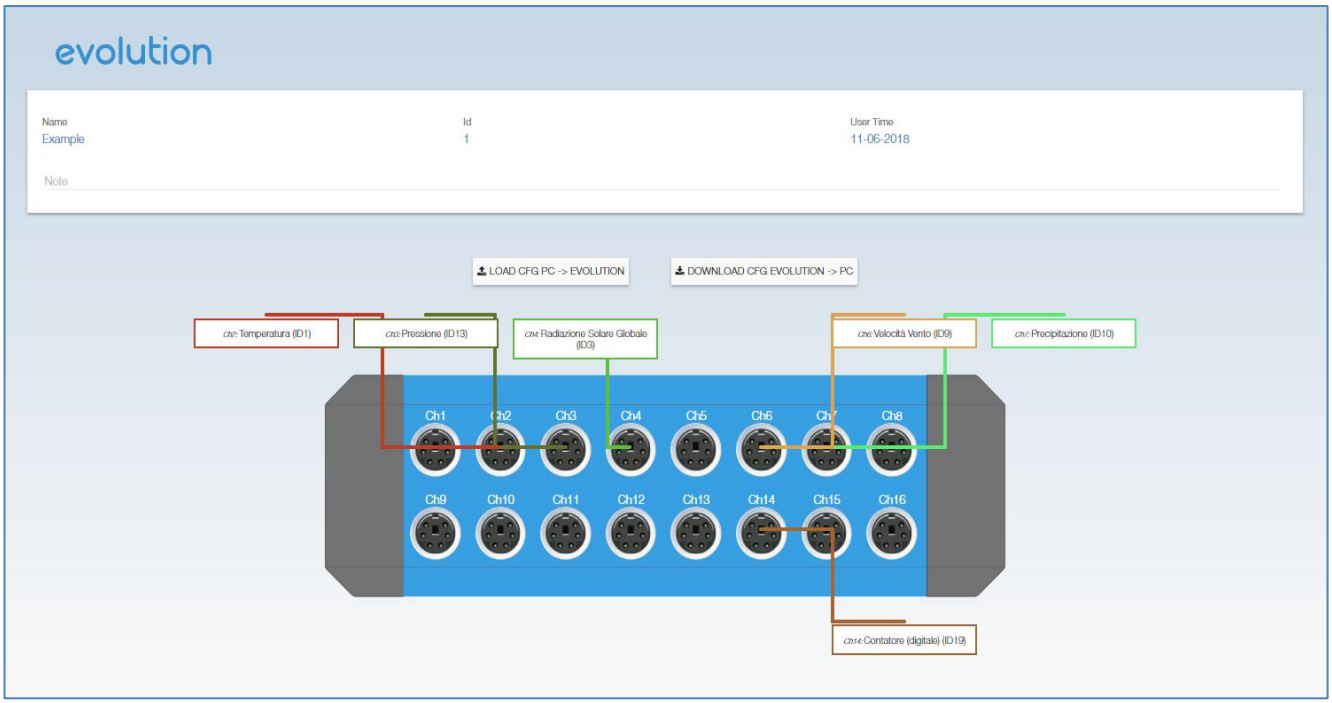
The sending of the alarm and the return is a process independent to the others and therefore asynchronous.



Press the key SAVE PAGE to finish.

8.4 Step four: Check configuration and connections

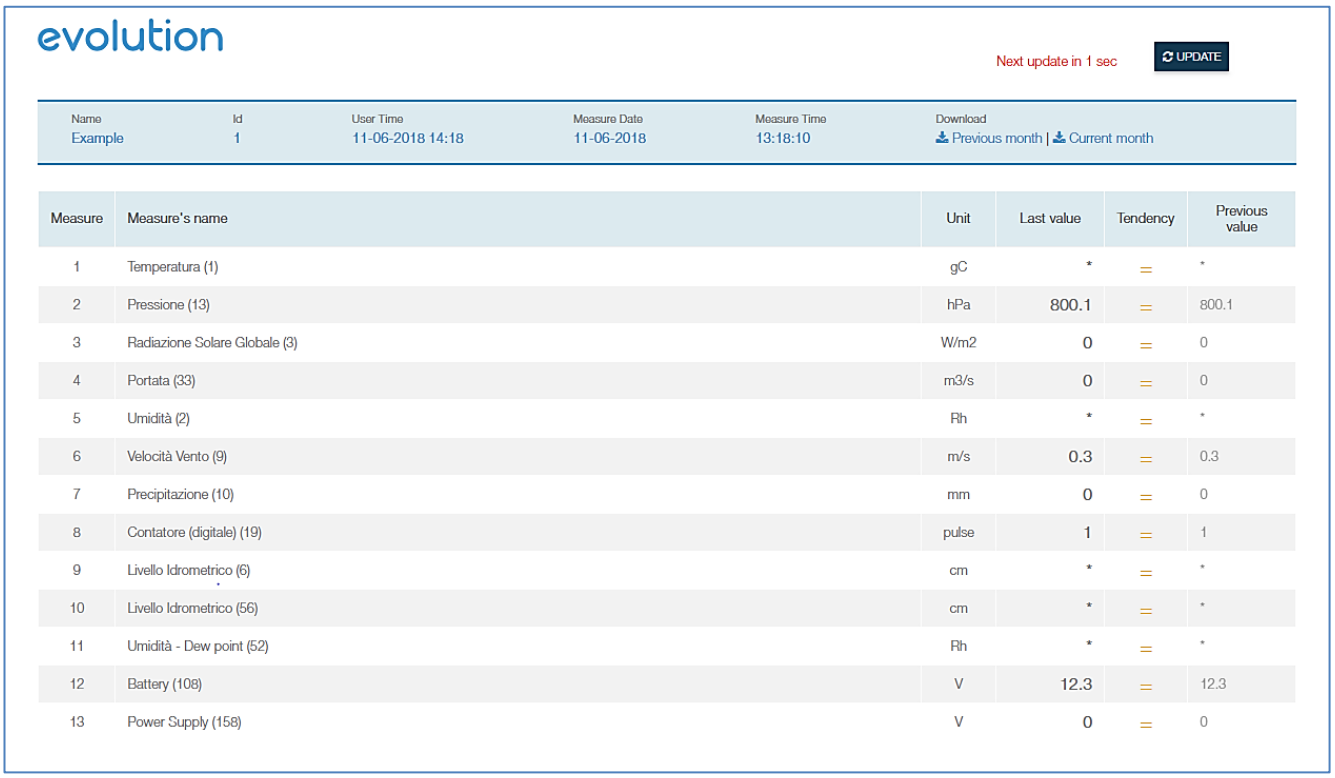
Opening *CFG Service* page, the configuration is summarized from the point of view of the connections showing with specific labels where to connect the sensors in the physical channels



8.5 Step five: Functional check

The final check can be carried out after having switched off the machine connected all the sensors and switched back on, if not already done before.

Accessing the data page you will have immediacy of the correctness of the operations carried out for the configuration of each channel, and, if everything is in place, the datalogger is ready to be put on the field.



Measure	Measure's name	Unit	Last value	Tendency	Previous value
1	Temperatura (1)	gC	*	=	*
2	Pressione (13)	hPa	800.1	=	800.1
3	Radiazione Solare Globale (3)	W/m2	0	=	0
4	Portata (33)	m3/s	0	=	0
5	Umidità (2)	Rh	*	=	*
6	Velocità Vento (9)	m/s	0.3	=	0.3
7	Precipitazione (10)	mm	0	=	0
8	Contatore (digitale) (19)	pulse	1	=	1
9	Livello Idrometrico (6)	cm	*	=	*
10	Livello Idrometrico (56)	cm	*	=	*
11	Umidità - Dew point (52)	Rh	*	=	*
12	Battery (108)	V	12.3	=	12.3
13	Power Supply (158)	V	0	=	0

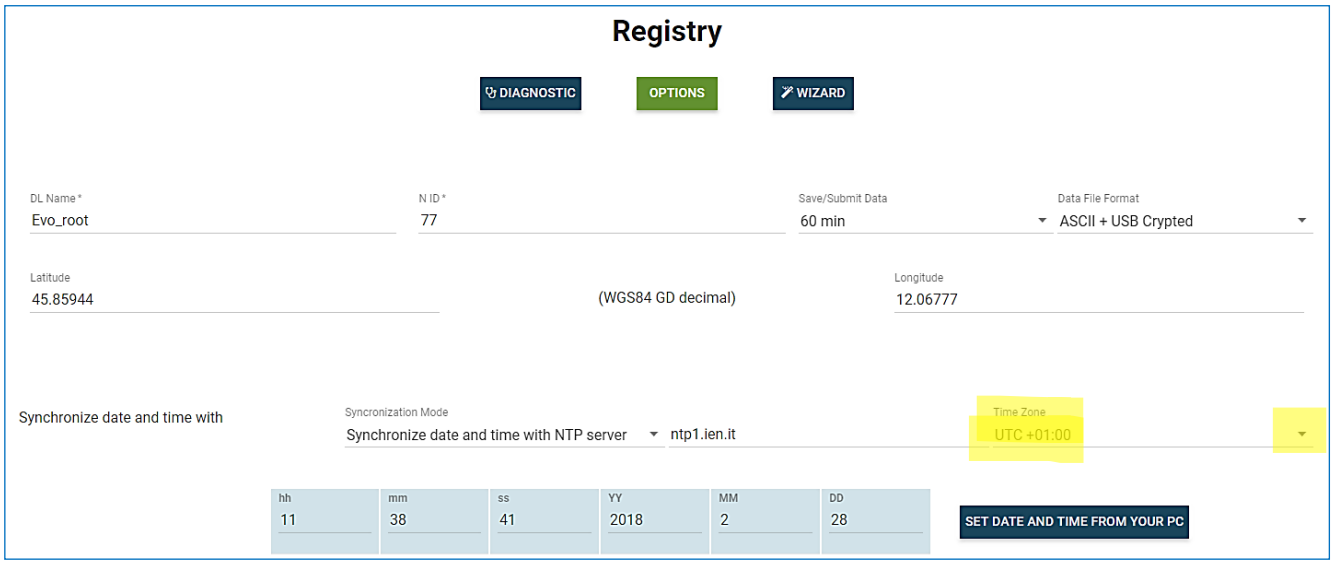
As can be seen in the figure above, the measures represented are 13 instead of 11 as configured, since the two measures for battery voltage and primary power supply, are always present and physically wired, but are available only as instantaneous and no averages.

9. Abbreviated Procedures and Tricks

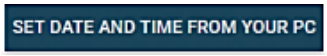
The data logger has some features that are not immediately accessible, but just as useful, described in the manual and referred to here for greater speed of use:

9.1 Set the TIME ZONE

Go to the REGISTRY section of the datalogger web interface:



Select the time zone to apply from the drop-down menu.

Press 

Press 

The time zone change occurs at midnight and, in the case of synchronization via NTP, it will be automatically aligned.

9.2 Change SMTP of email sending (use of personalized email)

To make this change it is necessary to have access credentials as administrator of the datalogger, via Telnet connection (the credentials must be requested from NESA Srl via email to tecnico@nesasrl.it specifying the need and providing the serial number of the datalogger).

- Opena Telnet o Putty connection on TCP-Ip, to the Ip address of datalogger (default 192.168.1.115)
- At login request, type the username and password provided (the credentials must be requested from NESA Srl via email to tecnico@nesasrl.it specifying the need and providing the serial number of the datalogger)
- To the command prompt type : *cd home/root/.mailrc* These info will appear:



- Change the set info as per your preference
- To test the new configuration, type:

echo "email body" | mailx –v –s "email object –a /file/folder of file/emailaddress 2>"name of result file"

Example:

echo "Email from datalogger" | mailx –v –s "File txt" –a /mnt/nand/parametri.txt name@address.com 2>result.txt



If the message is set correctly, the message mail accepted for delivery will appear



```
250-paganini37 hello [88.149.188.185], pleased to meet you
250-HELP
250-AUTH LOGIN PLAIN
250-SIZE 2000000000
250-ENHANCEDSTATUSCODES
250-PIPELINING
250-8BITMIME
250 OK
>>> AUTH LOGIN
334 VXNlcm5hbWU6
>>> c210cEBuZXNhc3JsLm10
334 UGFzc3dvcmQ6
>>> TjJzMXNybEA=
235 2.7.0 ... authentication succeeded
>>> MAIL FROM:<info@nesasrl.it>
250 2.1.0 <info@nesasrl.it> sender ok
>>> RCPT TO:<dalcin@nesasrl.it>
250 2.1.5 <dalcin@nesasrl.it> recipient ok
>>> DATA
354 OK
>>> .
250 2.0.0 wpYDeHowuMlJ9wpYDe9zvN mail accepted for delivery
>>> QUIT
221 2.0.0 paganini37 cmsmtp closing connection
root@evolution:~#
```

In other case . . . **message not sent**

```
~
root@evolution:~# echo "Email da Evolution" | mailx -v -s "File parametri" -a /m
nt/nand/parametri.txt dalcin@nesasrl.it
Resolving host authsmtp.nesasrl.it . . . done.
Connecting to 81.88.48.66:25 . . . connected.
220 paganini35 cmsmtp authsmtp ESMTP server ready
>>> EHLO localhost.localdomain
250-paganini35 hello [88.149.188.185], pleased to meet you
250-HELP
250-AUTH LOGIN PLAIN
250-SIZE 2000000000
250-ENHANCEDSTATUSCODES
250-PIPELINING
250-8BITMIME
250 OK
>>> AUTH LOGIN
334 VXNlcm5hbWU6
>>> c210cEBuZXNhc3JsLm10
334 UGFzc3dvcmQ6
>>> TjJzMXNybEBf
535 5.7.0 ...authentication rejected
smtp-server: 535 5.7.0 ...authentication rejected
"/home/root/dead.letter" 75/4304
. . . message not sent.
root@evolution:~#
```

9.3 COM ports using

COM0: reserved for RS485/SDI12

COM1: Reserved to the modem or other input devices

COM2: Service port

USB: The first USB port is identified as COM3

RS485/SDI: this port can be used or as RS485 or as SDI12 Master

9.4 TEST serial Modem with AT commands

Using the keyboard, it is possible to carry out a connection test of the serial Modem connected on COM1 to the data logger, by carrying out a connection test after having inserted the SIM in the modem and having configured the transmission parameters (see paragraph 2.3.2.4).

- Press the key  to enter in the display menu
- Scroll the menu with the arrows   until you see TEST MODEM
- Press again the key 

The data logger will perform a series of connection tests until completion, showing the conclusion of the correct test with an OK and stopping with the message ERROR if it is not possible to proceed.

NOTE:

- The most common problem for which the test fails is you forgot to remove the lock PIN code from the SIM. Therefore, make sure that you have proceeded to remove the block before inserting it into the modem.
- Be sure you have also correctly set the APN of the chosen provider.

9.5 Restore datalogger after an error on configuration

If the datalogger seems not run, the display seems freeze, there is an high possibility that you made a wrong configuration. It is possible to try a recovery following this procedure:

- Open a web connection to the datalogger and enter in its web pages, under **CFG SERVICE** tab.
- Press the button "**DOWNLOAD CFG**" and wait at least 3-5minutes following the video instruction.
- This procedure downloads the .NSA file configuration file on your PC and will regenerate the configuration fixing standard common errors. At the end the datalogger will reboot.
- Remember to close the browser as you lose the connection at reboot and open again just after if you need to check or change configuration.
- This is useful especially when, after a wrong configuration, you delete all the preconfigured measures and restart with an empty configuration



10. Diagnostic Function

Following the switching on of the datalogger as described in the paragraph 1.2, the following messages appear in the display order:

■■■■■■■■■■■■■■■■■■■■■■■■ :characters of the LCD test

Waiting ... Fw Ver. 1.14 :it shows the version of the firmware loaded in the first row

In the second line the messages appear in sequence:

evolution	: datalogger model
ip = 192.168.1.115	: ip of the datalogger
01>Parameters Ok	: control of the integrity of the parameters of calibration
02>Time zone Ok	: application of the time zone
03>LCD Setup Ok	: control of the display device
04>Expa OK	: control and activation of the analog expansion device
05>Syslogd OK	:control of the machine log activation process
06>Ad 24 OK	:functional check of the analog/digital converter
07>Julian seconds Ok . . .	: verification of the RTC and application of the second Julian
08>Digital input Ok	: control and activation of digital inputs
09>Object acq Ok	: upload and check the configuration of the logger process
10>Constant Ok	: loading and verification of the system's executive parameters
Running	: diagnostic message completed successfully
Waiting for instant data	: waiting for the first acquired measurements

The bootstrap phase, for particularly complex configurations, can exceed one minute.

If some checks are not successful, the machine stops and displays the last correct check made on the display.

Depending on the configuration, the configured measures will start cyclically alternating at the display.



10.1 Table of possible drawbacks

Problem	*Possible causes*	*Solution*
It does not switch on.	Power supply failure, low battery, incorrect power connection.	Check the wiring and the presence of at least 12Vdc of power supply between the + Vbatt and Gnd terminals in the power connector.
It crashes during start-up in one of the tests.	Internal hardware error.	Repeat the ignition of the machine. If the problem persists, contact the Nesa technical office.
The measurements appear on the display but there is no data.	Wrong configuration or incorrect connection of the sensors	Check the configuration, check the correct insertion of the sensor connection connectors or check the wiring of the sensors. Check that the sensor is not damaged
Only some measurements appear on the display, but not all.	Damaged sensors, or plugs not properly inserted; incorrect configuration.	Check the sensors, their wiring and connection, then check the configuration.
it is not possible to connect the datalogger on TCP-Ip port.	Unsuitable network cable (not cross), incorrect configuration of the IP address class of your computer, presence of firewalls or other devices to block each connection.	Check the type of Ethernet cable and its correct insertion into the PC sockets and datalogger. Check that the IP address of your computer is congruent with that of the datalogger (default 192.168.1.115, subnet mask 255.255.255.0)
I always see the same data on the web page	Browser cache not updated	Clear the browser cache from the tools menu and set the search for the most updated pages every time the web page is opened.



The configuration that is loaded does not match the one set	Browser cache not updated	Clear the browser cache from the tools menu and set the search for the most updated pages every time the web page is opened.
While the sensors are working, asterisks "*" appear in the data track.	Possible error in the configuration of the minimum and maximum validation parameters. Sensor out of scale or incorrectly working.	Check the set valuesfor the minimum and maximum measurement in the datalogger configuration. Check that the sensor works correctly.
After switching on, the display remains off.	Standby enabled, possible display breakdown, possible hardware breakdown.	Switch the machine off and on again or try connecting to the web server via LAN cable. Contact the Nesa technical office for repairs if necessary.
The Time is incorrect	The time zone is not selected, the synchronization is not made	Check the setup as per chapter 9.1 and wait until next middle night for automatic synchronization
I don’t receive email	The SMTP setup is incorrect, or the limit of email sent from default address is overloaded	Change the SMTP configuration a per chapter 9.2
I’m not sure which COM port to be use		See Chapter 9.3
Can I test the modem connection	Be sure of have connected a serial modem with AT commands on COM1	Use the procedure as per chapter 9.4
I did some mistakes on configuration and now datalogger is not well running or seems freeze	Some mistakes can interrupt the normal process and need to be restored	Use the procedure described at 9.5



11. Technical specifications

General

- 32-bit platform and open-source Linux operating system
- Built-in internet web server Apache + PhP interpreter with web application on board for datalogger setup and data display (instant values and dynamic charts, geo-localization) using web browser application, direct XLS download data. Web interface in source code.
- Multi users for different configuration on the same machine
- Data push to different FTP clients (FTP areas) by GPRS, lan cable, satellite, radio using a programmable time rate, data output on RS232, RS485, Modbus.
- Display LCD 24x2
- User keyboard for basic configuration (5 buttons)
- Battery charger inside the logger, up to 5Ah
- Battery Monitor, external power supply monitor (example solar panel) inside
- 3 levels of watch dog for security
- Data acquisition according to UNI IEC 13005

Inputs

- At least n.8/16 analogues inputs (Pt100, µV, 0÷2 Vdc, 0÷100 mV) and n.5 frequency (max 5000 Hz, ON/OFF) inputs.
- Possibility of analogue inputs extension for additional analogue inputs with specific module or connecting more datalogger on the same IP network.
- Possibility of single input expansion device on RS485 if required. Bridge connection of several units (up to 128 units) by RS485 to datalogger. User's programmable for connection of digital (max 5000 Hz) /logic On/Off status, analogue (Pt100, µV, 0÷2 Vdc, 0÷100 mV) signals.
- At least N.2 RS232 ports (RS485 if required). Possibility of n.2 additional RS232 ports (RS485 if required).
- At least N.3 USB ports (2 host, 1 slave)
- N.1 SDI-12 sensors
- N.1 Lan Ethernet 10/100 Base T RJ45
- Possibility to connect GPS device when required and to send geographical position information to FTP clients
- Possibility of IP camera connection if required and jpg files storing and transmission. Each frame (jpg) should be stored inside datalogger memory and sent to FTP clients



- Possibility to activate the internal measure of Temperature, Humidity and Pressure (sensors inside the datalogger)
- Possibility to activate WiFi connection to the IP network

Data storage

- Data storage: Instant values. Statistical calculations: Average/min/max/standard deviation. Time Max/Time Min. Totals: sum or integration time. Wind elaborations: wind direction trigonometric average (sine/cosine method), standard deviation and Turbulence, vertical temperature difference and Pasquill class stability E,F and G. Others customizable by user. All measures programmable independently for each channel over 1 min. up to 24 hrs time base.
- Serial communication programmable by user.
- At least 128 MB internal memory. Additional minimum 8 Gb external memory using USB industrial pen-drive with data backup function or external HD, SSD
- Built-in FTP server to access to the storage data
- Data output format: ASCII txt file, Excel, email, AES encrypted, (CSV, XML, SYNOP, METAR on request)

Outputs

- At least n. 4 analogue outputs (0÷2 Vdc) addressed to sensors measurement
- At least n.4 open collector outputs with programmable logics to activate external devices when required
- SMS messages over programmable events (alarms). Storing feature on datalogger memory of event date/time.

Data communication

- Supported protocols: HTTP, HTTPS, FTP, FTPS, Modbus RTU and TCP-IP, Telnet, SMTP, Socket, I2Cbus, SDI-12, SNP.

System security

- System recovery and data back-up by USB key without PC&Software
- Clock synchronisation using NTP (Network Time Protocol) or GPS
- External modules for inputs electrical protection
- Protection against polarity reverse and Transient voltage suppressor

Power supply

- Power supply by solar panel and battery. Power consumption not more than 1W

For more details and updates, see the specific datasheet



Appendix A: Data record track extract

The evolution series datalogger stores or transmits a text file in **ASCII NESA** format as standard. It is always possible to construct a customised format (see paragraph 7). Here we will explain the structure of the NESA format which, in its minimal form, is as follows:

S, ID_SENS, TIME, DATE, ID_MIS1, Type_ELAB_MIS1, DATA, ID_MIS1, Type_ELAB_MIS2, DATA, … , ID_MIS1, Type_ELAB_MISn, DATA, … , ID_MISm, Type_ELAB_MISn, DATA, #

The various fields of the track have the following definitions:

ID_SENS: it is a long integer and is unique for every datalogger terminal produced by Nesa srl

TIME: time of record in format hh, mm, ss

DATE: record date in the format dd, mm, yyyy

ID_MISm: D of the m-th measurement associated with the station/sensor.

Example: 1 = Temperature
2 = Humidity
3 = Pressure
4 = Wind Direction
5 = Wind Speed
6 = …..

Type_ELAB_MISn: ID of the n-th processing provided by the control unit associated with the m-th measurement acquired.

Example: 1 = Instant
2 = Medium
3 = Minimum
4 = Maximum
5 = Min. Minimum
6 = Min. Maximum
7 = ….

DATA: data associated with the n-th processing provided by the control unit associated with the m-th measurement acquired. The nature of the data and its formatting depend on the type of signal acquired. The record ends with #.



If there are **several records** in the same file, they are **stored on different lines**, so at the end of each line there are the characters CR (0xA) and LF (0xD).

If there are **several measurements of the same time**, the identifier of the second measurement is entered with an **offset of 50 (fifty)** added to the identification of the previous one: for example, if there are three temperatures in a configuration, the first has identifier 1, the second will have identifier 51 and the third will have identifier 101.

In data storage, if the **measurement is outside the acquisition interval**, an ***** (**asterisk**) character will be inserted in the path instead of the data.

Example:

S,000001,00,05,00,12,03,2006,1,1,16.8,1,2,16.8,#

S,000001,00,10,00,12,03,2006,1,1,16.8,1,2,16.9,#

In the example above we can see that the file consists of two records at different times, coming from the TMF terminal n.000001: two pieces of temperature data were sent, the instantaneous and the medium one.



Appendix B Tables of measurements and processing

ID	Measurement	Measurement unit
1	Temperature	°C
2	Humidity	RH%
3	Global Solar Radiation	W/m^2
4	Wind Direction	GN
5	Evaporation	mm
6	Hydrometric Level	cm
7	Phreatic Level	cm
8	Battery Voltage	Volt
9	Wind Speed	m/s
10	Rainfall	mm
11	Net Solar Radiation	W/m^2
12	Depth Of Snow	cm
13	Pressure	hPa
14	Voltage	mV
15	Evapotranspiration	mm
16	Leaf Wetting	min
17	pH	pH
18	Conductivity	uS
19	Digital counter	pulses
20	Cracking	mm
21	Inclinometer	Degrees
22	Load cell	KN
23	Redox	mV
24	Oxygen saturation	%
25	Turbidity	NTU
26	Extensimeter	mm
27	Linear displacement	mm
28	Frequency	Hz
29	CH4	ppm
30	THC	ppm
31	NMHC	ppm
32	Current	mA
33	Flow rate	m^3/s
34	CO	ppm
35	NO	ppb
36	NO_X	ppb
37	NO_2	ppb
38	O_3	ppb
39	SO_2	ppb
40	Energy	KJ/m^2

ID	Processing
1	Instant
2	Medium
3	Minimum
4	Maximum
5	Min of the Minimum
6	Min of the Maximum
7	Accumulated
8	Standard Deviation
9	Mean Square Deviation
10	Delta M1-M2 relative to T1-T2
11	Daily Average
12	Day Minimum
13	Day Maximum
14	State 0=OK 1=Pre 2=All
15	State Measurement Value



41	Auxiliary Measurement	---
42	Temp. Auxiliary	°C
43	Visibility	m
44	Duration of Sunshine	W/m^2
45	UV	W/m^2
46	n. a.	---
47	n. a.	---
48	Latitude	Lat
49	Longitude	Long



Appendix C: Warranty Terms

NESA S.R.L. WARRANTY CONDITIONS

NESA Srl guarantees its products for all factory defects, malfunctioning and breakage of the components, under the following conditions:

A) Warranty coverage

All defects in the factory, malfunctions found during use and ascertained by our laboratory as attributable to an original defect, malfunctions of use that have occurred during the warranty period. Aesthetic defects of the machine are covered by the warranty only for a time of 3 working days from the date of order placing.

The warranty does not cover products in the following cases:

1) The product is tampered with or previously repaired by laboratories not authorised by NESA Srl
2) Natural causes such as lightning, fires, floods or extraordinary events such as theft, vandalism, explosions, etc.
3) Defects due to shipping of the product. Damages are attributable to the responsible carrier.
4) Malfunction due to improper use or non-guaranteed use conditions.
5) Normal wear of the components used (e.g. chemical or mechanical deterioration)

The warranty does also not include:

1) Any damage to equipment or objects of various kinds linked to our faulty product.
2) Non-operation of a system, whose causes are attributable to incorrect operation of our product.
3) Cost of removal and shipment of the product.
4) Labour related to the removal and verification of our product.
5) Damage caused by failure to comply with the normal maintenance rules shown in the user manuals

B) Duration of the warranty

NESA guarantees the warranty of all the catalogue items purchased for **24 months** (unless otherwise specified in the certificate) from the delivery date shown in the transport document (DDT).
The warranty is to be understood as ex-works in the sense that any production defect will be repaired free of charge, while the cost of dismantling the equipment and shipping for shipment will be borne by the customer unless otherwise agreed.

C) Who benefits from the guarantee

Replacement or repair under warranty can only be requested by the holder of the order processed by NESA. Second parties or third parties do not hold the warranty and must therefore contact their retailer.

D) How to obtain a replacement or repair under warranty

To obtain the replacement of a product or to proceed with a request for a repair it is necessary to fill in the form attached below and to forward it by fax to the number 0423-985305 or e-mail info@nesasrl.it. The timing of the repair and the methods of replacement are linked to the availability of our workshop and our warehouse. The material sent for repair must **obligatorily contain an "accompanying letter"** (hereinafter reported) appropriately filled out indicating the faults found. The product must be properly packed using the original or equivalent packaging in order to preserve the instrumentation during transportation. In addition, for repairs of products out of warranty (over 24 months), NESA assures the warranty of 3 months from the date of delivery.

IMPORTANT NOTE: *By filling out the accompanying letter enclosed in all its parts, you are entitled to repair or replacement of the product. In addition, if the supply refers to SYSTEMS AND PLANTS (i.e. a set of individual products assembled) and Nesa sends the appropriate form, the warranty only takes effect if the form of regular execution of work properly stamped and signed is returned in the original.*



SEND BACK UNDER WARRANTY

DATA OF THE PRODUCT OR OF THE SYSTEM(compiled by the seller)

Product code(s): .....

or

System Code (Enter purchase order reference): .....

Nesa DDT: ..... date (delivery date): .....

Warranty expiry date: .....

Date: ..... Legible stamp and signature: .....

ACCOMPANYING LETTER of warranty repair request (completed by the purchaser)

Company name: .....

Contact person: .....

phone: .....

E-mail: .....

Detected Malfunction:

.....

.....

.....

We hereby request the repair or replacement under warranty for the above product. Having read the warranty conditions of NESA regarding the warranty on its products, I declare that the product has not previously been tampered with in any of its parts.

Date: ..... Legible stamp and signature: .....

Completion of this form in all its parts and sending by fax to the number: +39 0423 985305 are essential to start the repair process. The shipment of the product, together with this form, must be made to our office: NESA S.r.l. Via Sartori, 6/8 -31020 Vidor - TV

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